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MISO DPP 2016 February West Area Phase 3 Study

Prepared for

MISO

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Executive Summary

This report presents results of a System Impact Study (SIS) performed to evaluate the interconnection of the DPP 2016 February Phase 3 West Area Group (DPP West Area) generating facilities. The results for 2022 scenario are summarized below.

1.1 Project List

The DPP West Area study group has twenty three (23) generation projects with a combined nameplate rating of 4686 MW. The DPP West Area generating facilities are listed in Table ES-1. The modeling details and projects' slider diagrams are shown in Appendix B.

Table ES-1: Generating Facilities in DPP 2016 February West Area Group

MISO Project #	Service Type	TO	County, State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW	NR MW
J432	ERIS	XEL	Brookings, SD	Brookings 345 kV	Wind	98	98	15.29	98	0
J460	ERIS	GRE	Lincoln, MN	Brookings-H081 345 kV line	Wind	200	200	31.2	200	0
J475	NRIS	MEC	Poweshiek, IA	Montezuma 345 kV	Wind	200	200	31.2	200	200
J485	NRIS	RPU	Olmsted, MN	West Side 161 kV	CT	46.85	0	46.85	46.85	46.85
J488	ERIS	OTP	Grant, Roberts, SD	Big Stone-Ellendale 345 kV	Wind	151.8	151.8	23.68	151.8	0
J493	ERIS	OTP	Yellow Medicine, MN	Big Stone-Brookings 345 kV	Wind	150	150	23.4	150	0
J495	NRIS	ITCM	Winnebago, IA	Ledyard-Colby 345 kV line	Wind	200	200	31.2	200	50
J498	NRIS	MEC	Boone, Greene, IA	Grimes-Lehigh 345 kV	Wind	340	340	53.04	340	340
J499	NRIS	MEC	Adair, Madison, IA	Fallow-Grimes 345 kV	Wind	340	340	53.04	340	340
J500	NRIS	MEC	Adair, IA	New substation at intersection of Boone-Atchison 345 kV and Rolling Hills-Madison Co 345 kV	Wind	500	500	78	500	500
J504	NRIS	ITCM	Linn, IA	Bertram-Duane Arnold 161 kV (0.5 mi from Duane Arnold)	Solar	50	25	50	50	50
J506	ERIS	MEC	Plymouth, IA	Raun-Lakefield Jct 345 kV	Wind	200	200	31.2	200	0
J510	ERIS	OTP	Deuel, SD	Scandinavia Township	CT	266 sum / 284.5 win	0	284.5	284.5	0
J514	NRIS	ITCM	Marshall, IA	Marshalltown substation	CC	30 sum / 65 win	32.5	65	65	30

MISO Project #	Service Type	TO	County, State	Point Of Interconnection	Fuel Type	Max Output	SH MW	SPK MW	Stability MW	NR MW
J523	NRIS	ITCM	Mower, MN	Adams 161 kV	Solar	50	25	50	50	30
J524	NRIS	MEC	Webster, IA	Webster 161 kV	Solar	100	50	100	100	60
J526	ERIS	OTP	Deuel, SD	Brooking Co-Big Stone South 345 kV	Wind	300	300	46.80	300	0
J527	NRIS	MEC	Adair, IA	Booneville-Cooper 345 kV	Wind	250	250	39	250	250
J528	NRIS	MEC	Adair, IA	Rolling Hills-Madison 345 kV	Wind	200	200	31.2	200	70
J529	NRIS	MEC	Palo Alto, IA	Obrien-Kossuth 345 kV	Wind	250	250	39	250	250
J530	NRIS	MEC	Iowa, IA	Montezuma-Hills 345 kV	Wind	250	250	39	250	250
J534	NRIS	MEC	Humboldt, Kossuth, IA	Kossuth-Webster 345 kV	Wind	250	250	39	250	100
J535	NRIS	MEC	Sac, IA	J411-Lehigh 345 kV	Wind	210	210	32.76	210	84

1.2 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 February West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation projects J498, J499, J500, and J529, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

The reactive power requirement analysis results are summarized as following:

- J498, J499, J500, and J529 are exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection Service as of the System Impact Study report date. The total cost of network upgrades in the interconnection plan required for each generation project is listed in Table ES-2. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies.

Table ES-2: Total Cost of Network Upgrades for DPP 2016 February West Area Generation Projects

Project Num	ERIS Network Upgrades (\$)								NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		SNU (\$)	Total Cost (Exclude TOIF) (\$)
	MISO Thermal & Voltage	Voltage Stability	Short-circuit	OTP LPC	CIPCO Affected System	MPC Affected System	PJM Affected System	SPP Affected System		TO Network Upgrades	TO - Owned Direct assigned		
J432	\$293,239	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$5,150,000	\$2,420,000	\$0	\$5,443,239
J460	\$3,532,864	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$11,127,000	\$2,482,000	\$0	\$14,659,864
J475	\$815,248	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$11,000,000	\$2,622,000	\$664,600	\$0	\$14,437,248
J485	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,796,900	\$589,700	\$0	\$1,796,900
J488	\$4,596,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$781,960	\$1,073,300	\$0	\$5,377,960
J493	\$10,145,953	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$4,050,492	\$1,073,340	\$0	\$14,196,445
J495	\$18,068,846	\$0	\$0	\$0	\$3,538,324	\$0	\$0	\$0	\$0	\$10,119,531	\$750,218	\$0	\$28,188,377
J498	\$10,410,164	\$0	\$0	\$0	\$0	\$0	\$2,365,383	\$0	\$0	\$10,799,800	\$634,600	\$0	\$21,209,964
J499	\$3,316,714	\$0	\$0	\$0	\$0	\$0	\$2,334,722	\$0	\$0	\$11,750,200	\$634,600	\$0	\$15,066,914
J500	\$2,173,735	\$0	\$0	\$0	\$0	\$0	\$2,651,697	\$0	\$0	\$23,714,000	\$634,600	\$0	\$25,887,735
J504	\$18,425	\$0	\$0	\$0	\$823,028	\$0	\$0	\$0	\$0	\$273,780	\$353,826	\$0	\$292,205
J506	\$741,740	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$9,916,000	\$825,000	\$0	\$10,657,740
J510	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$7,682,433	\$663,687	\$0	\$7,682,433
J514	\$2,385,696	\$0	\$0	\$0	\$683,385	\$0	\$0	\$0	\$0	\$15,000	NA	\$0	\$2,400,696
J523	\$170,221	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,200,000	NA	\$0	\$1,370,221
J524	\$6,487,206	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,000,000	NA	\$0	\$7,487,206
J526	\$23,504,167	\$0	\$0	\$0	\$0	\$0	\$2,142,670	\$0	\$0	\$11,250,000	NA	\$0	\$34,754,167
J527	\$1,034,106	\$0	\$0	\$0	\$0	\$0	\$1,656,241	\$0	\$0	\$10,681,000	\$825,000	\$0	\$11,715,106
J528	\$827,198	\$0	\$0	\$0	\$0	\$0	\$43,530	\$0	\$0	\$10,681,000	\$825,000	\$0	\$11,508,198
J529	\$1,048,974	\$0	\$0	\$0	\$0	\$0	\$1,723,417	\$0	\$0	\$11,328,700	\$634,600	\$0	\$12,377,674
J530	\$639,035	\$0	\$0	\$0	\$155,263	\$0	\$1,763,923	\$0	\$0	\$10,748,000	\$825,000	\$0	\$11,387,035
J534	\$1,190,330	\$0	\$0	\$0	\$0	\$0	\$1,723,417	\$0	\$0	\$9,467,000	\$825,000	\$0	\$10,657,330
J535	\$1,065,139	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$13,188,000	\$825,000	\$0	\$14,253,139
Total (\$)	\$92,465,000	\$0	\$0	\$0	\$5,200,000	\$0	\$16,405,000	\$0	\$11,000,000	\$179,342,796	\$17,559,071	\$0	\$282,807,796

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

1.4 Per Project Summary

This section provides estimated cost of Network Upgrades on a per project basis for the 2022 scenario. The shared cost of Network Upgrades for all the generation projects are listed below.

The Interconnection Customers are required to mitigate the constraints observed from the 2022 scenario.

1.4.1 J432 Summary

Network Upgrade	Cost	J432	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$4,129	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$37,590	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$29,917	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$57,662	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$163,941	Reactive Power NU
Total Cost Per Project for ERIIS for each Project		\$293,239	

1.4.2 J460 Summary

Network Upgrade	Cost	J460	NUs Type
601054 HAZEL CK3 345 601053 HAZEL CK4 230 605753 HAZEL CKT 13.8 6	\$4,900,000	\$2,844,327	SH MISO
620327 HANKSON4 230 620329 WAHPETN4 230 1	\$1,050,000	\$99,674	SH MISO
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	\$300,000	\$8,049	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$74,730	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$56,326	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$112,589	Reactive Power NU

Network Upgrade	Cost	J460	NUs Type
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$337,169	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$3,532,864	

1.4.3 J475 Summary

Network Upgrade	Cost	J475	NUs Type
J530 POI-Hills 345 kV	\$300,000	\$115,118	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$53,435	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$22,955	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$612,929	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$10,811	Reactive Power NU
Irvine-Powershiek 161 kV	\$11,000,000	\$11,000,000	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project		\$11,815,248	

1.4.4 J485 Summary

Network Upgrade	Cost	J485	NUs Type
Not Required	\$0	\$0	
Total Cost Per Project for Actual NRIS Elections for each Project		\$0	

1.4.5 J488 Summary

Network Upgrade	Cost	J488	NUs Type
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$785,521	SH MISO
620314 BIGSTON4 230 620325 BROWNSV4 230 1	\$350,000	\$78,318	SH MISO
620314 BIGSTON4 230 652503 BLAIR 4 230 1	\$200,000	\$48,343	SH MISO
620327 HANKSON4 230 620329 WAHPETN4 230 1	\$1,050,000	\$248,664	SH MISO
620327 HANKSON4 230 620363 FORMAN 4 230 1	\$400,000	\$400,000	SH MISO
620362 OAKES 4 230 620363 FORMAN 4 230 1	\$950,000	\$950,000	SH MISO
620362 OAKES 4 230 661098 ELLENDLMVP4 230 1	\$1,650,000	\$1,650,000	SH MISO
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	\$300,000	\$6,072	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$56,218	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$42,960	Reactive Power NU

Network Upgrade	Cost	J488	NUs Type
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$84,860	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$245,043	Reactive Power NU
Total Cost Per Project for ERIIS for each Project		\$4,596,000	

1.4.6 J493 Summary

Network Upgrade	Cost	J493	NUs Type
616002 GRE-JOHNJCT7 115 620216 ORTONVL7 115 1	\$16,850,000	\$5,603,545	SH MISO
616002 GRE-JOHNJCT7 115 652555 MORRIS 7 115 1	\$8,725,000	\$2,901,539	SH MISO
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$821,453	SH MISO
620314 BIGSTON4 230 620325 BROWNSV4 230 1	\$350,000	\$90,564	SH MISO
620314 BIGSTON4 230 652503 BLAIR 4 230 1	\$200,000	\$50,552	SH MISO
620327 HANKSON4 230 620329 WAHPETN4 230 1	\$1,050,000	\$233,879	SH MISO
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	\$300,000	\$6,243	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$57,039	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$44,968	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$87,186	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$248,985	Reactive Power NU
Total Cost Per Project for ERIIS for each Project		\$10,145,953	

1.4.7 J495 Summary

Network Upgrade	Cost	J495	NUs Type
Hazleton-Washburn 161 kV	\$16,700,000	\$16,700,000	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$7,422	SH MISO
Lundquist-Deere Northeast 161 kV	\$100,000	\$100,000	SH MISO
Deere Northeast-Washburn 161 kV	\$100,000	\$100,000	SH MISO
Liberty-Dundee 161 kV	\$5,000,000	\$3,538,324	SH CIPCO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$71,687	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$38,168	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$106,516	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$945,053	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each		\$21,607,170	

Network Upgrade	Cost	J495	NUs Type
Project			

1.4.8 J498 Summary

Network Upgrade	Cost	J498	NUs Type
Boone Jct-Sub T Fort Dodge 161 kV	\$7,000,000	\$7,000,000	SH MISO
Poweshiek-Reasnor 161 kV	\$290,000	\$76,715	SH MISO
Grimes-Sycamore 345 kV #2	\$1,200,000	\$519,976	SH MISO
Grimes-Beaver Creek 345 kV	\$600,000	\$318,003	SH MISO
Granger Tap-108th & 54th 161 kV	\$2,000,000	\$889,769	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$34,333	SH MISO
Franklin-Wall Lake 161 kV	\$200,000	\$135,553	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$56,844	PJM AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,166,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$142,539	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$175,383	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$290,525	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$470,728	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$499,180	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$12,775,548	

1.4.9 J499 Summary

Network Upgrade	Cost	J499	NUs Type
Poweshiek-Reasnor 161 kV	\$290,000	\$88,935	SH MISO
Grimes-Sycamore 345 kV #2	\$1,200,000	\$680,024	SH MISO
Granger Tap-108th & 54th 161 kV	\$2,000,000	\$1,110,231	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$43,684	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$56,062	PJM AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,138,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$140,660	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$207,536	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$372,767	Reactive Power NU

Network Upgrade	Cost	J499	NUs Type
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$578,935	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$234,601	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$5,651,436	

1.4.10 J500 Summary

Network Upgrade	Cost	J500	NUs Type
Poweshiek-Reasnor 161 kV	\$290,000	\$124,351	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$60,570	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$2,488,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$163,697	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$340,912	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$533,626	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$802,791	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$311,486	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$4,825,432	

1.4.11 J504 Summary

Network Upgrade	Cost	J504	NUs Type
Mqoketa-Wyoming 161 kV	\$200,000	\$26,012	SH CIPCO
Liberty-Dundee 161 kV	\$5,000,000	\$797,016	SH CIPCO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$4,232	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$8,546	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$5,646	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$841,453	

1.4.12 J506 Summary

Network Upgrade	Cost	J506	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$12,066	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$97,214	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$100,935	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$167,016	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$364,509	Reactive Power NU
Total Cost Per Project for ERIS for each Project		\$741,740	

1.4.13 J510 Summary

Network Upgrade	Cost	J510	NUs Type
Not Required		\$0	
Total Cost Per Project for ERIS for each Project		\$0	

1.4.14 J514 Summary

Network Upgrade	Cost	J514	NUs Type
Marshalltown-Blairstown 115 kV	\$600,000	\$600,000	SH MISO
Blairstown-Prairie Crk 115 kV	\$1,700,000	\$1,700,000	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$1,199	SH MISO
Mqoketa-Wyoming 161 kV	\$200,000	\$18,725	SH CIPCO
Liberty-Dundee 161 kV	\$5,000,000	\$664,660	SH CIPCO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$11,800	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$8,084	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$16,929	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$47,685	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$3,069,081	

1.4.15 J523 Summary

Network Upgrade	Cost	J523	NUs Type
Adams 345-161-13.8 kV xfmr	\$100,000	\$100,000	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$6,580	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$7,134	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$56,507	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$170,221	

1.4.16 J524 Summary

Network Upgrade	Cost	J524	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$21,174	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$3,437	SH MISO
Webster-Sub T Ft Dodge 161 kV	\$6,250,000	\$6,250,000	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$22,816	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$28,506	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$47,635	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$113,639	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$6,487,206	

1.4.17 J526 Summary

Network Upgrade	Cost	J526	NUs Type
601054 HAZEL CK3 345 601053 HAZEL CK4 230 605753 HAZEL CKT 13.8 6	\$4,900,000	\$2,055,673	SH MISO
615529 GRE-PANTHER4 230 619940 HUC-MCLEOD 4 230 1	\$1,000,000	\$1,000,000	SH MISO
616002 GRE-JOHNJCT7 115 620216 ORTONVL7 115 1	\$16,850,000	\$11,246,455	SH MISO
616002 GRE-JOHNJCT7 115 652555 MORRIS 7 115 1	\$8,725,000	\$5,823,461	SH MISO
Big Stone 230-115-13.8 kV xfmr	\$3,250,000	\$1,643,026	SH MISO
620314 BIGSTON4 230 620325 BROWNSV4 230 1	\$350,000	\$181,117	SH MISO
620314 BIGSTON4 230 652503 BLAIR 4 230 1	\$200,000	\$101,106	SH MISO
620327 HANKSON4 230 620329 WAHPETN4 230 1	\$1,050,000	\$467,782	SH MISO
620329 WAHPETN4 230 658109 FERGSFL4 230 1	\$100,000	\$100,000	SH MISO
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	\$300,000	\$12,428	SH MISO

Network Upgrade	Cost	J526	NUs Type
Twin Branch–Argenta 345 kV	\$208,000	\$51,564	PJM AFS
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,962,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$129,106	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$113,681	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$89,320	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$173,588	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$496,528	Reactive Power NU
Total Cost Per Project for ERIIS for each Project		\$25,646,837	

1.4.18 J527 Summary

Network Upgrade	Cost	J527	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$30,834	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,554,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$102,241	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$168,141	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$270,924	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$408,583	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$155,623	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,690,347	

1.4.19 J528 Summary

Network Upgrade	Cost	J528	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$24,689	SH MISO
Twin Branch–Argenta 345 kV	\$208,000	\$43,530	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$134,778	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$216,876	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$327,174	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$123,681	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$870,728	

1.4.20 J529 Summary

Network Upgrade	Cost	J529	NUs Type
Bondurant-Montezuma 345 kV	\$300,000	\$13,533	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,617,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$106,417	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$105,646	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$100,644	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$188,821	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$640,330	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,772,391	

1.4.21 J530 Summary

Network Upgrade	Cost	J530	NUs Type
J530 POI-Hills 345 kV	\$300,000	\$184,882	SH MISO
Parnell-J438 POI 161 kV	\$100,000	\$100,000	SH MISO
Mqoketa-Wyoming 161 kV	\$200,000	\$155,263	SH CIPCO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,655,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$108,923	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$47,119	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$307,033	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,558,221	

1.4.22 J534 Summary

Network Upgrade	Cost	J534	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$101,961	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$14,780	SH MISO
Benton–T-094 345 kV (N5106)	\$19,000,000	\$1,617,000	PJM AFS
Benton–T-094 345 kV (N5312.3)	\$1,000,000	\$106,417	PJM AFS
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$106,804	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$111,693	Reactive Power NU

Network Upgrade	Cost	J534	NUs Type
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$205,752	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$649,340	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$2,913,747	

1.4.23 J535 Summary

Network Upgrade	Cost	J535	NUs Type
Grimes-Beaver Creek 345 kV	\$600,000	\$158,862	SH MISO
Bondurant-Montezuma 345 kV	\$300,000	\$16,532	SH MISO
Franklin-Wall Lake 161 kV	\$200,000	\$64,447	SH MISO
25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000	\$106,658	Reactive Power NU
25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000	\$140,805	Reactive Power NU
100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000	\$227,591	Reactive Power NU
150 Mvar switched cap bank at Blackhawk 345 kV (636199) and 50 Mvar switched cap bank at Midport 161 kV (636202)	\$6,000,000	\$350,243	Reactive Power NU
Total Cost Per Project for Actual NRIS Elections for each Project		\$1,065,139	

1.5 Study Compliance with NERC FAC-002-2 Standard

This DPP 2016 February West Area Phase 3 study was completed in compliance with NERC FAC-002-2:

R1.1: The reliability impact of the new interconnection, or materially modified existing interconnection, on affected system(s).

Section 2 covers steady-state analysis results which include thermal and voltage constraints impacted by the DPP West Area generating facilities. Thermal and voltage upgrades required to interconnect the new generating facilities are also identified.

Section 3.1 covers reliability impact of the new generating facilities in the CIPCO and MPC affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.2 covers reliability impact of the new generating facilities per OTP LPC. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.3 covers reliability impact of the new generating facilities in the PJM affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 3.4 covers reliability impact of the new generating facilities in the SPP affected systems. Network Upgrades required to interconnect the new generating facilities are also identified.

Section 4 covers transient stability analysis results.

Section 5 covers voltage stability (PV) analysis on the MWEX System Operating Limit (SOL). No additional Network Upgrades are required for MWEX voltage stability in the 2022 scenario.

Section 6 covers short circuit reliability impact of the new generating facilities.

Section 7 covers Deliverability reliability impact of the new NRIS generating facilities.

R1.2: Adherence to applicable NERC Reliability Standards; regional and Transmission Owner planning criteria; and Facility interconnection requirements.

Sections 2.3-2.5, Section 3, and Section 4 all cover NERC Reliability Standard TPL-001-4.

Section 3.1 covers CIPCO and MPC's system planning criteria.

Section 3.2 covers OTP LPC.

Section 3.3 covers PJM system planning criteria.

Section 3.4 covers SPP system planning criteria.

Section 5 (voltage stability analysis) covers individual system planning criteria (ATC).

Section 7 covers MISO system planning criteria.

R1.3: Steady-state, short-circuit, and dynamics studies, as necessary, to evaluate system performance under both normal and contingency conditions.

Section 2 covers MISO steady-state assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.1 covers CIPCO and MPC's affected system assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.2 covers OTP's LPC assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.3 covers PJM affected system assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 3.4 covers SPP affected system assessment including NERC category P0 to P7 contingencies (TPL-001-4).

Section 4 covers transient stability studies under NERC category P0 to P7 contingencies (TPL-001-4).

Section 6 covers short-circuit assessment.

Section 7 covers MISO deliverability study (steady-state assessment) including NERC category P0 to P7 contingencies (TPL-001-4).

R1.4: Study assumptions, system performance, alternatives considered, and coordinated recommendations. While these studies may be performed independently, the results shall be evaluated and coordinated by the entities involved.

Section 2.2, Section 2.3, Section 2.4, Section 2.5, Section 4.2, Section 4.3, and Section 4.4 cover study assumptions and system performance criteria.

Jointly coordinated recommendations can be found in Sections 3.1 (MISO, CIPCO, and MPC), Section 3.2 (MISO and OTP), Section 3.3 (MISO and PJM), Section 3.4 (MISO and SPP), and Section 5 (MISO and ATC). Results in Section 2, 3, 4, 6, and 7 have also been reviewed by PJM, SPP, CIPCO, and MPC.

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Introduction

Twenty three (23) generation projects, listed in Appendix 0, Table A-1, have requested to interconnect to the MISO transmission network in the West Area and have advanced to the Definitive Planning Phase (DPP) 2016 February Phase 3 study (DPP West Area). Generating facilities J432, J460, J488, J493, J506, J510, and J526 have requested Energy Resource Interconnection Service (ERIS), and other generating facilities have requested both ERIS and Network Resource Interconnection Service (NRIS).

This report presents the study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generating facilities in the DPP West Area Phase 3 study.

The study was performed under the direction of MISO by Siemens PTI and an ad hoc study group. The ad hoc study group was formed to review the study scope, methodology, models and results. The ad hoc study group consisted of representatives from the interconnection customers and the following utility companies – Ameren, American Transmission Company, Basin Electric Power, Cedar Falls Utilities, Central Iowa Power Cooperative, City of Springfield (IL) Water Light & Power, Columbia (MO) Water and Light, Commonwealth Edison, Corn Belt Power Cooperative, Dairyland Power, Great River Energy, ITC Midwest, Lincoln Electric System, Manitoba Hydro, MidAmerican Energy Company, MISO, Minnesota Power, Minnkota Power, Missouri River Energy Services, Montana-Dakota Utilities Co., Muscatine Power & Water, Nebraska Public Power District, Northwestern Public Service, Omaha Public Power District, Otter Tail Power, PJM, Southern Illinois Power Cooperative, Southern Minnesota Municipal Power Agency, SPP, Western Area Power Administration, and Xcel Energy.

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Steady-State Analysis

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generating facilities in the DPP 2016 February Phase 3 West Area group (DPP West Area) to the transmission system.

2.1 Study Procedure

2.1.1 Computer Programs

Steady-state analyses were performed using PSS[®]E version 33.11 and PSS[®]MUST version 12.0.

2.1.2 Study Methodology

Study cases representing 2022 summer shoulder (SH) and summer peak (SPK) system conditions with generating facilities in the DPP 2016 February Phase 3 West Area group (DPP West Area) were developed from the MTEP17 steady state models with LBA dispatch. Benchmark cases were created by removing the DPP West Area generating facilities from the study cases. Nonlinear (AC) contingency analysis was performed on the benchmark and study cases, and the incremental impact of the DPP West Area generating facilities was evaluated by comparing the steady-state performance of the transmission system in the benchmark and study cases. Network upgrades were identified to mitigate any constraints.

2.2 Model Development

2.2.1 Study Cases

Power flow study cases representing 2022 summer shoulder and summer peak conditions were developed from the MTEP17 steady state models with LBA dispatch.

The study cases for DPP 2016 February Phase 3 West Area study were created as follows:

- MISO Prior queued generation projects and their associated Network Upgrades (NU) were modeled.
- DPP 2016 February generation projects in the West Area (DPP West Area, Table A-1) were modeled and dispatched.
- DPP 2016 February generation projects in the Central Area (Table A-4), Michigan Area (Table A-5), and Wisconsin Area (Table A-6) were modeled and dispatched.
- For MISO generation projects, their output was sunk to the MISO Classic (Appendix A.3, Table A-8), where generation was scaled uniformly;
- PJM generation projects were modeled and dispatched. The generation output was sunk to the PJM market (Appendix A.4, Table A-9), where generation was scaled uniformly.

- SPP generation projects were modeled and dispatched. The generation output was sunk to the SPP market (Appendix A.5, Table A-10), where generation was scaled uniformly.
- Network Upgrades based on DPP 2015 August West J298 withdrawn restudy were modeled.
- Due to slow voltage recovery at Hickory Creek 345 kV bus under several disturbances, the Hickory Creek–Cardinal 345 kV project (MVP project 3127 with in-service date of 12/31/2023) is required and was included in the 2022 models; The Hickory Creek–Cardinal 345 kV project has an in-service date of 12/31/2023.
- In summer shoulder steady state and stability cases, Big Stone generator was turned on and dispatched at Pmin (163 MW), and Coyote generator was turned on and dispatched at Pmin (217 MW).
- Models were further reviewed by the Ad Hoc study members (transmission owners and customers). Model corrections and changes were made based on the comments and feedback. These modeling changes are listed in Appendix A.2.
- Adjusted Square Butte DC to match the total output of the Bison (Bison 1 to 6) and Oliver County (Oliver County 1 and 2) wind farms.
- Adjusted CU DC to match the total output of Coal Creek generation units #1 and #2.
- Set MHEX interface transfer level at 492 MW in summer shoulder and 1406 MW in summer peak cases.

The study power flow cases were solved with transformer tap adjustment enabled, area interchange disabled, phase shifter adjustment enabled and switched shunt adjustment enabled.

The interface transfer levels in the 2022 study cases are summarized in Table 2-1.

Table 2-1: Interface Transfer Levels in 2022 Steady State Study Cases

Interface	2022 SH Case (MW)	2022 SPK Case (MW)
MHEX	492	1407
MWEX	1203	636
Arrowhead – Stone Lake 345 kV	458	236

2.2.2 Benchmark Cases

Benchmark cases are used to benchmark system performance without the DPP West Restudy generating facilities, and were created by removing the DPP West Area generating facilities (Table A-1) from the study cases. The MISO Classic was used for power balance, where generation was scaled uniformly.

2.3 Contingency Criteria

A variety of contingencies were considered for steady-state analysis:

- NERC Category P0 with system intact (no contingencies)

- NERC Category P1 contingencies
 - o NERC Category P1 contingencies, at buses with a nominal voltage of 69 kV and above, in the following areas: CWLD (area 333), AMMO (area 356), AMIL (area 357), CWLP (area 360), SIPC (area 361), WEC (area 295), WEC MI (area 296), XCEL (area 600), MP (area 608), SMMPA (area 613), GRE (area 615), OTP (area 620), ITCM (area 627), MPW (area 633), MEC (area 635), MDU (area 661), BEPC-MISO (area 663), MHEB (area 667), DPC (area 680), ALTE (area 694), WPS (area 696), MGE (area 697), UPPC (area 698), CE(area 222), NPPD (area 640), OPPD (area 645), LES (area 650), WAPA (area 652), BEPC-SPP (area 659), AECI (area 330), MIPU(area 540), KCPL (area 541), KACY (area 542), INDN (area 545).
 - o Multiple-element NERC Category P1 contingencies, in Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P1 contingencies are listed in Appendix A.6.
- NERC Category P2-P7 contingencies
 - o Selected NERC Category P2-P7 contingencies provided by the Ad Hoc Study Group, in the study region of Dakotas, Illinois, Iowa, Minnesota, Missouri, and Wisconsin. These specified Category P2-P7 contingencies are listed in Appendix A.6.

For all contingency and post-disturbance analyses, cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment disabled (fixed) and switched shunt adjustment enabled.

2.4 Monitored Elements

The study area is defined in Table 2-2. Facilities in the study area were monitored for system intact and contingency conditions. Under NERC category P0 conditions (system intact) branches were monitored for loading above the normal (PSS[®]E rate A) rating. Under NERC category P1-P7 conditions, branches were monitored for loading as shown in the column labeled "Post-Disturbance Thermal Limits".

Table 2-2: Monitored Elements

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
AECI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
AMIL	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.05/0.95
AMMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.05/0.95
ATCLLC	69 kV and above	95% of Rate A	95% of Rate B	1.05/0.95	1.10/0.90
BEPC-MISO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
BEPC-SPP	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
CWLP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
CE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
DPC	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90

Owner / Area	Monitored Facilities	Thermal Limits ¹		Voltage Limits ²	
		Pre-Disturbance	Post-Disturbance	Pre-Disturbance	Post-Disturbance
GMO	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
GRE	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.92
INDN	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
ITCM	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.95	1.10/0.93
KACY	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
KCPL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
LES	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MDU	57 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
MEC	69 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.96/0.95	1.05/0.96/0.95 ³
MHEB	69 kV and above	100% of Rate A	100% of Rate B	1.12/1.1/1.07/1.05/1.04/0.99/0.97/0.96/0.95	1.15/1.10/0.94/0.90
MP	69 kV and above	100% of Rate A	100% of Rate B	1.05/1.00	1.10/0.95
MPW	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.06/0.92
NPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OPPD	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
OTP	40 kV and above	100% of Rate A	100% of Rate B	1.07/1.05/0.97	1.10/0.92
PPI	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.95
SIPC	69 kV and above	100% of Rate A	100% of Rate B	1.07/0.95	1.09/0.91
SMPMA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
WAPA	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.10/0.90
XEL	69 kV and above	100% of Rate A	100% of Rate B	1.05/0.95	1.05/0.92

Notes

- 1: PSS®E Rate A, Rate B or Rate C
- 2: Limits dependent on nominal bus voltage
- 3: For facilities in Cedar Falls Utilities or Ames Municipal Utilities, post-contingency voltage limits are 1.05/0.94 for >200 kV, and 1.05/0.93 for others.

2.5 Performance Criteria

A branch is considered as a thermal injection constraint if the branch is loaded above its applicable normal or emergency rating for the post-change case, and any of the following conditions are met:

- 1) the generator (NR/ER) has a larger than 20% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, or

- 2) the megawatt impact due to the generator is greater than or equal to 20% of the applicable rating (normal or emergency) of the overloaded facility, or
- 3) the overloaded facility or the overload-causing contingency is at generator's outlet, or
- 4) for any other constrained facility, where none of the study generators meet one of the above criteria in 1), 2), or 3), however, the cumulative megawatt impact of the group of study generators (NR/ER) is greater than 20% of the applicable rating, then only those study generators whose individual MW impact is greater than 5% of the applicable rating and has DF greater than 5% (OTDF or PTDF) will be responsible for mitigating the cumulative MW impact constraint.

A bus is considered a voltage constraint if both of the following conditions are met. All voltage constraints must be resolved before a project can receive interconnection service.

- 1) the bus voltage is outside of applicable normal or emergency limits for the post-change case, and
- 2) the change in bus voltage is greater than 0.01 per unit.

All DPP 2016 February Phase 3 West Area study generators must mitigate thermal injection constraints and voltage constraints in order to obtain unconditional Interconnection Service.

Further, all generators requesting Network Resource Interconnection Service (NRIS) must mitigate constraints found by using the deliverability algorithm, to meet the system performance criteria for NERC category P0-P1 events, if the constraint demonstrates an incremental flow caused by the generator equal to or greater than 5% of the generator's maximum dispatch level in each case.

2.6 Reactive Power Requirements for Non-Synchronous Generation (FERC Order 827)

For non-synchronous generation projects in the DPP 2016 February West Area study group, if they do not have signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

Except for generation projects J498, J499, J500, and J529, all other non-synchronous generation projects in this study group are required to meet the reactive power requirements per FERC Order 827.

Collector system and shunt compensation of DPP West projects are modeled, which are listed in Appendix A.1, Table A-3. An analysis was performed to study the reactive power requirements (FERC Order 827) for the non-synchronous generation projects in the DPP 2016 February West study group. The analysis was performed as follows:

Step 1: Verify whether total dynamic reactive power (reactive power from generators and dynamic compensation devices) in the plant can meet the dynamic reactive power range of 0.95 leading to 0.95 lagging at the generator terminal bus. The verification in Step 1 was performed when generator data was submitted and modeled.

Step 2: Verify whether total reactive power (reactive power from generators, dynamic compensation devices, and static compensation devices) in the plant can meet the dynamic

reactive power range of 0.95 leading to 0.95 lagging at the high-side of the generator substation. The testing procedure in Step 2 is described in the following:

- Lock the high-side of the generator substation at 1.0 pu voltage by adding a fictitious SVC. This is to ensure that the test result is not affected by the system condition.
- Lock the reactive power output of the generator to the maximum limit (Q_{max}). Make sure all shunt compensation devices within the substation are at the maximum capacitive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 lagging requirement.
- Lock the reactive power output of the generator to the minimum limit (Q_{min}). Make sure all shunt compensation devices within the substation are at the maximum reactive output. Adjust transformer tap to ensure bus voltages within the substation are within 0.95 – 1.05 pu range. Measure real power and reactive power from the generator plant to the high-side of the generator substation. Calculate the power factor to verify if it meets the 0.95 leading requirement.

Appendix C lists reactive power requirement analysis results for the DPP West generation projects. The results are summarized as following:

- J498, J499, J500, and J529 are exempted from the FERC Order 827 reactive power requirements.
- All other non-synchronous generation projects can meet the reactive power requirements per FERC Order 827.

2.7 2022 Contingency Analysis Results

The incremental impact of the proposed interconnection on individual facilities was evaluated by comparing flows and voltages between benchmark case (without DPP West Area projects) and study case (with DPP West Area projects). Analysis was performed in the 2022 summer shoulder and summer peak scenarios using PSS[®]E and PSS[®]MUST.

2.7.1 2022 Summer Shoulder Contingency Analysis Results

2.7.1.1 System Intact Conditions

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table D-1, and voltage constraints are listed in Table D-2.

2.7.1.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies.

Short Circuit Ratio (SCR) is a measure of system strength relative to the rating of the renewable plant and is used by manufacturers to screen for weak grid risks. Composite SCR (CSCR) considers the grid strength as seen by all electrically close converters and is used for renewable plants. CSCR is calculated as the ratio of the composite short-circuit MVA at the points of interconnection (POI) of all renewable plants in a given area to their combined MW rating. When Big Stone South - J488 POI 345 kV is out of service, CSCR of the renewable generation (J436, J437, and J488) is around 1.217 (Table 2-3). Further study is

recommended to provide evidence that the J488 wind farm can operate at the CSCR of 1.217.

Table 2-3: Composite SC MVA and SCR of J488

System Condition	Composite SC MVA	Composite SCR
P0: System Intact	1475	3.265
P1: Big Stone South-POI of J488	550	1.217

Four category P2-P7 contingencies (Table D-7) were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these non-converged contingencies.

For P1 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table D-3, and voltage constraints are listed in Table D-4.

For P2-P7 contingencies in the 2022 summer shoulder scenario, thermal constraints are listed in Table D-5, and voltage constraints are listed in Table D-6.

For all non-converged contingencies (Table D-7), the dc thermal results are listed in Table D-8.

2.7.1.3 Worst Thermal Constraints in 2022 Summer Shoulder

Table 2-4 lists worst thermal constraints in the 2022 summer shoulder scenario.

2.7.1.4 Worst Voltage Constraints in 2022 Summer Shoulder

Table 2-5 lists worst voltage constraints in the 2022 summer shoulder scenario.

Table 2-4: 2022 Shoulder Thermal Constraints, Maximum Screened Loading

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
75730 J530 POI 345 636400 HILLS 3 345 1	1010.0	MEC	1126.1	111.5	CEII Redacted	P1	J475,J530
75730 J530 POI 345 636400 HILLS 3 345 1	1010.0	MEC	1133.2	112.2	CEII Redacted	P2-P7	J475,J530
601002 ADAMS 3 345 631046 ADAMS 5 161 605739 ADAMS1 9 13.8 9	334.6	XEL	337.9	101.0	CEII Redacted	P2-P7	J523
601006 SPLT RK3 345 652537 WHITE 3 345 1	717.1	XEL WAPA	902.3	125.8	CEII Redacted	P1	J432,J460,J488,J493, J526
601006 SPLT RK3 345 652537 WHITE 3 345 1	717.1	XEL WAPA	902.3	125.8	CEII Redacted	P2-P7	J432,J460,J488,J493, J526
601031 BRKNGCO3 345 652537 WHITE 3 345 2	788.8	XEL WAPA	946.7	120.0	CEII Redacted	P2-P7	J432,J488,J493,J526
601054 HAZEL CK3 345 601053 HAZEL CK4 230 605753 HAZEL CKT 13.8 6	386.0	XEL	428.9	111.1	CEII Redacted	P2-P7	J460,J526
602008 MINVALT4 230 652550 GRANITF4 230 1	259.0	XEL WAPA	260.8	100.7	CEII Redacted	P2-P7	J526
615529 GRE-PANTHER4 230 619940 HUC-MCLEOD 4 230 1	321.9	GRE	421.7	131.0	CEII Redacted	P2-P7	J526

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
616002 GRE-JOHNJCT7 115 620216 ORTONVL7 115 1	108.0	MRES GRE OTP	142.8	132.2	CEII Redacted	P1	J526,J493
616002 GRE-JOHNJCT7 115 620216 ORTONVL7 115 1	108.0	MRES GRE OTP	146.9	136.0	CEII Redacted	P2-P7	J526
616002 GRE-JOHNJCT7 115 652555 MORRIS 7 115 1	92.0	GRE WAPA OTP	131.0	142.4	CEII Redacted	P1	J526,J493
616002 GRE-JOHNJCT7 115 652555 MORRIS 7 115 1	92.0	GRE WAPA OTP	135.0	146.7	CEII Redacted	P2-P7	J526
620214 BIGSTON7 115 620314 BIGSTON4 230 620316 BIGSTON9 13.8 1	262.9	OTP	275.6	104.8	CEII Redacted	P1	J493
620214 BIGSTON7 115 620314 BIGSTON4 230 620316 BIGSTON9 13.8 1	262.9	OTP	294.7	112.1	CEII Redacted	P2-P7	J488,J493,J526
620314 BIGSTON4 230 620214 BIGSTON7 115 620316 BIGSTON9 13.8 1	291.2	OTP	301.2	103.4	CEII Redacted	P2-P7	J488,J493,J526
620314 BIGSTON4 230 620325 BROWNSV4 230 1	285.0	OTP MDU	290.7	102.0	CEII Redacted	P1	J526

Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
620314 BIGSTON4 230 620325 BROWNSV4 230 1	285.0	OTP MDU	322.3	113.1	CEII Redacted	P2-P7	J488,J526
620314 BIGSTON4 230 652503 BLAIR 4 230 1	438.0	OTP BEPC	555.1	126.7	CEII Redacted	P1	J488,J493,J526
620314 BIGSTON4 230 652503 BLAIR 4 230 1	438.0	OTP BEPC	538.0	122.8	CEII Redacted	P2-P7	J488,J493,J526
620327 HANKSON4 230 620329 WAHPETN4 230 1	368.2	OTP	479.3	130.2	CEII Redacted	P1	J488,J493,J526,J460
620327 HANKSON4 230 620329 WAHPETN4 230 1	368.2	OTP	487.3	132.3	CEII Redacted	P2-P7	J488,J493,J526
620327 HANKSON4 230 620363 FORMAN 4 230 1	311.0	OTP	365.5	117.5	CEII Redacted	P1	J488
620327 HANKSON4 230 620363 FORMAN 4 230 1	311.0	OTP	365.6	117.6	CEII Redacted	P2-P7	J488
620329 WAHPETN4 230 658109 FERGSFL4 230 1	351.0	OTP MRES	353.3	100.7	CEII Redacted	P2-P7	J526
620362 OAKES 4 230 620363 FORMAN 4 230 1	361.3	OTP	435.2	120.5	CEII Redacted	P1	J488
620362 OAKES 4 230 620363 FORMAN 4 230 1	361.3	OTP	435.2	120.5	CEII Redacted	P2-P7	J488
620362 OAKES 4 230 661098	362.6	OTP	439.0	121.1	CEII Redacted	P1	J488

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
ELLENDLMVP4 230 1		MDU					
620362 OAKES 4 230 661098 ELLENDLMVP4 230 1	362.6	OTP MDU	438.9	121.1	CEII Redacted	P2-P7	J488
631004 M-TOWN 7 115 631013 BLRSTWN7 115 1	106.0	ITCM	114.4	108.0	CEII Redacted	P1	J514
631013 BLRSTWN7 115 631017 PRAR CK7 115 1	81.0	CIPCO ITCM	87.5	108.0	CEII Redacted	P0	J514
631013 BLRSTWN7 115 631017 PRAR CK7 115 1	81.0	CIPCO ITCM	113.5	140.2	CEII Redacted	P1	J514
631047 LIME CK L1 5 161 631160 LIME CK L2 5 161 Z	335.0	ITCM	443.5	132.4	CEII Redacted	P2-P7	J495
631051 HAZLTON L2 5 161 636210 WASHBRN5 161 1	196.0	ITCM MEC	314.0	160.2	CEII Redacted	P2-P7	J495
631079 BNE JCT5 161 636020 SUB T FD 5 161 1	199.0	MEC CIPCO	241.3	121.3	CEII Redacted	P1	J498
631079 BNE JCT5 161 636020 SUB T FD 5 161 1	199.0	MEC CIPCO	259.4	130.3	CEII Redacted	P2-P7	J498
631083 TRAER 5 161 631086 DYSART 5 161 1	258.0	ITCM	293.7	113.8	CEII Redacted	P1	J514
631083 TRAER 5 161 631086 DYSART 5 161 1	258.0	ITCM	290.1	112.4	CEII Redacted	P2-P7	J514

Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
631085 PARNEL 5 161 51113 J438 POI 161 1	240.0	MEC ITCM	246.6	102.7	CEII Redacted	P1	J530
631085 PARNEL 5 161 51113 J438 POI 161 1	240.0	MEC ITCM	244.6	101.9	CEII Redacted	P2-P7	J530
631088 ARNOLD 5 161 631094 FAIRFAX5 161 1	301.0	ITCM	324.1	107.7	CEII Redacted	P2-P7	J504
631113 POWESHK5 161 631117 REASNOR5 161 1	311.0	ITCM	324.4	104.3	CEII Redacted	P1	J500,J499,J498,J527
631113 POWESHK5 161 631117 REASNOR5 161 1	311.0	ITCM	354.1	113.9	CEII Redacted	P2-P7	J500,J499,J498,J527
631142 ARNOLD 3 345 631148 MORGN VLLY 3 345 1	846.0	ITCM	910.9	107.7	CEII Redacted	P2-P7	J504
631148 MORGN VLLY 3 345 636420 TIFFIN 3 345 1	846.0	ITCM MEC	870.2	102.9	CEII Redacted	P2-P7	J504
635200 RAUN 3 345 645451 S3451 3 345 1	956.0	MEC OPPD	978.6	102.4	CEII Redacted	P1	J498
635600 GRIMES 3 345 635700 SYCAMOR3 345 2	1209.0	MEC	1257.9	104.0	CEII Redacted	P2-P7	J499,J498
635600 GRIMES 3 345 636003 BVR CRK 3 345 1	1196.0	MEC	1197.5	100.1	CEII Redacted	P2-P7	J498,J524,J534,J535

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
635605 GRNGRTP5 161 635614 100 54 5 161 1	360.0	MEC	396.8	110.2	CEII Redacted	P2-P7	J499,J498
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	926.0	MEC	959.8	103.6	CEII Redacted	P0	J514,J500,J499,J498, J529,J432,J460,J488, J493,J495,J506,J524, J526,J527,J528,J534, J535
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	1010.0	MEC	1040.6	103.0	CEII Redacted	P1	J500,J499,J527,J528
635680 BONDRNT3 345 635730 MNTZUMA3 345 1	1010.0	MEC	1079.3	106.9	CEII Redacted	P2-P7	J500,J499,J527,J528
636001 WEBSTER5 161 636020 SUB T FD 5 161 1	210.0	MEC	223.4	106.4	CEII Redacted	P2-P7	J524
636203 LUNDQST5 161 636204 DR NE 5 161 1	215.0	MEC	263.3	122.5	CEII Redacted	P2-P7	J495
636204 DR NE 5 161 636210 WASHBRN5 161 1	215.0	MEC	238.4	110.9	CEII Redacted	P2-P7	J495

Steady-State Analysis

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator
			(MVA)	(%)			
636230 FRANKLN5 161 636235 WALL LK5 161 1	201.0	MEC	203.2	101.1	CEII Redacted	P2-P7	J498,J535
652503 BLAIR 4 230 652550 GRANITF4 230 1	410.0	WAPA	454.4	110.8	CEII Redacted	P1	J460,J493,J526
652503 BLAIR 4 230 652550 GRANITF4 230 1	410.0	WAPA	461.3	112.5	CEII Redacted	P2-P7	J488,J493,J526

Table 2-5: 2022 Shoulder Worst Voltage Constraints

Bus			Area	Vlow	Vhi	Benchmark	StudyCase	Delta (> 0.01 p.u.)	Contingency Details	Cont Type
						VCONT	VCONT			
51113	J438 POI	161	635 MEC	1.00	1.05	1.0199	0.9961	-0.0238	CEII Redacted	P1
75730	J530 POI	345	635 MEC	1.00	1.05	1.0172	0.9986	-0.0186	CEII Redacted	P1
344923	5LATHROP	161	356 AMMO	1.00	1.05	0.9989	0.9835	-0.0154	CEII Redacted	P0
631140	SALEM 3	345	627 ALTW	0.93	1.10	0.9834	0.9261	-0.0573	CEII Redacted	P1
635730	MNTZUMA3	345	635 MEC	1.00	1.05	1.0212	0.9961	-0.0251	CEII Redacted	P1
636199	BLACKHAWK 3	345	635 MEC	0.96	1.05	0.9871	0.9362	-0.0509	CEII Redacted	P1
636199	BLACKHAWK 3	345	635 MEC	0.94	1.05	0.9932	0.9308	-0.0624	CEII Redacted	P2-P7
636240	BUTLER 5	161	635 MEC	0.95	1.05	0.9863	0.9460	-0.0403	CEII Redacted	P1
640065	AXTELL 3	345	640 NPPD	0.95	1.05	0.9721	0.9460	-0.0261	CEII Redacted	P0
640065	AXTELL 3	345	640 NPPD	0.90	1.10	0.9237	0.8836	-0.0401	CEII Redacted	P1
640325	REDWILO3	345	640 NPPD	0.95	1.05	0.9584	0.9187	-0.0397	CEII Redacted	P0
640325	REDWILO3	345	640 NPPD	0.90	1.10	0.9005	0.8538	-0.0467	CEII Redacted	P1

2.7.2 2022 Summer Peak Contingency Analysis Results

2.7.2.1 System Intact Conditions

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table D-9, Table D-10).

2.7.2.2 Post Contingency Conditions

The results in this Section are for analysis of conditions following NERC Category P1-P7 contingencies. All category P1 contingencies were converged.

For P1 contingencies in the 2022 summer peak scenario, no thermal or voltage constraints were identified (Table D-11, Table D-12).

Five category P2-P7 contingencies (Table D-15) were not converged, and their dc thermal results are listed in Table D-16. The five contingencies were not converged in the benchmark or study cases. No mitigation plan is required for the study projects for these contingencies.

For P2-P7 contingencies in the 2022 summer peak scenario, no thermal or voltage constraints were identified (Table D-13, Table D-14).

2.8 Network Upgrades Identified in MISO ERIS Analysis

Based on the MISO steady state analyses, the highest loading of each facility that violates criteria in the 2022 scenario (SH, SPK) is shown in Table 2-6. Potential thermal network upgrades are also included.

Based on the MISO steady state analyses, the worst voltage constraints are shown in Table 2-7. Potential reactive power network upgrades are also included.

When Big Stone South - J488 POI 345 kV is out of service, CSCR of the renewable generation (J436, J437, and J488) is around 1.217. Further study is recommended to provide evidence that the J488 wind farm can operate at the CSCR of 1.217.

To mitigate voltage constraint identified at Lathrop 161 kV bus (Ameren area), 1×25 Mvar switched shunt capacitor bank at 161 kV bus is required to be installed.

To mitigate voltage constraints identified in the areas of Hickory Creek and Salem (ITCM area), the Hickory Creek-Cardinal 345 kV project (MVP project 3127 with in-service date of 12/31/2023) is required. Also transformer tap should be locked at neutral position for the two 345-161 kV transformers at Salem.

To mitigate voltage constraints identified in the areas of Blackhawk, Montezuma, J530 POI, and J438 POI (MEC area), 150 Mvar switched shunt capacitor bank at Blackhawk 345 kV bus, 1×50 Mvar switched shunt capacitor bank at Midport 161 kV bus, 100 Mvar switched shunt capacitor bank at Montezuma 345 kV bus, 1×25 Mvar switched shunt capacitor bank at J438 POI 161 kV bus are required to be installed.

Costs of the thermal network upgrades identified in the MISO steady state analysis for the 2022 scenario are listed in Table 2-8.

Table 2-6: Network Upgrades for Thermal Constraints in 2022 Scenario, Maximum Screened Loading

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J475,J530	J530 POI-Hills 345 kV	1010.0	MEC	1126.1	111.5	CEII Redacted	P1	2022 SH	MEC: Structure replacements
J475,J530	J530 POI-Hills 345 kV	1010.0	MEC	1133.2	112.2	CEII Redacted	P2-P7	2022 SH	MEC: Structure replacements
J523	Adams 345-161-13.8 kV xfmr	334.6	XEL	337.9	101.0	CEII Redacted	P2-P7	2022 SH	Replace the bushings on the Adams transformer
J432,J460,J488,J493,J526	Split Rock-White 345 kV	717.1	XEL WAPA	902.3	125.8	CEII Redacted	P1	2022 SH	Line is currently rated 1075 MVA for SN/SE no mitigation required
J432,J460,J488,J493,J526	Split Rock-White 345 kV	717.1	XEL WAPA	902.3	125.8	CEII Redacted	P2-P7	2022 SH	Line is currently rated 1075 MVA for SN/SE no mitigation required
J432,J488,J493,J526	Brookings Co-White 345 kV #2	788.8	XEL WAPA	946.7	120.0	CEII Redacted	P2-P7	2022 SH	XEL: Line is limited by WAPA terminal limits. NSP's equipment able to handle the required loading. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.
J460,J526	Hazel Creek 345-230-13.8 kV xfmr #6	386.0	XEL	428.9	111.1	CEII Redacted	P2-P7	2022 SH	Upgrade Hazel Creek TR9 to 672 MVA
J526	Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	260.8	100.7	CEII Redacted	P2-P7	2022 SH	Minn Valley - Granite Falls is rated to 407 MVA SN/SE. No mitigation required

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J526	Panther-McLeod 230 kV	321.9	GRE	421.7	131.0	CEII Redacted	P2-P7	2022 SH	Remediate 4 Structures
J526,J493	Johnson Jct-Ortonville 115 kV	108.0	MRES GRE OTP	142.8	132.2	CEII Redacted	P1	2022 SH	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M
J526	Johnson Jct-Ortonville 115 kV	108.0	MRES GRE OTP	146.9	136.0	CEII Redacted	P2-P7	2022 SH	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M
J526,J493	Johnson Jct-Morris 115 kV	92.0	GRE WAPA OTP	131.0	142.4	CEII Redacted	P1	2022 SH	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 OTP: OTP's 1 mile portion will be required to be fully rebuilt. \$550,000 WAPA: NU is not required unless it is identified as constraint in affected system

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
									study.
J526	Johnson Jct-Morris 115 kV	92.0	GRE WAPA OTP	135.0	146.7	CEII Redacted	P2-P7	2022 SH	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 OTP: OTP's 1 mile portion will be required to be fully rebuilt. \$550,000 WAPA: NU is not required unless it is identified as constraint in affected system study.
J493	Big Stone 115-230-13.8 kV xfmr	262.9	OTP	275.6	104.8	CEII Redacted	P1	2022 SH	Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.
J488,J493,J526	Big Stone 115-230-13.8 kV xfmr	262.9	OTP	294.7	112.1	CEII Redacted	P2-P7	2022 SH	Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.
J488,J493,J526	Big Stone 230-115-13.8 kV xfmr	291.2	OTP	301.2	103.4	CEII Redacted	P2-P7	2022 SH	Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J526	Big Stone-Browns Valley 230 kV	285.0	OTP MDU	290.7	102.0	CEII Redacted	P1	2022 SH	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000
J488,J526	Big Stone-Browns Valley 230 kV	285.0	OTP MDU	322.3	113.1	CEII Redacted	P2-P7	2022 SH	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000
J488,J493,J526	Big Stone-Blair 230 kV	438.0	OTP BEPC	555.1	126.7	CEII Redacted	P1	2022 SH	OTP: Big Stone & Blair terminal equipment requires replacement. \$200,000 BEPC: NU is not required unless it is identified as constraint in affected system study.
J488,J493,J526	Big Stone-Blair 230 kV	438.0	OTP BEPC	538.0	122.8	CEII Redacted	P2-P7	2022 SH	OTP: Big Stone & Blair terminal equipment requires replacement. \$200,000 BEPC: NU is not required unless it is identified as constraint in affected system study.
J488,J493,J526, J460	Hankinson-Wahpeton 230 kV	368.2	OTP	479.3	130.2	CEII Redacted	P1	2022 SH	Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.
J488,J493,J526	Hankinson-Wahpeton 230 kV	368.2	OTP	487.3	132.3	CEII Redacted	P2-P7	2022 SH	Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J488	Hankinson-Forman 230 kV	311.0	OTP	365.5	117.5	CEII Redacted	P1	2022 SH	Line clearance mitigation is required. Forman substation terminal equipment requires replacement.
J488	Hankinson-Forman 230 kV	311.0	OTP	365.6	117.6	CEII Redacted	P2-P7	2022 SH	Line clearance mitigation is required. Forman substation terminal equipment requires replacement.
J526	Wahpeton-Fergus Falls 230 kV	351.0	OTP MRES	353.3	100.7	CEII Redacted	P2-P7	2022 SH	Line clearance mitigation is required. Wahpeton substation terminal equipment requires replacement.
J488	Oakes-Forman 230 kV	361.3	OTP	435.2	120.5	CEII Redacted	P1	2022 SH	Line clearance mitigation is required. Oakes and Forman substation terminal equipment requires replacement.
J488	Oakes-Forman 230 kV	361.3	OTP	435.2	120.5	CEII Redacted	P2-P7	2022 SH	Line clearance mitigation is required. Oakes and Forman substation terminal equipment requires replacement.
J488	Oakes-Ellendale 230 kV	362.6	OTP MDU	439.0	121.1	CEII Redacted	P1	2022 SH	OTP: Line clearance mitigation is required. Oakes substation terminal equipment requires replacement. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0
J488, J488,	Oakes-Ellendale 230 kV	362.6	OTP MDU	438.9	121.1	CEII Redacted	P2-P7	2022 SH	OTP: Line clearance mitigation is required. Oakes substation terminal equipment requires replacement. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J514	Marshalltown-Blairstown 115 kV	106.0	ITCM	114.4	108.0	CEII Redacted	P1	2022 SH	Sag limit at 162F. Replace 4 structures and replace bus conductor at Blairstown. New rating of 120 MVA.
J514,	Blairstown-Prairie Crk 115 kV	81.0	CIPCO ITCM	87.5	108.0	CEII Redacted	P0	2022 SH	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Replace 14 structures to reach a minimum rating of 114 MVA
J514	Blairstown-Prairie Crk 115 kV	81.0	CIPCO ITCM	113.5	140.2	CEII Redacted	P1	2022 SH	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Replace 14 structures to reach a minimum rating of 114 MVA
J495	Lime Creek 161 kV bus tie	335.0	ITCM	443.5	132.4	CEII Redacted	P2-P7	2022 SH	Not a valid constraint. Bus is rated at 448 MVA
J495	Hazleton-Washburn 161 kV	196.0	ITCM MEC	314.0	160.2	CEII Redacted	P2-P7	2022 SH	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$16.7 M ITCM: Terminal limit is 325 MVA.
J498	Boone Jct-Sub T Fort Dodge 161 kV	199.0	MEC CIPCO	241.3	121.3	CEII Redacted	P1	2022 SH	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$7,000,000
J498	Boone Jct-Sub T Fort Dodge 161 kV	199.0	MEC CIPCO	259.4	130.3	CEII Redacted	P2-P7	2022 SH	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
									substation terminal equipment upgrades. \$7,000,000
J514	Traer-Dysart 161 kV	258.0	ITCM	293.7	113.8	CEII Redacted	P1	2022 SH	New rating based on revising rating methodology is 297/297
J514	Traer-Dysart 161 kV	258.0	ITCM	290.1	112.4	CEII Redacted	P2-P7	2022 SH	New rating based on revising rating methodology is 297/297
J530	Parnell-J438 POI 161 kV	240.0	MEC ITCM	246.6	102.7	CEII Redacted	P1	2022 SH	MEC: Structure replacements. \$100,000 ITCM: ITCM Limit is 335 MVA at Parnell
J530	Parnell-J438 POI 161 kV	240.0	MEC ITCM	244.6	101.9	CEII Redacted	P2-P7	2022 SH	MEC: Structure replacements. \$100,000 ITCM: ITCM Limit is 335 MVA at Parnell
J504	Arnold-Fairfax 161 kV	301.0	ITCM	324.1	107.7	CEII Redacted	P2-P7	2022 SH	New rating based on revising rating methodology is 335/335
J500,J499,J498, J527	Poweshiek-Reasnor 161 kV	311.0	ITCM	324.4	104.3	CEII Redacted	P1	2022 SH	New rating based on revising rating methodology is 353/353
J500,J499,J498, J527	Poweshiek-Reasnor 161 kV	311.0	ITCM	354.1	113.9	CEII Redacted	P2-P7	2022 SH	Replacing structures would make the line conductor 366 MVA but terminal limited to 354 MVA. To go beyond 354 MVA requires 2 breaker replacements and achieves 366 MVA.
J504	Arnold-Morgan Valley 345 kV	846.0	ITCM	910.9	107.7	CEII Redacted	P2-P7	2022 SH	New rating based on revising rating methodology is 1043/1043

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J504	Morgan Valley-Tiffin 345 kV	846.0	ITCM MEC	870.2	102.9	CEII Redacted	P2-P7	2022 SH	MEC: No Mitigation Required. Existing MEC rating is 961 MVA. \$0 ITCM: ITCM future facilities at Morgan Valley will be constructed to a rating greater than what is required for the constraint. \$0
J498	Raun-S3451 345 kV	956.0	MEC OPPD	978.6	102.4	CEII Redacted	P1	2022 SH	MEC: No mitigation required. Existing MEC emergency rating is 1152 MVA. \$0 OPPD: NU is not required unless it is identified as constraint in affected system study.
J499,J498	Grimes-Sycamore 345 kV #2	1209.0	MEC	1257.9	104.0	CEII Redacted	P2-P7	2022 SH	Sycamore substation terminal equipment upgrades.
J498,J524,J534, J535	Grimes-Beaver Creek 345 kV	1196.0	MEC	1197.5	100.1	CEII Redacted	P2-P7	2022 SH	Structure replacements.
J499,J498	Granger Tap-108th & 54th 161 kV	360.0	MEC	396.8	110.2	CEII Redacted	P2-P7	2022 SH	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades.
J514,J500,J499, J498,J529,J432, J460,J488,J493, J495,J506,J524, J526,J527,J528, J534,J535	Bondurant-Montezuma 345 kV	926.0	MEC	959.8	103.6	CEII Redacted	P0	2022 SH	Structure replacements

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J500,J499,J527, J528	Bondurant-Montezuma 345 kV	1010.0	MEC	1040.6	103.0	CEII Redacted	P1	2022 SH	Structure replacements
J500,J499,J527, J528	Bondurant-Montezuma 345 kV	1010.0	MEC	1079.3	106.9	CEII Redacted	P2-P7	2022 SH	Structure replacements
J524	Webster-Sub T Ft Dodge 161 kV	210.0	MEC	223.4	106.4	CEII Redacted	P2-P7	2022 SH	Reconductor with structure replacements as appropriate.
J495	Lundquist-Deere Northeast 161 kV	215.0	MEC	263.3	122.5	CEII Redacted	P2-P7	2022 SH	MEC: Replace limiting substation terminal equipment at both ends of line.
J495	Deere Northeast-Washburn 161 kV	215.0	MEC	238.4	110.9	CEII Redacted	P2-P7	2022 SH	MEC: Replace limiting substation terminal equipment at both ends of line.
J498,J535	Franklin-Wall Lake 161 kV	201.0	MEC	203.2	101.1	CEII Redacted	P2-P7	2022 SH	Structure replacements
J460,J493,J526	Blair-Granite Falls 230 kV	410.0	WAPA	454.4	110.8	CEII Redacted	P1	2022 SH	NU is not required unless it is identified as constraint in affected system study.

Steady-State Analysis

Generator	Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Scenario	Mitigation
				(MVA)	(%)				
J488,J493,J526	Blair-Granite Falls 230 kV	410.0	WAPA	461.3	112.5	CEII Redacted	P2-P7	2022 SH	NU is not required unless it is identified as constraint in affected system study.

Table 2-7: Network Upgrades for Voltage Constraints in 2022 Scenario, Worst Voltage Constraints

Bus			Area	Vlow	Vhi	Benchmark	StudyCase	Delta (> 0.01 p.u.)	Cont Type	Mitigation
						VCONT	VCONT			
51113	J438 POI	161	635 MEC	1.00	1.05	1.0199	0.9961	-0.0238	P1	1*25 Mvar cap at J438 POI 161 kV (51113)
75730	J530 POI	345	635 MEC	1.00	1.05	1.0172	0.9986	-0.0186	P1	100 MVAR 345 kV Capacitor at Montezuma (Bus 635730)
344923	5LATHROP	161	356 AMMO	1.00	1.05	0.9989	0.9835	-0.0154	P0	25 Mvar switched cap bank at Lathrop 161 kV (344923)
631140	SALEM 3	345	627 ALTW	0.93	1.10	0.9834	0.9261	-0.0573	P1	lock taps at neutral position for two xfms at Salem (631140)
635730	MNTZUMA3	345	635 MEC	1.00	1.05	1.0212	0.9961	-0.0251	P1	100 Mvar switched cap bank at Montezuma 345 kV (635730)
636199	BLACKHAWK 3	345	635 MEC	0.96	1.05	0.9871	0.9362	-0.0509	P1	150 MVAR 345 kV capacitor at Black Hawk (Bus 636199), and 50 MVAR 161 kV capacitor at Midport (Bus 636202)
636199	BLACKHAWK 3	345	635 MEC	0.94	1.05	0.9932	0.9308	-0.0624	P2-P7	150 MVAR 345 kV capacitor at Black Hawk (Bus 636199), and 50 MVAR 161 kV capacitor at Midport (Bus 636202)
636240	BUTLER 5	161	635 MEC	0.95	1.05	0.9863	0.9460	-0.0403	P1	150 MVAR 345 kV capacitor at Black Hawk (Bus 636199), and 50 MVAR 161 kV capacitor at Midport (Bus 636202)

Table 2-8: Thermal Network Upgrades and Cost Identified in MISO Steady State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements	\$300,000
Adams 345-161-13.8 kV xfmr	XEL	Replace the bushings on the Adams transformer	\$100,000
Split Rock-White 345 kV	XEL WAPA	Line is currently rated 1075 MVA for SN/SE no mitigation required	\$0
Brookings Co-White 345 kV #2	XEL WAPA	XEL: Line is limited by WAPA terminal limits. NSP's equipment able to handle the required loading. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Hazel Creek 345-230-13.8 kV xfmr #6	XEL	Upgrade Hazel Creek TR9 to 672 MVA	\$4,900,000
Minn Valley-Granite Falls 230 kV	XEL WAPA	Minn Valley - Granite Falls is rated to 407 MVA SN/SE. No mitigation required	\$0
Panther-McLeod 230 kV	GRE	Remediate 4 Structures	\$1,000,000
Johnson Jct-Ortonville 115 kV	MRES GRE OTP	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M	\$16,850,000
Johnson Jct-Morris 115 kV	GRE WAPA OTP	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 OTP: OTP's 1 mile portion will be required to be fully rebuilt. \$550,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$8,725,000
Big Stone 115-230-13.8 kV xfmr	OTP	Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.	\$3,250,000
Big Stone-Browns Valley 230 kV	OTP MDU	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000	\$350,000
Big Stone-Blair 230 kV	OTP BEPC	OTP: Big Stone terminal equipment requires replacement. \$200,000 BEPC: NU is not required unless it is identified as constraint in affected system study.	\$200,000
Hankinson-Wahpeton 230 kV	OTP	Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.	\$1,050,000
Hankinson-Forman 230 kV	OTP	Line clearance mitigation is required. Forman substation terminal equipment requires replacement.	\$400,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	Line clearance mitigation is required. Wahpeton substation terminal equipment requires replacement.	\$100,000

Constraint	Owner	Mitigation	Cost (\$)
Oakes-Forman 230 kV	OTP	Line clearance mitigation is required. Oakes and Forman substation terminal equipment requires replacement.	\$950,000
Oakes-Ellendale 230 kV	OTP MDU	OTP: Line clearance mitigation is required. Oakes substation terminal equipment requires replacement. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$1,650,000
Marshalltown-Blairtown 115 kV	ITCM	Sag limit at 162F. Replace 4 structures and replace bus conductor at Blairtown. New rating of 120 MVA.	\$600,000
Blairtown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Replace 14 structures to reach a minimum rating of 114 MVA	\$1,700,000
Lime Creek 161 kV bus tie	ITCM	Not a valid constraint. Bus is rated at 448 MVA	\$0
Hazleton-Washburn 161 kV	ITCM MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$16.7 M ITCM: Terminal limit is 325 MVA.	\$16,700,000
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$7,000,000	\$7,000,000
Traer-Dysart 161 kV	ITCM	New rating based on revising rating methodology is 297/297	\$0
Parnell-J438 POI 161 kV	MEC ITCM	MEC: Structure replacements. \$100,000 ITCM: ITCM Limit is 335 MVA at Parnell	\$100,000
Arnold-Fairfax 161 kV	ITCM	New rating based on revising rating methodology is 335/335	\$0
Poweshiek-Reasnor 161 kV	ITCM	New ratings methodology uprates sag limit to 353 MVA. Replacing structures would make the line conductor 366 MVA but terminal limited to 354 MVA. To go beyond 354 MVA requires 2 breaker replacements and achieves 366 MVA.	\$290,000
Arnold-Morgan Valley 345 kV	ITCM	New rating based on revising rating methodology is 1043/1043	\$0
Morgan Valley-Tiffin 345 kV	ITCM MEC	MEC: No Mitigation Required. Existing MEC rating is 961 MVA. \$0 ITCM: ITCM future facilities at Morgan Valley will be constructed to a rating greater than what is required for the constraint. \$0	\$0
Raun-S3451 345 kV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1152 MVA. \$0 OPPD: NU is not required unless it is identified as constraint in affected system study.	\$0
Grimes-Sycamore 345 kV #2	MEC	Sycamore substation terminal equipment upgrades.	\$1,200,000
Grimes-Beaver Creek 345 kV	MEC	Structure replacements.	\$600,000
Granger Tap-108th & 54th 161 kV	MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades.	\$2,000,000
Bondurant-Montezuma 345 kV	MEC	Structure replacements	\$300,000
Webster-Sub T Ft Dodge 161 kV	MEC	Reconductor with structure replacements as appropriate.	\$6,250,000

Constraint	Owner	Mitigation	Cost (\$)
Lundquist-Deere Northeast 161 kV	MEC	MEC: Replace limiting substation terminal equipment at both ends of line.	\$100,000
Deere Northeast-Washburn 161 kV	MEC	MEC: Replace limiting substation terminal equipment at both ends of line.	\$100,000
Franklin-Wall Lake 161 kV	MEC	Structure replacements	\$200,000
Blair-Granite Falls 230 kV	WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0
Voltage constraint identified at Lathrop 161 kV bus	AMRN	25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000
Voltage constraints in the areas of Hickory Creek, Salem, and Rock Creek	ITCM	Hickory Creek-Cardinal 345 kV project (MVP project 3127); XFMR tap should be locked at neutral position for the two 345-161 kV xfmr's at Salem (631140)	\$0
Voltage constraint in the area of Blackhawk	MEC	150 Mvar switched cap bank at Blackhawk 345 kV (636199) (\$4 M) and 1×50 Mvar switched cap bank at Midport 161 kV (636202) (\$2 M)	\$6,000,000
Voltage constraint in the area of Montezuma, J530 POI	MEC	100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000
Voltage constraint in the area of J438 POI	MEC	1×25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000

Affected System Steady-State Analysis

Steady state analyses were performed to identify constraints on affected systems.

3.1 Affected System Analysis for CIPCO and MPC Companies

Per CIPCO and MPC's Affected System Planning Criteria, transmission facilities which meet all of the following three (3) conditions are considered as constraints:

- 1) the branch is loaded above its applicable normal or emergency rating for the post-change case, and
- 2) the generator has a larger than 3% DF on the overloaded facility under post contingent condition or 5% DF under system intact condition, and
- 3) the loading increase of the overloaded facility is greater than 1 MVA compared with that in the pre-change case under system intact or contingency conditions.

AC contingency analysis was performed for this CIPCO and MPC affected system analysis, using the DPP 2016 February West Phase 3 benchmark (pre-change) cases and study (post-change) cases for the 2022 summer peak and summer shoulder scenarios. All NERC category P0-P7 contingencies described in Section 2.3 were simulated. The CIPCO and MPC affected systems were monitored.

The additional thermal constraints in the affected systems identified in this CIPCO and MPC affected system analysis are listed in Appendix E.1. These include the thermal constraints for the 2022 summer peak and shoulder scenarios. Whether these constraints will be mitigated or not will be determined by the affected transmission owning company's planning criteria.

The highest loading of CIPCO and MPC affected system thermal constraints and potential network upgrades are listed in Table 3-1 for the 2022 summer shoulder scenario. No CIPCO or MPC's thermal constraints were identified in the 2022 summer peak scenario.

Table 3-1: Additional Thermal Constraints Identified in CIPCO/MPC Affected System Analysis and their Highest Loadings

Constraint	Rating	Owner	Worst Loading		Generator	Scenario	Mitigation	Cost (\$)
			(MVA)	(%)				
Blairstown-Prairie Crk 115 kV	81	CIPCO ITCM	113.1	139.6	J514	2022 SH	CIPCO: No Mitigation Needed. \$0 ITCM: CIPCO LPC. \$0	\$0
Blairstown-Prairie Crk 115 kV	81	CIPCO ITCM	141.8	175.1	J514	2022 SH	CIPCO: No Mitigation Needed. \$0 ITCM: CIPCO LPC. \$0	
Prairie Crk-Squared 115 kV	164	CIPCO ITCM	189.8	115.7	J514	2022 SH	CIPCO: No Mitigation Needed. \$0 ITCM: CIPCO LPC. \$0	\$0
Boone Jct-Sub T Fort Dodge 161 kV	199	MEC CIPCO	207.9	104.5	J529,J524,J534,J535	2022 SH	CIPCO: No Mitigation Needed. \$0 MEC: CIPCO LPC. \$0 ITCM: CIPCO LPC. \$0	\$0
Boone Jct-Sub T Fort Dodge 161 kV	199	MEC CIPCO	229.8	115.5	J498,J529,J524,J534,J535	2022 SH	CIPCO: No Mitigation Needed. \$0 MEC: CIPCO LPC. \$0 ITCM: CIPCO LPC. \$0	
Mqoketa-Wyoming 161 kV	167	CIPCO	176.9	105.9	J514,J504,J530	2022 SH	CIPCO: Replace structures to achieve rating of 177 MVA. \$200,000	\$200,000
Liberty-Dundee 161 kV	327	CIPCO ITCM	330.6	101.1	J514,J495,J504	2022 SH	CIPCO: If feasible, Reconductor line with high temp conductor. \$5,000,000 ITCM: CIPCO LPC	\$5,000,000

3.2 OTP Local Planning Criteria Analysis

3.2.1 Introduction

The OTP LPC study is performed for the DPP 2016 February West Area Phase 3 study to adhere to OTP Local Planning Criteria. This LPC analysis is required in addition to MISO's standard DPP analysis.

Three additional scenarios are analyzed in this OTP LPC study:

1. LPC Summer Peak
2. LPC Summer Shoulder
3. LPC Light Load No Wind (LLNW)

3.2.2 Model Development

3.2.2.1 OTP LPC Study Cases

The OTP LPC study cases representing 2022 summer shoulder and summer peak conditions were developed from the DPP 2016 February Phase 3 corresponding summer shoulder (SH) and summer peak (SPK) cases, with the following additional changes:

- Dispatch and lock G736, J436, J437, and J442 generation at 90% of its Pmax in shoulder case and 15.6% of its Pmax in peak case. These generators have GIA status as completed.
- Power mismatch was balanced by generation in the MISO Classic.

The OTP LPC study case representing 2022 light load no wind condition was developed from the OTP LPC summer shoulder case, with the following additional changes:

- Light load profile was applied on OTP loads
- Wind generation in Table E-3 was turned off.
- Wind generation in DPP 2016 February Phase 3 study was turned off, but their GSUs, collector systems, shunt compensation devices, substation transformers, and interconnection tie lines are still in service.
- Power mismatch was balanced by generation in the MISO Classic.

3.2.2.2 OTP LPC Benchmark Cases

OTP LPC benchmark cases representing summer shoulder and summer peak conditions were created by removing the DPP West Area generating facilities from the study cases. The MISO Classic was used for power balance, where generation was scaled uniformly.

OTP LPC benchmark case representing light load no wind condition was created by removing the DPP West Area generating facilities (GSUs, collector systems, shunt compensation devices, substation transformers, and interconnection tie lines) from the study case.

3.2.3 OTP LPC Study Areas

The OTP LPC study areas include OTP (620), Xcel (600), MP (608), GRE (615), WAPA (652), BEPC-SPP (659), MDU (661), BEPC-MISO (663), and MH (667). The contingencies in these study areas (Section 2.3) are all simulated, the applicable thermal and voltage performance criteria are applied (Sections 2.4 and 2.5).

3.2.4 OTP LPC Contingency Analysis Results

3.2.4.1 Summer Shoulder Contingency Analysis Results

For NERC category P0 (system intact) conditions, thermal constraints are listed in Table E-4. No voltage constraints were identified (Table E-5).

For NERC category P1 contingencies, thermal constraints are listed in Table E-6. No voltage constraints were identified (Table E-7).

For NERC category P2-P7 contingencies, thermal constraints are listed in Table E-8. No voltage constraints were identified (Table E-9).

3.2.4.2 Summer Peak Contingency Analysis Results

For NERC category P0 (system intact) conditions, no thermal or voltage constraints were identified (Table E-10, Table E-11).

For NERC category P1 contingencies, no thermal or voltage constraints were identified (Table E-12, Table E-13).

For NERC category P2-P7 contingencies, no thermal or voltage constraints were identified (Table E-14, Table E-15).

3.2.4.3 Light Load No Wind Contingency Analysis Results

The purpose of light load no wind (LLNW) study is to identify whether generating facilities in DPP 2016 February Phase 3 study will aggravate high voltage issue identified in areas of Big Stone and Ellendale.

All bus voltages are below applicable high voltage limits. Under a few NERC Category P1-P7 contingencies, several bus voltages are above 1.05 p.u. but below their applicable high voltage limits (Appendix E.2.4).

3.2.5 Network Upgrades Identified in OTP LPC Analysis

The highest loading of OTP local planning criteria thermal constraints and potential network upgrades are listed in Table 3-2. Besides the thermal constraints and NUs identified in MISO ACCC analysis, no additional thermal constraints or NUs were identified in the OTP LPC analysis.

No low voltage constraints were identified in the OTP LPC analysis.

No high voltage constraints were identified in the light load no wind (LLNW) study. Under a few contingencies, several bus voltages are above 1.05 p.u. but below their applicable high voltage limits.

Table 3-2: Thermal Constraints Identified in OTP LPC Analysis and their Highest Loadings

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Split Rock-White 345 kV	717.1	XEL WAPA	910.1	126.9	CEII Redacted	P1	J432,J460,J488, J493,J526	Mitigated by NU identified in MISO ACCC. Line is currently rated 1075 MVA for SN/SE no mitigation required	\$0
Split Rock-White 345 kV	717.1	XEL WAPA	910.2	126.9	CEII Redacted	P2-P7	J432,J460,J488, J493,J526	Mitigated by NU identified in MISO ACCC. Line is currently rated 1075 MVA for SN/SE no mitigation required	
Brookings Co-White 345 kV #2	788.8	XEL WAPA	958.4	121.5	CEII Redacted	P2-P7	J432,J488,J493, J526	Mitigated by NU identified in MISO ACCC. XEL: Line is limited by WAPA terminal limits. NSP's equipment able to handle the required loading. \$0 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$0
Hazel Creek 345-230-13.8 kV xfmr #6	386.0	XEL	430.3	111.5	CEII Redacted	P2-P7	J460,J526	Mitigated by NU identified in MISO ACCC. Upgrade Hazel Creek TR9 to 672 MVA	\$4,900,000
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	262.3	101.3	CEII Redacted	P1	J526	Minn Valley - Granite Falls is rated to 407 MVA SN/SE. No mitigation required	\$0
Minn Valley-Granite Falls 230 kV	259.0	XEL WAPA	262.3	101.3	CEII Redacted	P2-P7	J526	Minn Valley - Granite Falls is rated to 407 MVA SN/SE. No mitigation required	

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Panther-McLeod 230 kV	321.9	GRE	423.2	131.5	CEII Redacted	P2-P7	J526	Mitigated by NU identified in MISO ACCC. Remediate 4 Structures	\$1,000,000
Wing River-Inman 230 kV	252.5	GRE	253.7	100.5	CEII Redacted	P2-P7	J526	Non-OTP facility. NU is not required for OTP LPC.	\$0
Johnson Jct-Ortonville 115 kV	108.0	MRES GRE OTP	144.7	134.0	CEII Redacted	P1	J526,J493	Mitigated by NU identified in MISO ACCC. OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M	\$16,850,000
Johnson Jct-Ortonville 115 kV	108.0	MRES GRE OTP	149.5	138.4	CEII Redacted	P2-P7	J526	Mitigated by NU identified in MISO ACCC. OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M	

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Johnson Jct-Morris 115 kV	92.0	GRE WAPA OTP	132.8	144.3	CEII Redacted	P1	J526,J493	Mitigated by NU identified in MISO ACCC. GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 OTP: OTP's 1 mile portion will be required to be fully rebuilt. \$550,000 WAPA: Affected system study	\$8,725,000
Johnson Jct-Morris 115 kV	92.0	GRE WAPA OTP	137.5	149.5	CEII Redacted	P2-P7	J526	Mitigated by NU identified in MISO ACCC. GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 OTP: OTP's 1 mile portion will be required to be fully rebuilt. \$550,000 WAPA: Affected system study	
Big Stone 115-230-13.8 kV xfmr	262.9	OTP	287.6	109.4	CEII Redacted	P1	J493	Mitigated by NU identified in MISO ACCC. Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.	
Big Stone 115-230-13.8 kV xfmr	262.9	OTP	297.5	113.2	CEII Redacted	P2-P7	J488,J493,J526	Mitigated by NU identified in MISO ACCC. Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.	\$3,250,000

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Big Stone 230-115-13.8 kV xfmr	291.2	OTP	293.8	100.9	CEII Redacted	P1	J493	Mitigated by NU identified in MISO ACCC. Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.	
Big Stone 230-115-13.8 kV xfmr	291.2	OTP	304.0	104.4	CEII Redacted	P2-P7	J488,J493,J526	Mitigated by NU identified in MISO ACCC. Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.	
Big Stone-Browns Valley 230 kV	285.0	OTP MDU	293.9	103.1	CEII Redacted	P1	J526	Mitigated by NU identified in MISO ACCC. OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000	\$350,000
Big Stone-Browns Valley 230 kV	285.0	OTP MDU	327.8	115.0	CEII Redacted	P2-P7	J488,J493,J526	Mitigated by NU identified in MISO ACCC. OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000	

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Big Stone-Blair 230 kV	438.0	OTP BEPC	563.4	128.6	CEII Redacted	P1	J488,J493,J526	Mitigated by NU identified in MISO ACCC. OTP: Big Stone & Blair terminal equipment requires replacement. \$200,000 BEPC: NU is not required unless it is identified as constraint in affected system study.	\$200,000
Big Stone-Blair 230 kV	438.0	OTP BEPC	553.8	126.4	CEII Redacted	P2-P7	J488,J493,J526	Mitigated by NU identified in MISO ACCC. OTP: Big Stone & Blair terminal equipment requires replacement. \$200,000 BEPC: NU is not required unless it is identified as constraint in affected system study.	
Hankinson-Wahpeton 230 kV	368.2	OTP	370.9	100.7	CEII Redacted	P0	J488,J493,J526	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.	
Hankinson-Wahpeton 230 kV	368.2	OTP	483.4	131.3	CEII Redacted	P1	J488,J493,J526, J460	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.	

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Hankinson-Wahpeton 230 kV	368.2	OTP	494.3	134.3	CEII Redacted	P2-P7	J488,J526	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.	\$1,050,000
Hankinson-Forman 230 kV	311.0	OTP	374.3	120.3	CEII Redacted	P1	J488	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Forman substation terminal equipment requires replacement.	\$400,000
Hankinson-Forman 230 kV	311.0	OTP	374.2	120.3	CEII Redacted	P2-P7	J488	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Forman substation terminal equipment requires replacement.	
Wahpeton-Fergus Falls 230 kV	351.0	OTP MRES	356.5	101.6	CEII Redacted	P2-P7	J526	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Wahpeton substation terminal equipment requires replacement.	\$100,000
Oakes-Forman 230 kV	361.3	OTP	448.3	124.1	CEII Redacted	P1	J488	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Oakes and Forman substation terminal equipment requires replacement.	\$950,000

Constraint	Rating	Owner	Worst Loading		Contingency	Cont Type	Generator	Mitigation	Cost (\$)
			(MVA)	(%)					
Oakes-Forman 230 kV	361.3	OTP	448.2	124.1	CEII Redacted	P2-P7	J488	Mitigated by NU identified in MISO ACCC. Line clearance mitigation is required. Oakes and Forman substation terminal equipment requires replacement.	
Oakes-Ellendale 230 kV	362.6	OTP MDU	452.3	124.7	CEII Redacted	P1	J488	Mitigated by NU identified in MISO ACCC. OTP: Line clearance mitigation is required. Oakes substation terminal equipment requires replacement. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$1,650,000
Oakes-Ellendale 230 kV	362.6	OTP MDU	452.2	124.7	CEII Redacted	P2-P7	J488	Mitigated by NU identified in MISO ACCC. OTP: Line clearance mitigation is required. Oakes substation terminal equipment requires replacement. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	
Blair-Granite Falls 230 kV	410.0	WAPA	457.3	111.5	CEII Redacted	P1	J460,J493,J526	NU is not required unless it is identified as constraint in affected system study.	\$0
Blair-Granite Falls 230 kV	410.0	WAPA	467.7	114.1	CEII Redacted	P2-P7	J488,J493,J526	NU is not required unless it is identified as constraint in affected system study.	

3.3 PJM Affected System AC Contingency Analysis

The PJM affected system analysis details (dated on 05/15/2018) can be found in Appendix E.3.

3.3.1 Steady State Study Results

3.3.1.1 Overload on Twin Branch–Argenta 345 kV line

To relieve the Twin Branch–Argenta 345 kV line overload:

- a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA. The result could prove that no additional upgrades are necessary, that some upgrades on the circuit are necessary, or that the entire 52.0 mile section of line would need to be rebuilt. PJM Network Upgrade N5240.
- b. MISO end rating (1691 MVA) is sufficient through 2016 February MISO DPP.

Cost Estimate: Estimated Cost for the Sag Study: \$208,000. If deemed necessary to rebuild section of line, Estimated Cost: \$104.0 Million.

The cost allocation is as follows:

Queue	MW contribution	% of Cost	Cost (\$208,000)
J498	17.44	27.33	56,844
J499	17.2	26.95	56,062
J526	15.82	24.79	51,564
J528	13.355	20.93	43,530

3.3.1.2 Overload on Benton–T-094 345 kV line

To relieve the Benton–T-094 345 kV overload:

- a) Reconductor or rebuild (depending on the existing structures) the portions of 345 kV lines between the Benton Harbor and Sagreto 345 kV substations. PJM Network Upgrade N5106.

Estimated Cost for rebuilding the line: \$19,000,000

Estimated Cost for reconductoring the line: \$13,500,000

The cost allocation is as follows:

Queue	MW contribution	% of Cost	Cost (\$19 M)
X1-020	15.099	8.41	1.597
J515	20.836	11.60	2.204
J498	20.48	11.40	2.166
J499	20.21	11.25	2.138
J500	23.52	13.09	2.488
J526	18.55	10.33	1.962
J527	14.69	8.18	1.554
J529	15.29	8.51	1.617
J530	15.65	8.71	1.655
J534	15.29	8.51	1.617

The new expected ratings upon completion of PJM Network upgrade N5106 are no longer sufficient; therefore, additional reinforcements are required in order to resolve this overload.

Additional Reinforcement: Replace the Benton H wavetrapp. Replace the Benton H Line Riser and Replace the Segreto Wave trap. PJM Network Upgrades N5312.1, N5312.2, and N5312.3. Cost: \$1,000,000.

MISO 2016 DPP February project J530 is the driver for the additional reinforcements.

The cost allocation for the additional reinforcements is shown below:

Queue	MW contribution	% of Cost	Cost (\$1 M)
J498	20.48	14.25	0.143
J499	20.21	14.07	0.141
J500	23.52	16.37	0.164
J526	18.55	12.91	0.129
J527	14.69	10.22	0.102
J529	15.29	10.64	0.106
J530	15.65	10.89	0.109
J534	15.29	10.64	0.106

3.3.1.3 Overload on Pleasant Prairie–Zion Station R- Zion EC 345 kV line

To relieve the Pleasant Prairie–Zion Station R- Zion EC 345 kV overload:

- The MISO-end of the Pleasant Prairie–Zion Station R 345 kV line SE rating is 1528 MVA and not sufficient. This constraint is expected to be relieved due to future planned MISO system conditions. If there are changes to this plan, any queue projects who contribute to this constraint may be responsible for mitigating the constraint.

- b) The ComEd SSTE rating of the Zion EC–Zion Station R 345 kV line is 1251 MVA and not sufficient. This constraint is expected to be relieved due to future planned MISO system conditions. If there are changes to this plan, any queue projects who contribute to this constraint may be responsible for mitigating the constraint.
- c) The following MISO 2016 February DPP projects contribute to these flowgate overloads: J505

Per PJM cost allocation rules, the DPP 2016 February West projects presently do not receive any cost allocation for this upgrade.

3.3.1.4 Study Summary

Multiple projects in the MISO DPP 2016 February West Area Phase 3 study contribute loading to the overloads in the PJM system. These projects are responsible for the cost of Network Upgrades per PJM cost allocation rules.

3.3.2 J475 Dynamic Stability Study Results

PJM performed a dynamic simulation analysis of J475 as part of the overall system impact study.

For all fault contingencies tested, the system with J475 included was transiently stable and the J475 generator was able to ride through the faults (except for faults where protective action trips the generator(s)). However, post fault bus voltages at the POI and Ottumwa, Hills, Montezuma, Sub 92, Louisa, Walcott 345 kV buses did not recover to pre-fault values for several stuck breaker faults. Model might need to be tuned to resolve the issue.

No stability or voltage recovery issues were caused due to the model on PJM system. Therefore, no mitigations were found to be required.

3.3.3 J530 Dynamic Stability Study Results

PJM performed a dynamic simulation analysis of J530 as part of the overall system impact study.

For all fault contingencies tested, the system with J530 included was transiently stable and the J530 generator was able to ride through the faults (except for faults where protective action trips the generator(s)). However, post fault J530 terminal voltage does not stabilize – continues ramping up even after 20s. Model might need to be tuned to resolve the issue.

No stability or voltage recovery issues are caused due to the model on PJM system. Therefore, no mitigations were found to be required.

3.4 SPP Affected System AC Contingency Analysis

Southwest Power Pool (SPP) conducted an Affected System Impact Study (ASIS) to evaluate potential impacts to the SPP Transmission System related to the interconnection of generators on the Mid-Continent Independent System Operation (MISO) Transmission System.

Power flow analysis has indicated that for the power flow cases studied, the SPP Transmission System is not impacted from the interconnection of the MISO DPP 2016 February West Phase 3 generation.

This Phase 3 analysis assumed the Gentleman Generation Station (GGS) – Thedford - Holt 345 kV (R-Plan) transmission line and MISO MVP projects were in service by 2020. Full Interconnection Service for these requests may need to be delayed until that time.

If the MISO Interconnection Requests proceed, additional analysis will be required through an Affected System Facility Study (ASFS). The purpose of the ASFS is to refine the scope, cost estimates, and expected time of construction of Network Upgrades.

MISO Generation Interconnection projects that come into service prior to completion of required Network Upgrades on the SPP transmission system will be subject to SPP's study process for Limited Operation. SPP updates the output limits on all Generator Interconnection Agreements when events occur on the Transmission System that are listed in the Limited Operation Impact Study for that generator to account for changing transmission and generation assumptions. SPP will coordinate project output limitations with MISO on an annual basis or more often as events are needed, and SPP will provide MISO the list of conditions that will be added to MISO Generator Interconnection Agreement.

The Limited Operation Amount determined by SPP impacts are listed in Table 3-3. If the study upgrade assumptions change, then the Limited Operation Amounts listed in Table 3-3 have the possibility of being revised.

The SPP affected system analysis results (dated on 6/12/2018) for this study are in Appendix E.4.

Table 3-3: Interconnection Service Availability based on Constraints

MISO Gen #	Requested ERIS (MW)	Requested NRIS (MW)	ERIS Availability (MW) ¹	NRIS Availability (MW) for the entire cluster (All Requests must reduce NRIS to this amount or lower to avoid NRIS cost assignment) ²	NRIS Availability (MW) for individual requests (Each Request must reduce NRIS to this amount or lower to avoid NRIS cost assignment regardless of what other customers do) ³
J432	98	N/A	98	N/A	N/A
J460	200	N/A	200	N/A	N/A
J475	200	200	200	200	200
J485	46.85	46.85	46.85	46.85	46.85
J488	151.8	N/A	151.8	N/A	N/A
J493	150	150 ⁴	150	0	0
J495	200	50	200	50	50
J498	340	340	340	340	340
J499	340	340	340	340	340
J500	500	500	500	500	500
J504	50	50	50	50	50
J506	200	N/A	200	N/A	N/A
J510	266 sum / 284.5 win	266 ⁵	266 sum / 284.5 win	0	0
J514	30 sum / 65 win	30	30 sum / 65 win	30	30
J523	50	30	50	30	30
J524	100	60	100	60	60
J526	300	N/A	300	N/A	N/A
J527	250	250	250	250	250
J528	200	70	200	70	70
J529	250	250	250	250	250
J530	250	250	250	250	250
J534	250	100	250	100	100
J535	210	84	210	84	84

Note 1: Maximum ERIS availability calculated per cluster study (all requests). This ERIS amount is dependent on all other requests with ERIS service reducing to this amount.

Note 2: Amount of NRIS service that is available at no NRIS upgrade cost assignment if all customers reduce their NRIS requests to this amount or lower. Reducing NRIS to this amount may not eliminate the risk of NRIS cost assignment unless all customers reduce NRIS to this amount or lower.

Note 3: Amount of NRIS service that is available at no NRIS upgrade cost assignment if other customers do NOT reduce their NRIS requests. Reducing NRIS to this amount or lower should result in no NRIS upgrade cost assignment

regardless of what other customers choose to do. An amount of "0" means that NRIS must be dropped completely in order to avoid the risk of cost assignment.

Note 4: J493 elected 0 MW NRIS at final NRIS election.

Note 5: J510 elected 0 MW NRIS at final NRIS election.

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Stability Analysis

Stability analysis was performed to evaluate the transient stability and impact on the region of the generating facilities in the DPP 2016 February West Area group.

4.1 Procedure

4.1.1 Computer Programs

Stability analysis was performed using PSS®E revision 33.11.

4.1.2 Study Methodology

Study stability package representing 2022 summer shoulder (SH) conditions with generating facilities in the DPP 2016 February Phase 3 West Area group (DPP West Area) was developed from MTEP17 stability package with LBA dispatch. Benchmark case was created by removing the DPP West Area generating facilities from the study case. Disturbances were simulated to evaluate the transient stability and impact on the region of the generating facilities. If a simulation for the study case violates MISO transient stability criteria or local TOs' planning criteria, the simulation was repeated on the benchmark case to assess the impact of the generating facilities on the violation.

4.2 Case Development

4.2.1 Study Case

Study case representing 2022 summer shoulder (SH) condition was developed from the MTEP17 stability package with LBA dispatch.

The stability study case for DPP 2016 February Phase 3 study was created as described in Section 2.2.1. The interface transfer levels are summarized in Table 4-1.

Table 4-1: Interface Transfer Levels in 2022 Stability Study Cases

Interface	2022 SH Case (MW)
MHEX	492
MWEX	1251
Arrowhead – Stone Lake 345 kV	477

4.2.2 Benchmark Case

The DPP West Area generating facilities as described in Table A-1 (Appendix 0) were removed from the study case. MISO Classic was used for power balance, where generation was scaled uniformly.

4.3 Disturbance Criteria

The stability simulations performed as part of this study considered all the regional and local contingencies listed in Table 4-2. Regional contingencies with the pre-defined switching sequences were selected from the MISO MTEP17 study; switching sequences for local contingencies were developed based on the generic clearing times shown in Table 4-3. The admittance for local single line-to-ground (SLG) faults were estimated by assuming that the Thevenin impedance of the positive, negative and zero sequence networks at the fault point are equal.

Table 4-2: Regional and Local Disturbance Descriptions

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Table 4-3: Generic Clearing Time Assumption

Voltage Level (kV)	Primary Clearing Time (cycle)	Backup Clearing Time (cycle)
345 kV	4	11
230 kV	5	13
161/138 kV	6	18
115 kV	6	20
69 kV	8	24

4.4 Performance Criteria

All generators must mitigate the stability constraints listed below in order to obtain any type of Interconnection Service:

- System instability
- Transient voltage constraint
- Damping violation

4.4.1 MISO Criteria

Stability simulation results are evaluated based on the following MISO criteria:

- All on-line generating units are stable
- No unexpected generator tripping
- Post-fault transient voltage limits: 1.2 per unit maximum, 0.7 per unit minimum.
- Per local TOs' planning criteria, specific transient voltage limits are applied to specific buses, areas or companies that have different requirements.

- All machine rotor angle oscillations must be positively damped with a minimum damping ratio of 0.81633% for disturbances with a fault or 1.6766% for line trips without a fault.

A bus is considered a transient voltage constraint if both of the following conditions are met. All transient voltage constraints must be resolved before a project can receive interconnection service.

1. the bus transient voltage is outside of specified transient voltage limits during transient period, and
2. the bus voltage is at least 0.01 per unit worse than the benchmark case voltage for the same contingency.

4.4.2 Local Planning Criteria

4.4.2.1 ATC Local Planning Criteria

ATC has the following local transient voltage recovery criteria. For facilities in the ATC footprint, transient voltage recovery is evaluated based on ATC's local planning criteria.

- Voltage recovery within 80 percent and 120 percent of nominal for between 2 and 20 seconds following the clearing of a disturbance.

4.4.2.2 ITCM Local Planning Criteria

ITCM has the following local transient voltage and damping criteria. For facilities in the ITCM footprint, transient voltages and dampings are evaluated based on ITCM's local planning criteria.

- Voltages at all busses on the Transmission Systems should not drop below 0.70 per unit after the first swing for more than 5 cycles. The duration for the minimum voltage dip starts after the first swing post clearing of fault.
- Voltage at all Transmission System buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-2), within 1.0 second of the clearing of the fault.
- Rotor angle oscillation damping ratios are not to be less than 0.03.

4.4.2.3 MEC Local Planning Criteria

MEC has the following local transient voltage and damping criteria. For facilities in the MEC footprint, transient voltages and dampings are evaluated based on MEC's local planning criteria.

- Generator bus transient voltage limits shall adhere to the high voltage duration and low voltage duration curve in Attachment 2 of NERC PRC-024, which is:
 - o Generator bus transient over voltage limits (after fault clearing): 1.2 pu voltage from 0.0 to and including 0.2 s; 1.175 pu voltage from 0.2 to and including 0.5

- s; 1.15 pu voltage from 0.5 to and including 1.0 s; 1.1 pu voltage for greater than 1.0 s.
- o Generator bus transient low voltage limits (after fault clearing): may be less than 0.45 pu voltage from 0 to 0.15 seconds; Voltage shall remain above 0.45 pu from 0.15 to 0.3 s; Voltage shall remain above 0.65 pu from 0.3 to 2.0 s; Voltage shall remain above 0.75 pu from 2.0 to 3.0 s; Voltage shall recover to 0.9 pu after 3 s.
- Load bus transient voltage limits:
 - o Load bus transient over voltage limits (after fault clearing): 1.6 pu voltage from 0.01 to and including 0.04 s; 1.2 pu voltage from 0.04 to and including 0.5 s; 1.1 pu voltage from 0.5 to and including 5 s; and 1.05 pu voltage for greater than 5 s. These voltage limits also apply to buses without loads or generators.
 - o Load bus transient low voltage limits (after fault clearing): may be less than 0.7 pu voltage from 0 to 2 s; Voltage shall remain above 0.7 pu from 2 to 20 s; Voltage shall recover to 0.9 pu after 20 s.
- Angular transient stability minimum damping ratio (ζ) should not be less than 0.03.

4.5 Stability Results

The contingencies listed in Table 4-2 were simulated using the 2022 summer shoulder study case. If a transient stability criteria violation was identified, the same disturbance was repeated in the benchmark case.

Appendix F.2 contains plots of generator rotor angles, generator power output, generator terminal voltages, bus voltages, and branch flows for each simulation. Simulations were performed with a 2.0 seconds steady-state run followed by the appropriate disturbance. Simulations were run for a 12-second duration.

4.5.1 2022 Stability Analysis Results

Stability study results summary is in Appendix F, Table F-1. The following stability related issues were identified.

4.5.1.1 Slow Voltage Recovery at Hickory Creek 345 kV Bus

Per ITCM LPC, voltage at all ITCM transmission system buses should recover to the applicable post-contingency steady-state voltage level (ITCM post-disturbance limits in Table 2-2) within 1.0 second of the clearing of the fault.

If the Hickory Creek–Cardinal 345 kV MVP project is not modeled in the case, under disturbances in Table 4-4, voltage at Hickory Creek 345 kV bus is still below post-contingency steady-state voltage limit (0.93 pu) after 1.0 second with the fault cleared. With addition of the Hickory Creek-Cardinal 345 kV MVP project in the stability model, no stability violations were identified under these disturbances (Table 4-4).

Without the Hickory Creek–Cardinal 345 kV MVP project, voltage constraint at Hickory Creek 345 kV bus was also identified in the steady state analysis.

**Table 4-4: Disturbances Causing Slow Voltage Recovery at
Hickory Creek 345 kV Bus**

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4.5.1.2 Out of Step (OOS) Relay at Forbes on M602F Tripping

The OOS relay at Forbes on M602F tripped for 3-phase fault '0890_w_xel_p12_pc3_at_king-eauclaire' (Table 4-5) in both study case and benchmark case. This OOS trip is initiated prior to fault clearing. During the fault OOS relay can operate if the apparent impedance initially remains between blinders long enough for the timer to time out and then moves into the trip area before the fault is cleared (Figure 4-1). M602F OOS relay settings may need to be adjusted when the GNTL 500 kV project goes into service.

The 3-phase fault '0890_w_xel_p12_pc3_at_king-eauclaire' was simulated again by blocking the OOS relay at Forbes on M602F to keep it from operating during the fault. No stability violations were identified.

**Table 4-5: Disturbance Causing OOS Relay at Forbes on M602F
Tripping**

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**Figure 4-1: Forbes M602F OOS Relay Apparent Impedance
under Disturbance “0890_w_xel_p12_pc3_at_king-eauclaire”**

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4.6 Network Upgrades Identified in Stability Analysis

The following Network Upgrades are required for mitigating constraints identified in the stability analysis:

- Hickory Creek-Cardinal 345 kV MVP project (MVP project 3127 with in-service date of 12/31/2023).

MWEX Voltage Stability Study

ATC performed steady state voltage stability analysis. Voltage stability analysis is required to determine if the initial conditions of the DPP system models under study are in a stable state as defined by Power-Voltage (PV) curves of the Minnesota Wisconsin Export Interface (MWEX) for the worst contingency.

The voltage stability analysis used 2022 shoulder load cases to compare the Pre-DPP and Post-DPP states.

As shown in Table 5-1, the **Pre-DPP scenario in the 2022 shoulder load (2022 SH) case violates ATC Planning Criteria by the nose voltage of the PV curve exceeding 0.95 p.u.** However, Post-DPP scenario meets the criteria and there is no aggravation of the criteria violation and sufficient margin is maintained. Network Upgrades related to voltage stability will **NOT** be assigned to the Interconnection Customers, based on the assumptions used in this analysis..

Table 5-1: MWEX Margins to Collapse in the 2022SH Cases

	Real Power Flows (MW)						
	AHD-SLK ¹	MWEX			Margin to Nose ^{2,3}		
Case	N-0 Initial Condition	N-0 I.C.	N-1 I.C.	N-1 Nose	(MW)	(%)	Notes
Pre-DPP	373	977.2	462.7	746.6	283.9	38%	Voltage Stable Sufficient Margin ⁴ V(nose) > 0.95 p.u. ⁵
Post-DPP	457.8	1203.3	590.6	698.4	107.8	15.4%	Voltage Stable Sufficient Margin ⁴

Notes:

- As described in the active MWEX Operating Guide, the AHD-SLK interface is a single element PTDF interface measured at the Minnesota Power 230 kV side of the Arrowhead 230 kV phase shifter.
- Margin to Nose (MW) is defined as the MWEX N-1 Nose minus the N-1 Initial Condition.
- Initial Condition flows were measured in the base cases with the worst contingency plus operation of various control systems as needed with all transformer taps, switched shunts, and PARs locked.
- ATC Planning Criteria requires a 10% voltage stability margin.
- ATC Planning Criteria requires $V_{nose} < V_{min}$. V_{min} is 0.95 p.u. at the MP Arrowhead 230 kV.

The results of this analysis show that the Post-DPP scenario in the 2022 shoulder load case is voltage stable. The calculated margin to nose is greater than 10%, as required by ATC Planning Criteria. Therefore there will be no Network Upgrade costs assigned to Interconnection Customers for this study.

The MWEX voltage stability study details can be found in Appendix G.

Short Circuit Analysis

6.1 Introduction

Siemens PTI and several transmission owning companies performed short circuit analysis for the DPP 2016 February West Area generation projects.

6.2 J432 Short Circuit Study Performed by Siemens PTI

The J432 short circuit study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J432 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.1.

6.3 J460 Short Circuit Study Performed by Siemens PTI

The J460 short circuit study was performed by Siemens PTI. The study results show that the 3PH fault current is 11577 A (increased by 715.2 A) and SLG fault current is 9751 A (increased by 1399.6 A) at the Hawks Nest 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J460 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.2.

6.4 J475 Short Circuit Study Performed by MEC

The J475 short circuit study was performed by MEC. The study results show that the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project.

Study details can be found in Appendix H.3.

6.5 J485 Short Circuit Study Performed by Siemens PTI

The J485 short circuit facilities study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J485 generation project does not cause any Transmission Owner short circuit constraints

Study details can be found in Appendix H.4.

6.6 J488 Short Circuit Study Performed by OTP

The J488 short circuit study was performed by OTP. With the proposed J488 project additions, the fault currents are estimated to be roughly 8.6 kA at the Twin Brooks 345 kV bus, 11.2 kA at the Big Stone South 345 kV bus, 19.7 kA at the Big Stone South 230 kV bus, and 33.6 kA at the Big Stone South 34.5 kV bus. Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required.

Study details can be found in Appendix H.5.

6.7 J493 and J510 Short Circuit Study Performed by OTP

The J493 and J510 short circuit study was performed by OTP.

With the proposed J493 project additions, the fault currents are estimated to be roughly 16 kA at the Big Stone South 230 kV bus and the faults on the 345 kV bus were below these values.

With the proposed J493 and J510 project additions, the fault currents are estimated to be roughly 16.6 kA at the Big Stone South 230 kV bus.

Based on the short circuit analysis performed, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded and there are no upgrades required.

Study details can be found in Appendix H.6.

6.8 J495 Short Circuit Study Performed by ITCM

The J495 short circuit study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J495. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50 kA interrupting capability.

Study details can be found in Appendix H.7.

6.9 J498 Short Circuit Facilities Study Performed by MEC

The J498 short circuit facilities study was performed by MEC. The study results show that the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project.

Study details can be found in Appendix H.8.

6.10 J499 Short Circuit Facilities Study Performed by MEC

The J499 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the

Transmission Owner's short circuit criteria, interconnection of the J499 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.9.

6.11 J500 Short Circuit Facilities Study Performed by MEC

The J500 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J500 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.10.

6.12 J504 Short Circuit Facilities Study Performed by ITCM

The J504 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J504. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. There are several existing breakers at DAEC 161 kV substation that are expected to be replaced as part of planned work at this location, as well as new breakers being installed with the bus expansion. 40 kA interrupting capability was assumed for the new and replaced breakers.

Study details can be found in Appendix H.11.

6.13 J506 Short Circuit Facilities Study Performed by Siemens PTI

The J506 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 12205.7 A (increased by 806.3 A) and SLG fault current is 9955.5 A (increased by 1698.8 A) at the J506 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J506 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.12.

6.14 J514 Short Circuit Facilities Study Performed by ITCM

The J514 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J514. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. Circuit breaker 2680 at Marshalltown 115 kV was previously identified as a constraint for J233, but has since been removed.

Study details can be found in Appendix H.13.

6.15 J523 Short Circuit Facilities Study Performed by ITCM

The J523 short circuit facilities study was performed by ITCM. Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J523. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

Study details can be found in Appendix H.14.

6.16 J524 Short Circuit Facilities Study Performed by Siemens PTI

The J524 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 19095 A (increased by 580.4 A) and SLG fault current is 19487 A (increased by 1244.2 A) at the Webster 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J524 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.15.

6.17 J526 Short Circuit Facilities Study Performed by OTP

The J526 short circuit facilities study was performed by OTP.

With the proposed J526 project additions, the fault currents are estimated to be roughly 11.7 kA at the Big Stone South 345 kV bus, 20.6 kA at the Big Stone South 230 kV bus, and 37.1 kA at the Big Stone South 34.5 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J526 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.16.

6.18 J527 Short Circuit Facilities Study Performed by Siemens PTI

The J527 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 17037 A (increased by 1119.2 A) and SLG fault current is 14696 A (increased by 2339.9 A) at the J527 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J527 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.17.

6.19 J528 Short Circuit Facilities Study Performed by Siemens PTI

The J528 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 16673 A (increased by 868.9 A) and SLG fault current is 14134 A (increased by 1757.2 A) at the J528 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J528 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.18.

6.20 J529 Short Circuit Facilities Study Performed by MEC

The J529 short circuit facilities study was performed by MEC. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J529 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.19.

6.21 J530 Short Circuit Facilities Study Performed by Siemens PTI

The J530 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 13118 A (increased by 1118.4 A) and SLG fault current is 10678 A (increased by 2793.4 A) at the J530 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J530 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.20.

6.22 J534 Short Circuit Facilities Study Performed by Siemens PTI

The J534 short circuit facilities study was performed by Siemens PTI. The study results show that the 3PH fault current is 13335 A (increased by 1119.5 A) and SLG fault current is 11425 A (increased by 2413.0 A) at the J534 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J534 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.21.

6.23 J535 Short Circuit Facilities Study Performed by Siemens PTI

The J535 short circuit facilities study was performed by Siemens PTI. The study results show that the change in fault current at buses in the Study Project area is relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J535 generation project does not cause any Transmission Owner short circuit constraints.

Study details can be found in Appendix H.22.

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Deliverability Study

7.1 Project Description

Interconnection requests requesting Network Resource Interconnection Services (NRIS) were considered for deliverability analysis.

7.2 Introduction

Generator interconnection projects have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS).

If the generator is determined as not fully deliverable, the customer can choose either to change his project to an Energy Resource (ER) project or proceed with the system upgrades that will make the generator fully deliverable.

Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottlenecked up. The wind generators are tested at 100 % of their maximum output level which then can be used to meet Resource Adequacy obligations, under Module E, of the MISO Transmission and Energy Market Tariff (TEMT).

7.3 Study Methodology

MISO Generator Deliverability Study whitepaper describing the algorithm can be found at https://cdn.misoenergy.org/Generator_Deliverability_Study_Methodology108139.pdf.

7.4 Determining the MW restriction

If one facility is overloaded based on the assessed “severe yet credible dispatch” scenario described in the study methodology, and the generator under study is in the “Top 30 DF List” (see white paper for detail), part or all of its output is not deliverable. The restricted MW is calculated as following:

$$(\text{MW restricted}) = (\text{worst loading} - \text{MW rating}) / (\text{generator sensitivity factor})$$

If the result is larger than the maximum output of the generator, 100% of this generator’s output is not deliverable.

The generator is also responsible for any NEW base case (pre-shift) overload or NEW “severe yet credible dispatch overload” where the generator is not in the “Top 30 DF List”, if the generator’s DF is greater than 5%. Please see white paper for detail. The formula above also applies to these situations.

7.5 2022 Deliverability Study Result

7.5.1 J475

J475 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	0 MW (0%)						
Next Upgrade for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Level of Service Attainable (MW)	Distribution Factor	Constraints in ERIS Analysis	Projects Associated With ERIS Constraint	Projects Associated With NRIS Constraint	Upgrade Cost Allocated to Project	Total Cost of Upgrade
Irvine - Powershiek 161 kV	200.0	5.11%	No	No	J475	\$11 M	\$11 M

7.5.2 J485

J485 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	46.85 MW (100%)
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7.5.3 J495

J495 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	50 MW (100%)
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7.5.4 J498

J498 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	340 MW (100%)
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7.5.5 J499

J499 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	340 MW (100%)
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7.5.6 J500

J500 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	500 MW (100%)
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7.5.7 J504

J504 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	50 MW (100%)
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7.5.8 J514

J514 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	30 MW (100%)
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7.5.9 J523

J523 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	30 MW (100%)
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7.5.10 J524

J524 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	60 MW (100%)
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7.5.11 J527

J527 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)
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7.5.12 J528

J528 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	70 MW (100%)
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7.5.13 J529

J529 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)
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7.5.14 J530

J530 Deliverable (NRIS) Amount in 2022 Case: (Conditional on ERIS upgrades and case assumptions)	250 MW (100%)
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7.5.15 J534

J534 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIIS upgrades and case assumptions)	100 MW (100%)
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7.5.16 J535

J535 Deliverable (NRIS) Amount in 2022 model: (Conditional on ERIIS upgrades and case assumptions)	84 MW (100%)
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Shared Network Upgrades Analysis

Shared Network Upgrade (SNU) test for Network Upgrades driven by higher queued interconnection project was performed for this System Impact Study. No SNUs were identified in this study.

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Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

9.1 Cost Assumptions for Network Upgrades

The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

9.2 ERIS Network Upgrades Proposed for DPP West Area Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the MISO ERIS analysis and the affected system analysis. The ERIS network upgrades include voltage support network upgrades and thermal network upgrades identified in the MISO steady-state analysis, voltage support network upgrades identified in the MWEX voltage stability analysis, stability and voltage collapse network upgrades identified in the MISO transient stability analysis, short circuit network upgrades identified in the MISO short circuit analysis, and network upgrades identified in the Local Planning Criteria analysis and affected system analysis. The total costs of ERIS network upgrades for the 2022 scenario are summarized in Table 9-1.

Table 9-1: Summary of ERIS Network Upgrades

Category of ERIS Network Upgrades	2022 Cost (\$)
Network Upgrades Identified in MISO Steady-State Analysis	\$92,465,000
Network Upgrades Identified in MWEX Voltage Stability analysis	\$0
Network Upgrades Identified in Stability Analysis	\$0
Network Upgrades Identified in Short Circuit Analysis	\$0
Additional Network Upgrades Identified in OTP LPC Analysis	\$0
Additional Network Upgrades Identified in CIPCO/MPC Affected System	\$5,200,000
Additional Network Upgrades Identified in PJM Affected System	\$16,405,000
Additional Network Upgrades Identified in SPP Affected System	\$0
Shared Network Upgrades	\$0
Total	\$114,070,000

ERIS network upgrades are listed below.

Table 9-2: Network Upgrades in MISO Steady-State Analysis

Constraint	Owner	Mitigation	Cost (\$)
J530 POI-Hills 345 kV	MEC	MEC: Structure replacements	\$300,000
Adams 345-161-13.8 kV xfmr	XEL	Replace the bushings on the Adams transformer	\$100,000
Hazel Creek 345-230-13.8 kV xfmr #6	XEL	Upgrade Hazel Creek TR9 to 672 MVA	\$4,900,000
Panther-McLeod 230 kV	GRE	Remediate 4 Structures	\$1,000,000
Johnson Jct-Ortonville 115 kV	MRES GRE OTP	OTP: Ortonville terminal equipment requires replacement. OTP's portion of the 0.56 mile line will be required to be fully rebuilt with a larger conductor. \$500,000 GRE: GRE owned switch at Johnson Jct needs to be replaced. \$350K MRES: The MRES owned section (majority of the line) will have to be rebuilt for rating increases. \$16M	\$16,850,000
Johnson Jct-Morris 115 kV	GRE WAPA OTP	GRE: Rebuild 15 miles of 115 kV line from Johnson Junction to Morris. \$8,175,000 OTP: OTP's 1 mile portion will be required to be fully rebuilt. \$550,000 WAPA: NU is not required unless it is identified as constraint in affected system study.	\$8,725,000
Big Stone 115-230-13.8 kV xfmr	OTP	Big Stone terminal equipment requires replacement. The Big Stone 230/115/13.8 kV transformer requires replacement.	\$3,250,000
Big Stone-Browns Valley 230 kV	OTP MDU	OTP: owns Big Stone and Browns Valley terminal equipment which is sufficient to handle the flow seen in the study. \$0 MDU: Conductor clearance mitigation. \$350,000	\$350,000
Big Stone-Blair 230 kV	OTP BEPC	OTP: Big Stone & Blair terminal equipment requires replacement. \$200,000 BEPC: NU is not required unless it is identified as constraint in affected system study.	\$200,000
Hankinson-Wahpeton 230 kV	OTP	Line clearance mitigation is required. Hankinson and Wahpeton substation terminal equipment requires replacement.	\$1,050,000
Hankinson-Forman 230 kV	OTP	Line clearance mitigation is required. Forman substation terminal equipment requires replacement.	\$400,000
Wahpeton-Fergus Falls 230 kV	OTP MRES	Line clearance mitigation is required. Wahpeton substation terminal equipment requires replacement.	\$100,000
Oakes-Forman 230 kV	OTP	Line clearance mitigation is required. Oakes and Forman substation terminal equipment requires replacement.	\$950,000

Constraint	Owner	Mitigation	Cost (\$)
Oakes-Ellendale 230 kV	OTP MDU	OTP: Line clearance mitigation is required. Oakes substation terminal equipment requires replacement. MDU: The Ellendale MVP 230 kV bus will be built to 796 MVA. \$0	\$1,650,000
Marshalltown-Blairstown 115 kV	ITCM	Sag limit at 162F. Replace 4 structures and replace bus conductor at Blairstown. New rating of 120 MVA.	\$600,000
Blairstown-Prairie Crk 115 kV	CIPCO ITCM	CIPCO: No Mitigation Needed. \$0 ITCM: Clearance limit at 153F. Replace 14 structures to reach a minimum rating of 114 MVA	\$1,700,000
Hazleton-Washburn 161 kV	ITCM MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$16.7 M ITCM: Terminal limit is 325 MVA.	\$16,700,000
Boone Jct-Sub T Fort Dodge 161 kV	MEC CIPCO	CIPCO: No Mitigation Needed. \$0 MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades. \$7,000,000	\$7,000,000
Parnell-J438 POI 161 kV	MEC ITCM	MEC: Structure replacements. \$100,000 ITCM: ITCM Limit is 335 MVA at Parnell	\$100,000
Poweshiek-Reasnor 161 kV	ITCM	New ratings methodology uprates sag limit to 353 MVA. Replacing structures would make the line conductor 366 MVA but terminal limited to 354 MVA. To go beyond 354 MVA requires 2 breaker replacements and achieves 366 MVA.	\$290,000
Morgan Valley-Tiffin 345 kV	ITCM MEC	MEC: No Mitigation Required. Existing MEC rating is 961 MVA. \$0 ITCM: ITCM future facilities at Morgan Valley will be constructed to a rating greater than what is required for the constraint. \$0	\$0
Raun-S3451 345 kV	MEC OPPD	MEC: No mitigation required. Existing MEC emergency rating is 1152 MVA. \$0 OPPD: NU is not required unless it is identified as constraint in affected system study.	\$0
Grimes-Sycamore 345 kV #2	MEC	Sycamore substation terminal equipment upgrades.	\$1,200,000
Grimes-Beaver Creek 345 kV	MEC	Structure replacements.	\$600,000
Granger Tap-108th & 54th 161 kV	MEC	MEC: Reconductor with structure replacements as appropriate and substation terminal equipment upgrades.	\$2,000,000
Bondurant-Montezuma 345 kV	MEC	Structure replacements	\$300,000
Webster-Sub T Ft Dodge 161 kV	MEC	Reconductor with structure replacements as appropriate.	\$6,250,000

Constraint	Owner	Mitigation	Cost (\$)
Lundquist-Deere Northeast 161 kV	MEC	MEC: Replace limiting substation terminal equipment at both ends of line.	\$100,000
Deere Northeast-Washburn 161 kV	MEC	MEC: Replace limiting substation terminal equipment at both ends of line.	\$100,000
Franklin-Wall Lake 161 kV	MEC	Structure replacements	\$200,000
Blair-Granite Falls 230 kV	WAPA	NU is not required unless it is identified as constraint in affected system study.	\$0
Voltage constraint identified at Lathrop 161 kV bus	AMRN	25 Mvar switched cap bank at Lathrop 161 kV (344923)	\$2,000,000
Voltage constraints in the areas of Hickory Creek, Salem, and Rock Creek	ITCM	Hickory Creek-Cardinal 345 kV project (MVP project 3127); XFMR tap should be locked at neutral position for the two 345-161 kV xfmrs at Salem (631140)	\$0
Voltage constraint in the area of Blackhawk	MEC	150 Mvar switched cap bank at Blackhawk 345 kV (636199) (\$4 M) and 1×50 Mvar switched cap bank at Midport 161 kV (636202) (\$2 M)	\$6,000,000
Voltage constraint in the area of Montezuma, J530 POI	MEC	100 Mvar switched cap bank at Montezuma 345 kV (635730)	\$5,000,000
Voltage constraint in the area of J438 POI	MEC	1×25 Mvar switched cap bank at J438 POI 161 kV (51113)	\$2,500,000

Table 9-3: Network Upgrades in MWEX Voltage Stability

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 9-4: Network Upgrades Required for Transient Stability

Constraint	Owner	Network Upgrades	Cost (\$)
Transient voltage recovery constraint ¹	ITCM	Hickory Creek-Cardinal 345 kV project (MVP project 3127); XFMR tap should be locked at neutral position for the two 345-161 kV xfmrs at Salem	\$0

Note 1: Due to this transient voltage recovery constraint, all projects in DPP 2016 February West Phase 3 study are contingent on the Hickory Creek-Cardinal 345 kV MVP project.

Table 9-5: Additional Network Upgrades in OTP LPC

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 9-6: CIPCO Affected System Network Upgrades

Constraint	Owner	Mitigation	Cost (\$)
Mqoketa-Wyoming 161 kV	CIPCO	CIPCO: Replace structures to achieve rating of 177 MVA. \$200,000	\$200,000
Liberty-Dundee 161 kV	CIPCO ITCM	CIPCO: If feasible, Reconductor line with high temp conductor. \$5,000,000 ITCM: CIPCO LPC	\$5,000,000

Table 9-7: PJM Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	DPP Projects
Twin Branch–Argenta 345 kV	PJM	a. A sag check will be required for the ACSR ~ 954 ~ 45/7 ~ RAIL - Conductor Section 1 to determine if the line section can be operated above its emergency rating of 1409 MVA.	\$208,000	J498, J499, J526, J528
Benton–T-094 345 kV (N5106)	PJM	Reconductor or rebuild (depending on the existing structures) the portions of 345 kV lines between the Benton Harbor and Sagreto 345 kV substations.	\$19,000,000	X1-020, J515, J498, J499, J500, J526, J527, J529, J530, J534
Benton–T-094 345 kV (N5312.3)	PJM	Replace the Benton H Line Riser and Replace the Segreto Wave trap.	\$1,000,000	J498, J499, J500, J526, J527, J529, J530, J534

Table 9-8: SPP Affected System Network Upgrades

Constraint	Owner	Mitigation	Total Cost (\$)	DPP Projects
Network Upgrades are not required			\$0	

Table 9-9: Network Upgrades in Short Circuit Analysis

Constraint	Owner	Mitigation	Cost (\$)
No Constraints			\$0

Table 9-10: Shared Network Upgrades

Shared Network Upgrades	Upgrade Cost \$
None	\$0

9.3 Cost Allocation Methodology

The costs of Network Upgrades (NU) for a set of generation projects (one or more sub-groups or entire group with identified NU) are allocated based on the MW impact from each project on the constrained facilities in the Post Case. For constraints identified in the shoulder peak scenario, the MW impact is calculated using the shoulder peak post-DPP case. The

MW impact on constraints identified in the summer peak scenario is calculated using the summer peak post-DPP case. With all Group Study generation projects dispatched in the Post Case, all thermal and voltage constraints will be identified and a distribution factor from each project on each constraint will be obtained.

Constraints which are mitigated by one or a subset of NU are identified. The MW contribution on these constraints from each generating facility is calculated in the Post Case without any network upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj.A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj.i MW contribution on constraint})} \right)$$

9.4 Cost Allocation

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the draft System Impact Study report date.

The Distribution Factor (DF) from each generating facility is calculated on the constraints identified in the steady-state analysis in the Post Case without any network upgrades. For a voltage support network upgrade required for mitigating voltage constraints identified in the steady-state AC contingency analysis, DFs are calculated under the most critical contingency on all branches (proxy branches for voltage support network upgrade) connecting to the high voltage side of the transformer, where the voltage support network upgrade is located. For a voltage support network upgrade required for mitigating MWEX voltage stability constraints identified in the voltage stability analysis, DFs are calculated under the most critical contingency on all branches (proxy branches) connecting to the high voltage side of the transformer, where the voltage collapse occurs.

For each thermal constraint, the maximum MW contribution (increasing flow) from each generating facility is calculated. MW contribution from one generating facility is set as zero if the MW contribution is less than 1 MW, or the constraint is not categorized as MISO ERIS constraint or affected system constraint for that specific generating facility.

For a voltage support network upgrade, the generators in same local area with positive MW impact will be responsible for mitigating the voltage or voltage stability constraints. No cutoff is applied to the MW contribution from each generating facility.

The calculated DF results and the MW contribution on each constraint are in Appendix I.1 for the 2022 scenario.

Finally, the cost allocation for each NU is calculated based on the MW contribution of each generating facility, as detailed in Appendix I.2 for the 2022 scenario.

Assuming all generating facilities in the DPP 2016 February West Area group advance, a summary of the costs for total NUs (NUs for ERIS, NRIS, and Interconnection Facilities) allocated to each generating facility is listed in Table 9-11.

Table 9-11: Summary of Total NU Costs Allocated to Each Generation Project

Project	Max Output (MW)	Total Cost of NU per Project (\$)	\$/MW	Share %
J432	98	\$5,443,239	\$55,543	1.79%
J460	200	\$14,659,864	\$73,299	4.82%
J475	200	\$14,437,248	\$72,186	4.74%
J485	46.85	\$1,796,900	\$38,354	0.59%
J488	151.8	\$5,377,960	\$35,428	1.77%
J493	150	\$14,196,445	\$94,643	4.66%
J495	200	\$31,726,701	\$158,634	10.42%
J498	340	\$23,575,348	\$69,339	7.74%
J499	340	\$17,401,636	\$51,181	5.72%
J500	500	\$28,539,432	\$57,079	9.38%
J504	50	\$1,115,233	\$22,305	0.37%
J506	200	\$10,657,740	\$53,289	3.50%
J510	284.5	\$7,682,433	\$27,003	2.52%
J514	65	\$3,084,081	\$47,447	1.01%
J523	50	\$1,370,221	\$27,404	0.45%
J524	100	\$7,487,206	\$74,872	2.46%
J526	300	\$36,896,837	\$122,989	12.12%
J527	250	\$13,371,347	\$53,485	4.39%
J528	200	\$11,551,728	\$57,759	3.79%
J529	250	\$14,101,091	\$56,404	4.63%
J530	250	\$13,306,221	\$53,225	4.37%
J534	250	\$12,380,747	\$49,523	4.07%
J535	210	\$14,253,139	\$67,872	4.68%
Total/Average	4686.15	\$304,412,796	\$61,707	100.00%

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Model Development for Steady-State and Stability Analysis

A.1 DPP 2016 February Generation Projects

Table A-1: DPP 2016 February West Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J432	SD	Brookings	XEL	Brookings 345 kV	98	Wind	ERIS
J460	MN	Lincoln	GRE	Brookings-H081 345 kV line	200	Wind	ERIS
J475	IA	Poweshiek	MEC	Montezuma 345 kV	200	Wind	NRIS
J485	MN	Olmsted	RPU	West Side 161 kV	46.85	CT	NRIS
J488	SD	Grant, Roberts	OTP	Big Stone-Ellendale 345 kV	151.8	Wind	ERIS
J493	MN	Yellow Medicine	OTP	Big Stone-Brookings 345 kV	150	Wind	ERIS
J495	IA	Winnebago	ITCM	Ledyard-Colby 345 kV line	200	Wind	NRIS
J498	IA	Boone, Greene	MEC	Grimes-Lehigh 345 kV	340	Wind	NRIS
J499	IA	Adair, Madison	MEC	Fallow-Grimes 345 kV	340	Wind	NRIS
J500	IA	Adair	MEC	New substation at intersection of Boone-Atchison 345 kV and Rolling Hills-Madison Co 345 kV	500	Wind	NRIS
J504	IA	Linn	ITCM	Bertram-Duane Arnold 161 kV (0.5 mi from Duane Arnold)	50	Solar	NRIS
J506	IA	Plymouth	MEC	Raun-Lakefield Jct 345 kV	200	Wind	ERIS
J510	SD	Deuel	OTP	Scandinavia Township	266 sum / 284.5 win	CT	ERIS
J514	IA	Marshall	ITCM	Marshalltown substation	30 sum / 65 win	CC	NRIS
J523	MN	Mower	ITCM	Adams 161 kV	50	Solar	NRIS
J524	IA	Webster	MEC	Webster 161 kV	100	Solar	NRIS
J526	SD	Deuel	OTP	Brooking Co-Big Stone South 345 kV	300	Wind	ERIS
J527	IA	Adair	MEC	Booneville-Cooper 345 kV	250	Wind	NRIS
J528	IA	Adair	MEC	Rolling Hills-Madison 345 kV	200	Wind	NRIS
J529	IA	Palo Alto	MEC	Obrien-Kossuth 345 kV	250	Wind	NRIS
J530	IA	Iowa	MEC	Montezuma-Hills 345 kV	250	Wind	NRIS
J534	IA	Humboldt, Kossuth	MEC	Kossuth-Webster 345 kV	250	Wind	NRIS
J535	IA	Sac	MEC	J411-Lehigh 345 kV	210	Wind	NRIS

Table A-2: Dynamic Modeling for DPP West Area Projects

MISO Project #	Dynamic Modeling	Generator Reactive Power Capability (power factor)
J432	49 Gamesa G9X 2.0 MW (GXX048)	± 0.95
J460	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J475	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J485	5 9.37 MW natural gas fired reciprocating engines	0.8 lagging -0.95 leading
J488	61 GE 2.5 MW WTG (GEWTG0501)	± 0.9
J493	45 V136 GS 3.45MW (GSCOR1)	0.902 lagging 0.918 leading
J495	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J498	170 Vestas V110 2.0 MW Mk 10D (VS3103)	± 0.95
J499	170 Vestas V110 2.0 MW Mk 10D (VS3103)	± 0.95
J500	250 Vestas V110 2.0 MW Mk 10D (VS3103)	± 0.95
J504	14 GE 1500V 4 MW Solar (PVGU1)	± 0.9
J506	100 Vestas V110 2.0 MW (VS3103)	± 0.95
J510	GE CT generator (GENROU)	0.85 lagging 0.95 leading
J514	J233 CC expansion, 2 CTGs, 1 STG (GENROU)	NA
J523	20 SMA SC 2500 ev 2.5 MVA inverters (REGCAU1)	± 0.9
J524	40 SMA SC 2500 ev 2.5 MVA inverters (REGCAU1)	± 0.9
J526	120 GE 2.5 MW WTG (GEWTG2)	± 0.9
J527	100 GE 2.5 MW WTG (GEWTG2)	± 0.9
J528	80 GE 2.5 MW WTG (GEWTG2)	± 0.9
J529	125 Vestas V110 2.0 MW (VS3103)	0.95 lagging 0.94 leading
J530	100 GE 2.5 MW WTG (GEWTG2)	± 0.9
J534	100 GE 2.5 MW WTG (GEWTG2)	± 0.9
J535	84 GE 2.5 MW WTG (GEWTG2)	± 0.9

**Table A-3: Collector System and Shunt Compensation
Modeling for DPP West Area Non-Synchronous Projects**

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J432	One 98 MW unit	Req = 0.010510 pu Xeq = 0.069124 pu Beq = 0.018199 pu	30 Mvar capacitor bank on 34.5 kV system
J460	Two 100 MW units	Circuit #1: R=0.008357 pu X=0.0101 B=0.05792 Circuit #2: R=0.019103 X=0.025036 B=0.04728	15 Mvar capacitor bank on each 34.5 kV collector system
J475	One 168 MW unit and one 32 MW unit	Circuit #1: R=0.00727 pu X=0.0116 pu B=0.17079 pu Circuit #2: R=0.03461 pu X=0.05527 pu B=0.05274 pu	3*15 Mvar capacitor bank and 1*5 Mvar reactor bank on one 34.5 kV system; 1*5 Mvar capacitor bank on another 34.5 kV system
J485	Five 9.37 MW units	None	None
J488	One 151.8 MW unit	R=0.01448 pu X=0.02156 pu B=0.0 pu	20 Mvar capacitor bank on 34.5 kV system
J493	One 150 MW unit	Req = 0.008928 pu Xeq = 0.039978 pu Beq = 0.06162 pu	10 Mvar capacitor bank on 34.5 kV system
J495	Two 100 MW units	Two equivalent collector systems, each has: Req = 0.02395 pu Xeq = 0.026499 pu Beq = 0.10306 pu	27 Mvar capacitor bank on each 34.5 kV collector system
J498	Two 170 MW units	None	None
J499	Two 170 MW units	None	None
J500	Two 166 MW units One 168 MW unit	None	None
J504	One 50 MW unit	Req = 0.001848 pu Xeq = 0.001092 pu Beq = 0.00232 pu	None

MISO Project #	Generator Modeling	Collector System Modeling	Shunt Compensation
J506	Two 100 MW units	Two equivalent collector systems, each has: Req = 0.026793 pu Xeq = 0.030726 pu Beq = 0.08777 pu	27.5 Mvar capacitor bank on each 34.5 kV collector system
J510	One 284.5 MW unit	None	None
J514	Two 232 MW units One 257 MW unit	None	None
J523	One 50 MW unit	Req = 0.007079 pu Xeq = 0.006495 pu Beq = 0.00475 pu	12 Mvar capacitor bank on 34.5 kV system
J524	One 100 MW unit	Req = 0.003379 pu Xeq = 0.003045 pu Beq = 0.00907 pu	None
J526	Two 150 MW units	Two equivalent collector systems, each has: R=0.0086 pu X=0.0110 pu B=0.1065 pu	10 Mvar capacitor bank on each 34.5 kV collector system
J527	Two 125 MW units	Two equivalent collector systems, each has: R=0.0104 pu X=0.0132 pu B=0.0884 pu	7 Mvar capacitor bank on each 34.5 kV collector system
J528	Two 100 MW units	Two equivalent collector systems, each has: R=0.0130 pu X=0.0166 pu B=0.0703 pu	7 Mvar capacitor bank on each 34.5 kV collector system
J529	Two 125 MW units	None	None
J530	Two 125 MW units	Two equivalent collector systems, each has: R=0.0104 pu X=0.0132 pu B=0.0884 pu	7 Mvar capacitor bank on each 34.5 kV collector system
J534	Two 125 MW units	Two equivalent collector systems, each has: R=0.0104pu X=0.0132pu B=0.0884pu	7 Mvar capacitor bank on each 34.5 kV collector system
J535	One 210 MW unit	R=0.0043pu X=0.0055pu B=0.2131pu	12 Mvar capacitor bank on 34.5 kV bus

Table A-4: DPP 2016 February Central Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J351	IN	St. Joseph	NIPS	Stillwell 345 kV	705 sum / 715 win	CC	NRIS
J468	IL	Douglas	AMIL	Sidney-Kansas West 345 kV line	202	Wind	NRIS
J515	IN	Benton, Warren	DEI	Cayuga 345 kV	400	Wind	ERIS

Table A-5: DPP 2016 February Michigan Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J533	MI	Gratiot	METC	Slate 345 kV	200	Wind	NRIS
J538	MI	Hillsdale	METC	Moore Rd-Dowling 138 kV	150	Wind	NRIS
J540	MI	Kalkaska	WPSC	Kalkaska County	48.6	CT	NRIS Only

Table A-6: DPP 2016 February ATC Area Projects

MISO Project Num	State	County	Trans. Owner	Point Of Interconnection	Max Output	Fuel Type	Service Type
J505	WI	Manitowoc	ATC	Mishicot-Kewaunee 138 kV line	100	Solar	NRIS

A.2 Model Review Comments

Table A-7: Model Review Comments

Company	Python/ Idev File Name	2022 SH	2022 SPK	2022 Stability
Xcel	G621_SH.py	x		x
Xcel	G621_PK.py		x	
MISO	RMV_J489.py	x	x	x
MISO	RMV_J525.py	x	x	x
J475	J475_update.py	x	x	x
MISO	Redispatch_Gen-GIA_SH.py	x		x
MISO	Redispatch_Gen-GIA_PK.py		x	
MEC	DELETE-J298.py	x	x	x
MEC	MEC DPP 2016FEB PH3 SH Updates.py	x		x
MEC	MEC DPP 2016FEB PH3 SUM Updates.py		x	
MISO	Gen_Retired_PK.py		x	
MISO	Disp-BS_Coyota_Pmin.py	x		x
MISO	HCK-CARDINAL-MVP.idv	x		x
J493	Disp_J493_SH.py	x		x
J493	Disp_J493_PK.py		x	
Ameren	TP-950.idv	x	x	x
Ameren	TP-980.idv	x	x	x
Ameren	Gateway.idv	x	x	x
Ameren	Beehive.idv	x	x	x
Ameren	Pike-Calumet Area.idv	x	x	x
Ameren	AMRN_Rating Chng.py	x	x	x
MDU	Remove duplicate Ellendale shunt reactor.py	x	x	x

A.3 MISO Classic as the Study Sink

Table A-8: MISO Classic as the Study Sink

Area #	Area Name	Area #	Area Name
207	HE	600	Xcel
208	DEI	608	MP
210	SIGE	613	SMMPA
216	IPL	615	GRE
217	NIPS	620	OTP
218	METC	627	ALTW
219	ITC	633	MPW
295	WEC	635	MEC
296	MIUP	661	MDU
314	BREC	663	BEPC-MISO
333	CWLD	680	DPC
356	AMMO	694	ALTE
357	AMIL	696	WPS
360	CWLP	697	MGE
361	SIPC	698	UPPC

A.4 PJM Market as PJM Projects Sink

Table A-9: PJM Market as PJM Projects Sink

Area #	Area Name	Area #	Area Name
201	AP	229	PPL
202	ATSI	230	PECO
205	AEP	231	PSE&G
209	DAY	232	BGE
212	DEO&K	233	PEPCO
215	DLCO	234	AE
222	CE	235	DP&L
225	PJM	236	UGI
226	PENELEC	237	RECO
227	METED	320	EKPC
228	JCP&L	345	DVP

A.5 SPP Market as SPP Projects Sink

Table A-10: SPP Market as SPP Projects Sink

Area #	Area Name	Area #	Area Name
515	SWPA	541	KCPL
520	AEPW	542	KACY
523	GRDA	544	EMDE
524	OKGE	545	INDN
525	WFEC	546	SPRM
526	SPS	640	NPPD
527	OMPA	645	OPPD
531	MIDW	650	LES
534	SUNC	652	WAPA
536	WERE	659	BEPC-SPP
540	GMO		

A.6 Contingency Files used in Steady-State Analysis

Table A-11: List of Contingencies used in Steady-State Analysis

Contingency File Name	Description	2022
Automatic single element contingencies	Single element outages at buses 60 kV and above in the study region	x
20170808_CHC-NLL_Eden2.con	Specified category P1, P2 contingencies in ATC, ITCM	x
CIPCO DPP-2016-FEB Phase 2 Additional P6.con	Specified category P6 contingencies in CIPCO	x
CC Bipole Events.con	Specified category P1, P7 contingencies in GRE Coal Creek	x
MEC-DPP2016FEB West Ph2 2022 Cat P1 09.15.2017.con	Specified category P1 contingencies in MEC	x
MEC-DPP2016FEB West Ph2 2022 Cat P2 09.15.2017.con	Specified category P2 contingencies in MEC	x
MEC-DPP2016FEB West Ph2 2022 Cat P5 09.15.2017.con	Specified category P5 contingencies in MEC	x
MEC-DPP2016FEB West Ph2 2022 Cat P7 09.15.2017.con	Specified category P7 contingencies in MEC	x
P1_AMES_MTEP17-2022TA.CON	Specified category P1 contingencies in AMES	x
P2-P7_AMES_MTEP17-2022TA.con	Specified category P2-P7 contingencies in AMES	x
P1_ATC_MTEP17-2022TA.con	Specified category P1 contingencies in ATC	x
P2-P7_ATC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ATC	x
P1_BEPC_MTEP17-2022TA.con	Specified category P1 contingencies in BEPC	x
P2-P7_BEPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in BEPC	x
P1_CBPC_MTEP17-2022TA.con	Specified category P1 contingencies in CBPC	x
P2-P7_CBPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CBPC	x
P1_CFU_MTEP17-2022TA.con	Specified category P1 contingencies in CFU	x
P2-P7_CFU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CFU	x
P1_CIPCO_MTEP17-2022TA.con	Specified category P1 contingencies in CIPCO	x
P2-P7_CIPCO_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CIPCO	x
P1_DPC_MTEP17-2022TA.con	Specified category P1 contingencies in DPC	x
P2-P7_DPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in DPC	x
P1_GRE_MTEP17-2022TA.con	Specified category P1 contingencies in GRE	x
P2-P7_GRE_MTEP17-2022TA.con	Specified category P2-P7 contingencies in GRE	x

Contingency File Name	Description	2022
P1_ITCM_MTEP17-2022TA.con	Specified category P1 contingencies in ITCM	x
P2-P7_ITCM_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ITCM	x
P1_MDU_MTEP17-2022TA.con	Specified category P1 contingencies in MDU	x
P2-P7_MDU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in MDU	x
P1_MP_MTEP17-2022TA.con	Specified category P1 contingencies in MP	x
P2-P7_MP_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MP	x
P1_MPC_MTEP17-2022TA.con	Specified category P1 contingencies in MPC	x
P2-P7_MPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in MPC	x
P1_MPW_MTEP17-2022TA.con	Specified category P1 contingencies in MPW	x
P2-P7_MPW_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MPW	x
P1_MRES_MTEP17-2022TA.con	Specified category P1 contingencies in MRES	x
P2-P7_MRES_MTEP17-2022TA.CON	Specified category P2-P7 contingencies in MRES	x
P1_OTP_MTEP17-2022TA.con	Specified category P1 contingencies in OTP	x
P2-P7_OTP_MTEP17-2022TA.con	Specified category P2-P7 contingencies in OTP	x
P1_RPU_MTEP17-2022TA.con	Specified category P1 contingencies in RPU	x
P2-P7_RPU_MTEP17-2022TA.con	Specified category P2-P7 contingencies in RPU	x
P1_SMPMA_MTEP17-2022TA.con	Specified category P1 contingencies in SMPMA	x
P2-P7_SMPMA_MTEP17-2022TA.con	Specified category P2-P7 contingencies in SMPMA	x
P1_XEL_MTEP17-2022TA.con	Specified category P1 contingencies in XEL	x
P2-P7_XEL_MTEP17-2022TA.con	Specified category P2-P7 contingencies in XEL	x
P1-4_ATC_MTEP17-2022TA.con	Specified category P1-4 contingencies in ATC	x
P1-4_DPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in DPC	x
P1-4_GRE_MTEP17-2022TA.con	Specified category P1-4 contingencies in GRE	x
P1-4_ITCM_MTEP17-2022TA.con	Specified category P1-4 contingencies in ITCM	x
P1-4_MDU_MTEP17-2022TA.con	Specified category P1-4 contingencies in MDU	x
P1-4_MP_MTEP17-2022TA.con	Specified category P1-4 contingencies in MP	x
P1-4_MPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in MPC	x
P1-4_OTP_MTEP17-2022TA.con	Specified category P1-4 contingencies in OTP	x
P1-4_SMPMA_MTEP17-2022TA.con	Specified category P1-4 contingencies in SMPMA	x
P1-4_XEL_MTEP17-2022TA.con	Specified category P1-4 contingencies in XEL	x

Contingency File Name	Description	2022
AMRN-P1-2 2021 4-6-2018.con	Specified category P1-2 contingencies in Ameren	x
AMRN-P1-3 2021 4-6-2018.con	Specified category P1-3 contingencies in Ameren	x
AMRN-P2-1 2021 4-6-2018.con	Specified category P2-1 contingencies in Ameren	x
AMRN-P2-2 2021 4-6-2018.con	Specified category P2-2 contingencies in Ameren	x
AMRN-P2-3 2021 4-6-2018.con	Specified category P2-3 contingencies in Ameren	x
AMRN-P2-4 2021 4-6-2018.con	Specified category P2-4 contingencies in Ameren	x
AMRN-P7 2021 4-6-2018.con	Specified category P7 contingencies in Ameren	x
P1_CWLD_MTEP17-2022TA.con	Specified category P1 contingencies in CWLD	x
P2-P7_CWLD_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CWLD	x
P1_CWLP_MTEP17-2022TA.con	Specified category P1 contingencies in CWLP	x
P2-P7_CWLP_MTEP17-2022TA.con	Specified category P2-P7 contingencies in CWLP	x
P1_PPI_MTEP17-2022TA.con	Specified category P1 contingencies in PPI	x
P2-P7_PPI_MTEP17-2022TA.con	Specified category P2-P7 contingencies in PPI	x
P1_SIPC_MTEP17-2022TA.con	Specified category P1 contingencies in SIPC	x
P2-P7_SIPC_MTEP17-2022TA.con	Specified category P2-P7 contingencies in SIPC	x
ComEd_RTEP_Cat_P1.con	Specified category P1 contingencies in ComEd	x
ComEd_RTEP_Cat_P2-P7.con	Specified category P2-P7 contingencies in ComEd	x
P1_AECI_MTEP17-2022TA.con	Specified category P1 contingencies in AECI	x
P2-P7_AECI_MTEP17-2022TA.con	Specified category P2-P7 contingencies in AECI	x
P1_CE_MTEP17-2022TA.con	Specified category P1 contingencies in ComEd	x
P2-P7_CE_MTEP17-2022TA.con	Specified category P2-P7 contingencies in ComEd	x
P1_WAPA_MTEP17-2022TA.con	Specified category P1 contingencies in WAPA	x
P2-P7_WAPA_MTEP17-2022TA.con	Specified category P2-P7 contingencies in WAPA	x
P1-4_AMRN_MTEP17-2022TA.con	Specified category P1-4 contingencies in Ameren	x
P1-4_CWLP_MTEP17-2022TA.con	Specified category P1-4 contingencies in CWLP	x
P1-4_SIPC_MTEP17-2022TA.con	Specified category P1-4 contingencies in SIPC	x
P1-4_WAPA_MTEP17-2022TA.con	Specified category P1-4 contingencies in WAPA	x

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Model Data

B.1 Power Flow Model Data

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B.2 Dynamic Model Data

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B.3 2022 Slider Diagrams

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Reactive Power Requirement Analysis Results (FERC Order 827)

Table C-1: Reactive Power Requirement Analysis Results

Project #	HV Side Bus #	Lagging Power Factor Results				Lagging Power Factor Results				Turbine Inherent Power Factor
		MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Lagging Power Factor at HV Side	Meet Lagging Power Factor Req.?	MW from plant to HV side (P)	MVAR from plant to HV side (Q)	Leading Power Factor at HV Side	Meet Leading Power Factor Req.?	
J432	61031	96.20	32.90	0.946	Y	95.68	-72.30	-0.798	Y	± 0.95
J460	61042	196.43	74.56	0.935	Y	196.00	-90.26	-0.908	Y	± 0.95
J475	65730	195.06	71.00	0.940	Y	194.70	-108.50	-0.874	Y	± 0.95
J488	60418	147.19	53.73	0.939	Y	144.90	-135.80	-0.730	Y	± 0.9
J493	60215	146.81	48.26	0.950	Y	146.24	-109.15	-0.801	Y	0.902 lagging 0.918 leading
J495	61536	192.76	84.72	0.915	Y	192.10	-99.06	-0.889	Y	± 0.95
J504	71089	49.74	15.08	0.957	Y	49.65	-37.37	-0.799	Y	± 0.9
J506	15010	192.32	83.38	0.917	Y	191.54	-102.48	-0.882	Y	± 0.95
J523	97996	48.99	26.23	0.882	Y	48.75	-37.07	-0.796	Y	± 0.9
J524	96996	98.05	39.51	0.928	Y	97.53	-76.36	-0.787	Y	± 0.9
J526	72032	292.76	119.00	0.926	Y	291.18	-210.46	-0.810	Y	± 0.9
J527	65201	243.98	98.98	0.927	Y	242.86	-169.66	-0.820	Y	± 0.9
J528	65301	195.16	79.84	0.926	Y	194.10	-140.50	-0.810	Y	± 0.9
J530	75731	243.98	98.98	0.927	Y	242.86	-169.66	-0.820	Y	± 0.9
J534	66001	243.98	98.98	0.927	Y	242.86	-169.66	-0.820	Y	± 0.9
J535	66202	205.82	90.16	0.916	Y	204.96	-136.07	-0.833	Y	± 0.9

2022 Contingency Analysis Results

D.1 2022 Summer Shoulder (SH) Constraints

Table D-1: 2022 SH System Intact Thermal Constraints

Table D-2: 2022 SH System Intact Voltage Constraints

Table D-3: 2022 SH Category P1 Thermal Constraints

Table D-4: 2022 SH Category P1 Voltage Constraints

Table D-5: 2022 SH Category P2-P7 Thermal Constraints

Table D-6: 2022 SH Category P2-P7 Voltage Constraints

Table D-7: 2022 SH Non-Converged Contingencies

Table D-8: 2022 SH Non-Converged Contingencies DCCC Results

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D.2 2022 Summer Peak (SPK) Constraints

Table D-9: 2022 SPK System Intact Thermal Constraints

Table D-10: 2022 SPK System Intact Voltage Constraints

Table D-11: 2022 SPK Category P1 Thermal Constraints

Table D-12: 2022 SPK Category P1 Voltage Constraints

Table D-13: 2022 SPK Category P2-P7 Thermal Constraints

Table D-14: 2022 SPK Category P2-P7 Voltage Constraints

Table D-15: 2022 SPK Non-Converged Category P2-P7 Contingencies

Table D-16: 2022 SPK Non-Converged Category P2-P7 Contingencies DCCC Results

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Affected System Contingency Analysis Results

E.1 CIPCO/MPC Companies Affected System Analysis Results

Table E-1: 2022 SH CIPCO/MPC Affected System Analysis Results

Table E-2: 2022 SPK CIPCO/MPC Affected System Analysis Results

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E.2 OTP LPC Analysis

E.2.1 OTP LPC Model Updates

Table E-3: Wind Generation Turned Off to Create OTP LPC Light Load No Wind Case

Bus Number	Bus Name	Machine Id	Area Num	Area Name
659194	NDPRAIRWND1W0.6900	1	659	BEPC-SPP
659273	WILTON 2 W0.6900	2	659	BEPC-SPP
659289	DAYCNTYWND W0.6900	1	659	BEPC-SPP
659290	POMONA W0.6900	1	659	BEPC-SPP
659291	HYDE W0.6900	1	659	BEPC-SPP
659294	WILTON 1 W0.6900	1	659	BEPC-SPP
659296	SDPRAIRWND1W0.6900	1	659	BEPC-SPP
659366	WILTON 3 W0.6900	3	659	BEPC-SPP
659411	CAMPBLCNTY W0.6900	1	659	BEPC-SPP
659453	BRADYGEN1 W0.6900	1	659	BEPC-SPP
659457	LINDAHLWNDGW0.6900	1	659	BEPC-SPP
661307	DM WILWW 0.5750	W	661	MDU
661317	CDR HLLW 0.5750	W	661	MDU
661986	FOXTAILW 0.6900	W	661	MDU
661987	G752-2 W 0.6600	W	661	MDU
661989	THNDRSPT G W0.6600	W	661	MDU
661999	TATANKAW 0.5750	W	661	MDU
6060	G736 0.6900	1	620	OTP
600162	J262 COURT W0.6000	W	620	OTP
600163	J263 COURT W0.6000	W	620	OTP
600165	LK AGASSIZ G0.6000	W	620	OTP
615132	G619TAMW 0.6000	W	620	OTP
620112	EDGE GEN 0.5750	1	620	OTP
620115	HRDLKGEN 0.6000	1	620	OTP
620115	HRDLKGEN 0.6000	2	620	OTP
620115	HRDLKGEN 0.6000	3	620	OTP
620115	HRDLKGEN 0.6000	4	620	OTP
620412	L1GEN2 9 0.5750	1	620	OTP
620413	L1GEN1 9 0.5750	1	620	OTP
620451	G474 GRNT CW41.600	W	620	OTP

Bus Number	Bus Name	Machine Id	Area Num	Area Name
657600	L1GEN3 9 0.5750	1	620	OTP
657601	L2GEN 9 0.5750	1	620	OTP
657737	ASH3GEN9 0.6900	1	620	OTP
657745	MPC02100GEN90.6900	1	620	OTP
657964	ASH2GEN9 0.5750	1	620	OTP
657964	ASH2GEN9 0.5750	2	620	OTP
657985	ASH1GEN9 0.5750	1	620	OTP
657985	ASH1GEN9 0.5750	2	620	OTP
562872	G15046_3 0.6900	1	652	WAPA
563302	G15096_3 0.6900	1	652	WAPA
563342	G15098_3 0.6900	1	652	WAPA
652004	G13_001IS_3 0.6900	1	652	WAPA
652007	G14_001IS_3 0.6900	1	652	WAPA
652173	G09_001IS_2 0.6900	1	652	WAPA
652345	G07_023IS_2 0.6900	W	652	WAPA
658019	MPS WIND G 0.6000	W1	652	WAPA
659031	G12_014IS_4 0.6900	1	652	WAPA
659127	G14_004IS_2 0.6900	1	652	WAPA
659141	G14_010IS_2 0.6900	1	652	WAPA
659407	ANTELPHILLSW0.6900	1	652	WAPA
660013	OAKTREE WIND0.7000	1	652	WAPA
660016	B_H WIND 0.7000	1	652	WAPA
50324	J442 G 0.6900	1	620	OTP
50420	J436 G 0.6900	1	620	OTP
50430	J437 G 0.6900	1	620	OTP

E.2.2 OTP LPC Summer Shoulder Study Results

Table E-4: OTP LPC SH System Intact Thermal Constraints

Table E-5: OTP LPC SH System Intact Voltage Constraints

Table E-6: OTP LPC SH Category P1 Thermal Constraints

Table E-7: OTP LPC SH Category P1 Voltage Constraints

Table E-8: OTP LPC SH Category P2-P7 Thermal Constraints

Table E-9: OTP LPC SH Category P2-P7 Voltage Constraints

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E.2.3 OTP LPC Summer Peak Study Results

Table E-10: OTP LPC SPK System Intact Thermal Constraints

Table E-11: OTP LPC SPK System Intact Voltage Constraints

Table E-12: OTP LPC SPK Category P1 Thermal Constraints

Table E-13: OTP LPC SPK Category P1 Voltage Constraints

Table E-14: OTP LPC SPK Category P2-P7 Thermal Constraints

Table E-15: OTP LPC SPK Category P2-P7 Voltage Constraints

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E.2.4 OTP LPC Light Load No Wind (LLNW) Study Results

Table E-16: OTP LPC LLNW System Intact Potential High Voltages

Table E-17: OTP LPC LLNW Category P1 Potential High Voltages

Table E-18: OTP LPC LLNW Category P2-P7 Potential High Voltages

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E.3 PJM Affected System Study Results

Below is the PJM affected system study report provided by PJM.

E.3.1 PJM Affected System Steady State Analysis

E.3.2 PJM Affected System J475 Dynamic Stability Analysis

E.3.3 PJM Affected System J530 Dynamic Stability Analysis

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E.4 SPP Affected System Study Results

Below is the SPP affected system study report provided by SPP.

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Transient Stability Results

F.1 2022 Summer Shoulder Stability Results Summary

Stability simulation was performed in the 2022 summer shoulder case.

If the identified stability related issues (Section 4.5.1) are resolved, all stability simulations will be transiently stable, transient period voltage criteria will be met, oscillations will be well damped. No additional Network Upgrades are required in stability analysis.

Stability study results summary is in Table F-1.

Table F-1: 2022 Summer Shoulder Stability Analysis Results Summary

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F.2 2022 Summer Shoulder Stability Plots

Plots of stability simulations for 2022 summer shoulder study case are in separate files which are listed below:

AppendixF2_2022SH_DPP 2016Feb-West_Phase 3_Plots.zip

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MWEX Voltage Study Details

Below is the MWEX voltage stability study report provided by ATC.

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Short Circuit Analysis

- H.1 J432 Short Circuit Study Performed by Siemens PTI
- H.2 J460 Short Circuit Study Performed by Siemens PTI
- H.3 J475 Short Circuit Study Performed by MEC
- H.4 J485 Short Circuit Study Performed by Siemens PTI
- H.5 J488 & J489 Short Circuit Study Performed by OTP
- H.6 J493 and J510 Short Circuit Study Performed by OTP
- H.7 J495 Short Circuit Study Performed by ITCM
- H.8 J498 Short Circuit Facilities Study Performed by MEC
- H.9 J499 Short Circuit Facilities Study Performed by MEC
- H.10 J500 Short Circuit Facilities Study Performed by MEC
- H.11 J504 Short Circuit Facilities Study Performed by ITCM
- H.12 J506 Short Circuit Facilities Study Performed by Siemens PTI
- H.13 J514 Short Circuit Facilities Study Performed by ITCM
- H.14 J523 Short Circuit Facilities Study Performed by ITCM
- H.15 J524 Short Circuit Facilities Study Performed by Siemens PTI
- H.16 J526 Short Circuit Facilities Study Performed by OTP
- H.17 J527 Short Circuit Facilities Study Performed by Siemens PTI
- H.18 J528 Short Circuit Facilities Study Performed by Siemens PTI

H.19 J529 Short Circuit Facilities Study Performed by MEC

H.20 J530 Short Circuit Facilities Study Performed by Siemens PTI

H.21 J534 Short Circuit Facilities Study Performed by Siemens PTI

H.22 J535 Short Circuit Facilities Study Performed by Siemens PTI

J432 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J432 generating facility (49 Gamesa 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the Xcel system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "NSP_03202017.GDB"

The J432 short circuit benchmark model was developed as follows:

- Added Big Stone South-Ellendale 345 kV line and Ellendale 345/230/34.5 kV transformer.
- Added J460 project and turned it online.

The J432 short circuit study model was developed starting from the above benchmark case as follows:

- The J432 generating facility with its sequence parameters was added at the POI on Brookings County 345 kV Substation. The associated 345 kV Gentie, 345-34.5-13.2 kV transformer ($Z=9\%$ at 60 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.69 kV pad mount transformer ($Z=11.6\%$ at 115.15 MVA, D-YG) were also added.
- The J432 generator was modeled as a generator with impedances listed below. These values represent the DFIG configuration with the maximum crowbar resistance connected in the rotor.
 - o Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): $0.71474+j0.3344$ pu
 - o Negative sequence impedance (Z_2): $0.04899+j0.1626$ pu
 - o Zero sequence impedance (Z_0): j999999 pu (zero sequence currents are blocked by the GSU transformers (Wye-Delta connection, therefore, this value does not have any impact on the short circuit results)

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J432 project) and benchmark case (without the J432 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J432 (study case) and without J432 (benchmark case) are listed in Table 1. With the addition of the J432 generating facility, the maximum increase in the 3PH short circuit current is 0.74% (126.1 A), and the maximum increase in the SLG short circuit current is 2.19% (382.1 A), which are both observed at the Brookings County 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

As shown in the Table 1, the changes in fault current at buses around the POI are relatively small. Based on the Transmission Owner's short circuit criteria, interconnection of the J432 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J432

Substation	Faulted Bus			Benchmark Case without J432 Project		Study Case with J432 Project		Ampere Change with J432 vs Benchmark		% Change with J432 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J432	61031	J432_POI	345	-	-	17514.8	16541.2	-	-	-	-
J432	61032	J432-COL	34.5	-	-	17756.7	11618.6	-	-	-	-
J432	61034	J432_COL	34.5	-	-	8518.4	8546.4	-	-	-	-
J432	61035	J432_TR_TER	13.2	-	-	0.0	21383.1	-	-	-	-
BROOKINGS COUNTY (NS	113200	BOK_345KV_1	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113201	BOK_345KV_2	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113202	BOK_IB_BSS	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113203	BOK_IB_HNL	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113204	BOK_IB_TR09	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113205	BOK_IB_TR10	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113206	BOK_IB_WH1	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
BROOKINGS COUNTY (NS	113207	BOK_IB_WH12	345	17465.9	16930.2	17848.0	17056.3	382.1	126.1	2.19%	0.74%
WHITE (WAPA)	225600	WHI_345KV_1	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225601	WHI_345KV_2	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225602	WHI_IB_BOK1	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225603	WHI_IB_BOK2	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225604	WHI_IB_SPK	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225605	WHI_IB_TR01	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	225606	WHI_IB_WTT	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
WHITE (WAPA)	300094	WHI_IB_DCR	345	17543.1	16953.6	17914.7	17078.6	371.6	125.0	2.12%	0.74%
DEER CREEK (WAPA)	300095	DCR_345KV	345	17140.7	16552.6	17476.8	16669.9	336.1	117.3	1.96%	0.71%
J460	61041	J460_POI	345	11512.8	12497.8	11577.8	12538.5	65.0	40.7	0.56%	0.33%
BROOKINGS COUNTY (NS	113208	BOK_115KV_1	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113209	BOK_115KV_2	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113210	BOK_IB_TR10	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113211	BOK_IB_YNK1	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
BROOKINGS COUNTY (NS	113212	BOK_IB_YNK2	115	28120.4	26806.8	28344.7	26889.9	224.3	83.1	0.80%	0.31%
J460	61042	J460	345	11150.9	11940.4	11210.8	11977.2	59.9	36.8	0.54%	0.31%
HAWKS NEST (WIND)	300068	HNL_345KV_1	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
HAWKS NEST (WIND)	300069	HNL_345KV_2	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
HAWKS NEST (WIND)	300070	HNL_BOK	345	9715.8	11548.8	9751.0	11576.7	35.2	27.9	0.36%	0.24%
RED PINE (WIND)	300071	HNL_GEN	345	9658.6	11499.7	9693.3	11527.4	34.7	27.7	0.36%	0.24%
WATERTOWN (WAPA)	221500	WTT_345KV	345	8739.3	9643.6	8760.8	9660.8	21.5	17.2	0.25%	0.18%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J432

Substation	Faulted Bus			Benchmark Case without J432 Project		Study Case with J432 Project		Ampere Change with J432 vs Benchmark		% Change with J432 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
BIG STONE SOUTH (OTP)	109600	BST_345KV	345	8289.1	7472.0	8302.6	7481.3	13.5	9.3	0.16%	0.12%
WHITE (WAPA)	225607	WHI_115KV	115	12163.7	10520.0	12191.8	10532.5	28.1	12.5	0.23%	0.12%
YANKEE (NSPM)	229500	YNK_115KV_1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229501	YNK_115KV_2	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229502	YNK_IB_BOK1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229503	YNK_IB_BOK2	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229504	YNK_IB_BRI	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229505	YNK_IB_RB1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229506	YNK_IB_TR01	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	229507	YNK_IB_TR02	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
YANKEE (NSPM)	300448	YNK_REAC1	115	13176.0	15367.3	13200.5	15385.3	24.5	18.0	0.19%	0.12%
WATERTOWN (WAPA)	221501	WTT_230KV	230	11808.2	13524.8	11825.4	13539.9	17.2	15.1	0.15%	0.11%
LYON COUNTY (NSPM)	168300	LYC_345KV_1	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168301	LYC_345KV_2	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168302	LYC_IB_BOK1	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168303	LYC_IB_CMT2	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
LYON COUNTY (NSPM)	168304	LYC_IB_TR09	345	10169.7	11844.0	10183.6	11856.6	13.9	12.6	0.14%	0.11%
BIG STONE SOUTH (OTP)	300906	BSS_230KV	230	16323.8	12559.0	16340.7	12569.7	16.9	10.7	0.10%	0.09%
BIG STONE (OTP)	109601	BST_230KV	230	17057.9	12766.6	17074.9	12777.1	17.0	10.5	0.10%	0.08%
SPLIT ROCK (NSPM)	206100	SPK_345KV_1	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206101	SPK_345KV_2	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206102	SPK_IB_NOB	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206103	SPK_IB_SOC	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%
SPLIT ROCK (NSPM)	206104	SPK_IB_WHI	345	13532.6	14710.5	13547.9	14721.6	15.3	11.1	0.11%	0.08%

J460 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J460 generating facility (100 Vestas V110 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the Xcel system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "NSP_03202017.GDB"

The J460 short circuit benchmark model was developed as follows:

- Added Big Stone South-Ellendale 345 kV line and Ellendale 345/230/34.5 kV transformer.
- Added J432 project and turned it online

The J460 short circuit study model was developed starting from the above benchmark case as follows:

- The J460 generating facility with its sequence parameters was added at the POI on Brookings County – Hawks Nest (H081) 345 kV Substation. The associated 345 kV Gentie, two 345-34.5-13.2 kV transformers ($Z=6\%$ at 100 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z+=7.8\%$, $Z0=7.3\%$ at 105 MVA, D-YG) were also added.
- The J460 generation was represented using the conventional generator model with the impedance of the machines set as follows per the data provided in the Vestas manual.
 - Positive saturated sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.1669 p.u.
 - Negative saturated sequence impedance ($Z2$): j0.1669 p.u. (same as positive sequence impedances)
 - Zero sequence saturated impedance ($Z0$): j0.1669 p.u.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J460 project) and benchmark case (without the J460 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J460 (study case) and without J460 (benchmark case) are listed in Table 1. With the addition of the J460 generating facility, the maximum increase in the 3PH short circuit current is 6.58% (715.2 A), and the maximum increase in the SLG short circuit current is 16.76% (1399.6 A), which are both observed at the Hawks Nest (H081) 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 11577 A (increased by 715.2 A) and SLG fault current is 9751 A (increased by 1399.6 A) at the Hawks Nest 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J460 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J460	61041	J460_POI	345	-	-	11577.8	12538.5	-	-	-	-
J460	61042	J460	345	-	-	11210.8	11977.2	-	-	-	-
J460	61043	J460_COL1	34.5	-	-	38909.0	29079.0	-	-	-	-
J460	61046	J460_COL2	34.5	-	-	38436.0	28683.7	-	-	-	-
J460	61047	J460_COL1	34.5	-	-	33279.5	26558.9	-	-	-	-
J460	61048	J460_COL2	34.5	-	-	27091.6	23334.5	-	-	-	-
J460	61049	J460_TER1	13.2	-	-	0.0	53996.6	-	-	-	-
J460	61050	J460_TER2	13.2	-	-	0.0	54574.1	-	-	-	-
HAWKS NEST (WIND)	300068	HNL_345KV_1	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
HAWKS NEST (WIND)	300069	HNL_345KV_2	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
HAWKS NEST (WIND)	300070	HNL_BOK	345	8351.4	10861.5	9751.0	11576.7	1399.6	715.2	16.76%	6.58%
RED PINE (WIND)	300071	HNL_GEN	345	8310.2	10818.4	9693.3	11527.4	1383.1	709.0	16.64%	6.55%
BROOKINGS COUNTY (NS)	113200	BOK_345KV_1	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113205	BOK_IB_TR10	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113206	BOK_IB_WH11	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113201	BOK_345KV_2	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113204	BOK_IB_TR09	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113207	BOK_IB_WH12	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113202	BOK_IB_BSS	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
BROOKINGS COUNTY (NS)	113203	BOK_IB_HNL	345	17011.8	16398.2	17848.0	17056.3	836.2	658.1	4.92%	4.01%
WHITE (WAPA)	225600	WHI_345KV_1	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	300094	WHI_IB_DCR	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225601	WHI_345KV_2	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225604	WHI_IB_SPK	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225606	WHI_IB_WTT	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225605	WHI_IB_TR01	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225603	WHI_IB_BOK2	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
WHITE (WAPA)	225602	WHI_IB_BOK1	345	17096.7	16425.7	17914.7	17078.6	818.0	652.9	4.78%	3.97%
J432	61031	J432_POI	345	16705.3	15910.4	17514.8	16541.2	809.5	630.8	4.85%	3.96%
DEER CREEK (WAPA)	300095	DCR_345KV	345	16729.8	16056.2	17476.8	16669.9	747.0	613.7	4.47%	3.82%
LYON COUNTY (NSPM)	168300	LYC_345KV_1	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168301	LYC_345KV_2	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168304	LYC_IB_TR09	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
LYON COUNTY (NSPM)	168302	LYC_IB_BOK1	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
LYON COUNTY (NSPM)	168303	LYC_IB_CMT2	345	9847.1	11563.6	10183.6	11856.6	336.5	293.0	3.42%	2.53%
BROOKINGS COUNTY (NS)	113210	BOK_IB_TR10	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113211	BOK_IB_YNK1	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113208	BOK_115KV_1	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113209	BOK_115KV_2	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
BROOKINGS COUNTY (NS)	113212	BOK_IB_YNK2	115	27824.4	26455.5	28344.7	26889.9	520.3	434.4	1.87%	1.64%
HAZEL CREEK (NSPM)	148702	HZC_IB_LYC	345	6884.9	7977.3	6974.8	8087.0	89.9	109.7	1.31%	1.38%
HAZEL CREEK (NSPM)	148700	HZC_345KV_1	345	6884.9	7977.3	6974.8	8087.0	89.9	109.7	1.31%	1.38%
CEDAR MOUNTAIN (GRE)	119104	CMT_IB_LYC2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119100	CMT_345KV_1	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119103	CMT_IB_HNA2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119101	CMT_345KV_2	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119105	CMT_IB_TR01	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
CEDAR MOUNTAIN (GRE)	119102	CMT_IB_HNA1	345	8864.5	11514.3	8941.6	11644.0	77.1	129.7	0.87%	1.13%
WATERTOWN (WAPA)	221500	WTT_345KV	345	8698.8	9561.6	8760.8	9660.8	62.0	99.2	0.71%	1.04%
LYON COUNTY (NSPM)	168305	LYC_115KV_1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168306	LYC_115KV_2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168312	LYC_IB_CLO	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168307	LYC_CAP	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168316	LYC_IB_TR09	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168311	LYC_IB_CAP2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168310	LYC_IB_CAP1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168315	LYC_IB_MSH	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168313	LYC_IB_LAY1	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168314	LYC_IB_LAY2	115	18361.7	17413.1	18563.3	17563.0	201.6	149.9	1.10%	0.86%
LYON COUNTY (NSPM)	168308	LYC_CAP1	115	17035.5	16216.2	17208.7	16346.0	173.2	129.8	1.02%	0.80%
LYON COUNTY (NSPM)	168309	LYC_CAP2	115	17035.5	16216.2	17208.7	16346.0	173.2	129.8	1.02%	0.80%
BIG STONE SOUTH (OTP)	109600	BST_345KV	345	8261.4	7429.0	8302.6	7481.3	41.2	52.3	0.50%	0.70%
WATERTOWN (WAPA)	221501	WTT_230KV	230	11771.8	13446.7	11825.4	13539.9	53.6	93.2	0.46%	0.69%
YANKEE (NSPM)	229500	YNK_115KV_1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229501	YNK_115KV_2	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229506	YNK_IB_TR01	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J460

Substation	Faulted Bus			Benchmark Case without J460 Project		Study Case with J460 Project		Ampere Change with J460 vs Benchmark		% Change with J460 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
YANKEE (NSPM)	229507	YNK_IB_TR02	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229504	YNK_IB_BRI	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229502	YNK_IB_BOK1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	300448	YNK_REAC1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229505	YNK_IB_RB1	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
YANKEE (NSPM)	229503	YNK_IB_BOK2	115	13133.8	15280.2	13200.5	15385.3	66.7	105.1	0.51%	0.69%
HAZEL CREEK (NSPM)	148705	HZC_IB_MNV	230	11431.9	11656.2	11497.3	11730.0	65.4	73.8	0.57%	0.63%
HAZEL CREEK (NSPM)	148703	HZC_230KV_1	230	11431.9	11656.2	11497.3	11730.0	65.4	73.8	0.57%	0.63%
WHITE (WAPA)	225607	WHI_115KV	115	12120.3	10468.6	12191.8	10532.5	71.5	63.9	0.59%	0.61%
MINNESOTA VALLEY (NS	176501	MNV_230KV_1	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176502	MNV_230KV_2	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176503	MNV_230KV_5	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176504	MNV_230KV_6	230	13600.1	12962.7	13667.1	13040.9	67.0	78.2	0.49%	0.60%
MINNESOTA VALLEY (NS	176500	MNVT_GRA_PTH	230	13506.2	12925.2	13571.7	13002.5	65.5	77.3	0.48%	0.60%
RED PINE (WIND)	300072	HNL_GEN1_034	34.5	15362.3	15293.5	15653.7	15384.2	291.4	90.7	1.90%	0.59%
RED PINE (WIND)	300073	HNL_GEN2_034	34.5	15362.3	15293.5	15653.7	15384.2	291.4	90.7	1.90%	0.59%
GRANITE FALLS (WAPA)	145100	GRA_230KV_1	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145101	GRA_230KV_2	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145102	GRA_IB_BAR	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145104	GRA_IB_KV8A	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145108	GRA_IB_WMR	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145105	GRA_IB_MNV1	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145107	GRA_IB_MOS	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145103	GRA_IB_KV3A	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145109	GRA_IB_WTT	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
GRANITE FALLS (WAPA)	145106	GRA_IB_MNV2	230	13892.4	13127.5	13957.1	13204.7	64.7	77.2	0.47%	0.59%
MARSHALL SW STATION	170600	MSH_115KV	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170603	MSH_IB_SEA	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170601	MSH_IB_LYC	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
MARSHALL SW STATION	170602	MSH_IB_NSV	115	9491.8	12041.8	9531.8	12102.6	40.0	60.8	0.42%	0.50%
BIG STONE SOUTH (OTP	300906	BSS_230KV	230	16282.6	12508.0	16340.7	12569.7	58.1	61.7	0.36%	0.49%
BIG STONE (OTP)	109601	BST_230KV	230	17015.1	12715.8	17074.9	12777.1	59.8	61.3	0.35%	0.48%
NORTH SEVENTH STREET	183200	NSV_115KV_1	115	8659.1	11161.9	8690.8	11212.5	31.7	50.6	0.37%	0.45%

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J475 and J555 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J475, a proposed 200 MW wind farm, and J555, a proposed 140 MW expansion, and nearby substations both with and without the Interconnection Customer interconnected. J475 and J555 were assumed to interconnect at the MidAmerican and ITC Midwest jointly-owned Montezuma Substation. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in DPP study cycles of the MISO generation interconnection process, where the system impact studies are not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 13,008 Amps without the Interconnection Customer included and 14,334 Amps with the full 340 MW from the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results, especially the 34.5 kV results that should be calculated by the Interconnection Customer. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican breakers become overdutied as a result of the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J475 and J555

Bus Number	Bus Name	English Name	Base kV	Area Num	Owner	SLG Fault Current Comparison					3 Ph Fault Current Comparison				
						Base SLG w/o wind farms J475/J555	SLG with new J475 wind farm	SLG Difference w/ J475 wind farm vs Base	SLG with J475 & J555 wind farm	SLG Difference w/ J475 & J555 wind farm vs Base	Base 3PH w/o wind farms J475/J555	3PH with new J475 wind farm	3PH Difference w/ J475 wind farm vs Base	3PH with J475 & J555 wind farm	3PH Difference w/ J475 & J555 wind farm vs Base
635730	MNTZUMA3	Montezuma	345	635	MEC	9,020	10,083	1,063	11,078	2,058	13,008	13,705	697	14,334	1,326
		J475 Collector Bus	34.5			19,371	32,795	13,424	32,856	13,485	16,744	27,713	10,969	27,756	11,012
		J555 Collector Bus	34.5			19,371	19,371	-	29,325	9,954	16,744	16,744	-	24,307	7,563
635680	BONDRNT3	Bondurant	345	635	MEC	17,016	17,131	115	17,223	207	19,055	19,188	133	19,299	244
631143	OTTUMWA3	Ottumwa	345	627	ITCM	9,031	9,193	162	9,322	291	10,891	11,055	164	11,194	303
	J530 POI	J530 POI	345	635	MEC	10,293	10,542	249	10,742	449	12,902	13,146	244	13,354	452
635700	SYCAMOR3	Sycamore	345	635	MEC	19,005	19,070	65	19,122	117	20,106	20,193	87	20,265	159
635690	GDMEC	GDMEC	345	635	MEC	18,566	18,638	72	18,697	131	18,624	18,711	87	18,783	159
344000	7ADAIR	Adair/Zachary 345	345	356	AMMO	8,920	8,934	15	8,946	26	12,465	12,494	29	12,517	52
631115	OTTUMWA5	Ottumwa	161	627	ITCM	27,848	27,929	81	27,993	145	23,967	24,035	68	24,091	124
636400	HILLS 3	Hills	345	635	MEC	16,822	16,902	80	16,966	144	19,937	20,054	117	20,151	214
		J530 Collector Bus	34.5			30,409	30,480	71	30,537	128	26,522	26,574	52	26,617	95
635600	GRIMES 3	Grimes	345	635	MEC	18,245	18,292	47	18,330	85	20,460	20,535	75	20,597	137
635701	SYCAMOR5	Sycamore	161	635	MEC	33,003	33,067	64	33,119	116	32,925	33,003	78	33,068	143
635650	SE POLK3	SE Polk	345	635	MEC	17,992	18,057	65	18,110	118	18,356	18,438	82	18,505	149
345435	7MAYWOOD	Maywood	345	356	AMMO	8,222	8,226	4	8,228	6	11,420	11,429	9	11,436	16
344006	5ADAIR3	Adair/Zachary 161	161	356	AMMO	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
631110	WAPELLO5	Wapello	161	627	ITCM	10,727	10,735	8	10,741	14	11,096	11,108	12	11,117	21
631118	LUCAS 5	Lucas Co	161	627	ITCM	3,221	3,222	1	3,222	1	3,339	3,340	1	3,341	2
631116	BRDGPRT5	Bridgeport	161	627	ITCM	15,772	15,794	22	15,811	39	17,111	17,144	33	17,170	59
631134	TRICNTY5	Tri County	161	627	ITCM	15,937	15,960	23	15,978	41	17,358	17,391	33	17,418	60
635880	PLSNT CRNR 5	Pleasant Corner	161	635	MEC	9,837	9,846	9	9,853	15	12,403	12,420	17	12,434	31
636420	TIFFIN 3	Tiffin	345	635	MEC	11,251	11,275	24	11,295	44	14,815	14,864	49	14,905	90
636620	WALCOTT3	Walcott	345	635	MEC	11,076	11,083	7	11,088	12	15,648	15,667	19	15,682	34
636640	LOUISA 3	Louisa	345	635	MEC	19,061	19,085	24	19,105	44	19,077	19,111	34	19,139	62
636645	SUB T 3	Sub T Haskins	345	635	MEC	11,232	11,252	20	11,269	37	15,712	15,759	47	15,797	85
636401	HILLS 5	Hills	161	635	MEC	19,245	19,278	33	19,304	59	19,685	19,723	38	19,755	70
631034	ROSEHLW5	Rose Hollow	161	627	ITCM	5,802	5,803	2	5,804	3	6,551	6,554	3	6,556	5

J485 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J485 generating facility on the adequacy of existing circuit breakers and related equipment in the Project area.

1.2 Short Circuit Model

An ASPEN short circuit database including positive, negative and zero sequence parameters of the project area was provided by MISO. This starting short circuit model is listed below:

- Short Circuit Model: "RPUcApe2014Base_2016.03.01.dxt"

The J485 short circuit study model was developed as follows:

- The J485 gas generating facility with its sequence parameter was added at the West Side 161 kV bus via a 13.8 kV/161 kV transformer ($Z=7.5\%$ at 39 MVA, $X/R=24$, D-YG)
- The J485 gas project was modeled at the West Side 13.8 kV bus with MVA base of 58.38 MVA. All impedance values are set using saturated reactance data.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J485 project) and benchmark case (without the J485 project). The following fault simulation options were used in the short circuit analysis.

- The prefault voltage was set to "from a linear network solution"

Three-phase-to-ground (3PH) and single-phase-to-ground (SLG) faults were simulated on buses within the Study Project area. The results of short circuit currents with J485 (study case) and without J485 (benchmark case) are listed in Table 1. With the addition of the J485 generating facility, the maximum increase in the 3PH short circuit current is 3.89% (643.1 amps), and the maximum increase in the SLG short circuit current is 13.91% (1632.1 amps), which are both observed at the West Side 161 kV substation bus.

1.4 Summary of J485 Short Circuit Analysis

The study results show that the 3PH fault current is 17159 kA (increased by 643.1 A) and SLG fault current is 13367 A (increased by 1632.1 A) at the West Side 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J485 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J485

ASPEN Bus Number	ASPEN Bus Name	ASPEN Base kV	LOCATION	AREA	ZONE	Benchmark Case without J485 Project		Study Case with J485 Project		Amps Change with J485 vs Benchmark		% Change with J485 vs Benchmark	
						3PH Fault SC Amps	SLG Fault SC Amps	3PH Fault SC Amps	SLG Fault SC Amps	3PH	SLG	3PH	SLG
16	WS 161	161	WESTSIDE	1	1	16515.8	11734.8	17158.9	13366.9	643.1	1632.1	3.89%	13.91%
98	Q9 T	161	IBM TAP	1	1	17794.7	13096.4	18337.7	14273.9	543.0	1177.5	3.05%	8.99%
61906	ML 161 TAP	161	MAPLE LE	1	1	16233.6	11439.9	16695.2	12412.6	461.6	972.7	2.84%	8.50%
61907	ML 161	161	MAPLE LE	1	1	16233.6	11439.8	16695.1	12412.6	461.5	972.8	2.84%	8.50%
110	IBM 161	161	IBM STAT	1	1	17536.3	12929.2	17993.9	13817.2	457.6	888.0	2.61%	6.87%
13	NH 161	161	NORTHERN	1	1	16699.4	12107.8	17112.2	12889.9	412.8	782.1	2.47%	6.46%
7	CC 161	161	CASCADE	1	1	18754.4	14433.5	19157.9	15116.2	403.5	682.7	2.15%	4.73%
14	CRT 161	161	CROSSTOW	1	1	18179.8	13912.0	18540.8	14496.0	361.0	584.0	1.99%	4.20%
5	SLS 161	161	SILVER L	1	1	18040.5	13804.6	18390.4	14365.0	349.9	560.4	1.94%	4.06%
61948	BYR 161	161	BYRON	1	1	20626.7	16352.2	21002.8	16955.3	376.1	603.1	1.82%	3.69%
8	ZR 161	161	ZUMBRO R	1	1	13178.6	8931.1	13394.3	9261.4	215.7	330.3	1.64%	3.70%
15	CHR 161	161	CHESTER	1	1	18106.0	13270.1	18385.3	13647.1	279.3	377.0	1.54%	2.84%
69547	ROCH 161	161	ROCHESTE	1	1	18074.4	13239.1	18351.8	13612.9	277.4	373.8	1.53%	2.82%
9	BV 161	161	BAMBER V	1	1	13681.0	9389.3	13870.4	9642.4	189.4	253.1	1.38%	2.70%
6	WC 161	161	WILLOW C	1	1	13109.0	8984.5	13272.7	9202.0	163.7	217.5	1.25%	2.42%
100	NR 161	161	NORTH RO	1	1	18282.8	17302.0	18407.0	17482.5	124.2	180.5	0.68%	1.04%
61950	BYR 345	345	BYRON	1	1	16573.2	12891.0	16675.2	13082.7	102.0	191.7	0.62%	1.49%
103	NR 345	345	NORTH RO	1	1	19418.7	15499.6	19513.8	15647.9	95.1	148.3	0.49%	0.96%
63032	PV 345	345	PLEASANT	1	1	16912.7	15416.4	16982.7	15480.9	70.0	64.5	0.41%	0.42%
63033	PV 161	161	PLEASANT	1	1	19807.0	4795.5	19875.0	4831.7	68.0	36.2	0.34%	0.75%
69549	WABACO 161	161	WABACO	1	1	10457.2	7464.1	10484.0	7490.9	26.8	26.8	0.26%	0.36%
60102	ADAMS 345	345	ADAMS	1	1	13874.1	11875.5	13900.4	11890.9	26.3	15.4	0.19%	0.13%

A short circuit analysis was performed to determine the impact of the proposed generation on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Otter Tail Power Company's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J488/J489 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Otter Tail equipment in the area are not exceeded.

With the proposed J488/J489 project additions, the fault currents are estimated to be roughly 8.6 kA at the Twin Brooks 345 kV bus, 11.2 kA at the Big Stone South 345 kV bus, 19.7 kA at the Big Stone South 230 kV bus, and 33.6 kA at the Big Stone South 34.5 kV bus. There does not appear to be any short circuit related upgrades needed for J488/J489.

A general protective relay coordination review will need to be performed by OTP when the projects approach completion and before energizing the generation source.



Interconnection Facility Study Report

Generation Interconnection
MISO Projects J493 and J510
Prepared for Midcontinent ISO

April 25, 2017

This report contains CEII and Trade Secret information and should not be released without consent from Otter Tail Power Company

2. Short Circuit Study

A short circuit analysis was performed to determine the impact of the proposed generation projects on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Transmission Owner's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J493 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Transmission Owner's equipment in the area are not exceeded.

With the proposed J493 project additions, the fault currents are estimated to be roughly 16 kA at the Big Stone South 230 kV bus and the faults on the 345 kV bus were below these values. There do not appear to be any short circuit related upgrades needed for J493.

With the proposed J493 and J510 project additions, the fault currents are estimated to be roughly 16.6 kA at the Big Stone South 230 kV bus. There do not appear to be any short circuit related upgrades needed for J493 and J510.

A general protective relay coordination review will need to be performed by Transmission Owner when the projects approach completion and before energizing the generation source.

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Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J495

7-17-2017

Summary

Generator Interconnection Project J495 is a 200 MW wind powered generating facility with proposed interconnection to the 345 kV bus at a new switch station tapping ITC Midwest's Ledyard to Colby 345 kV line. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 100 MW equivalent generators
- Two 34.5kV to 690V equivalent transformers
- Two 34.5kV to 345 kV step up transformers
- 1.3 mile 345 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

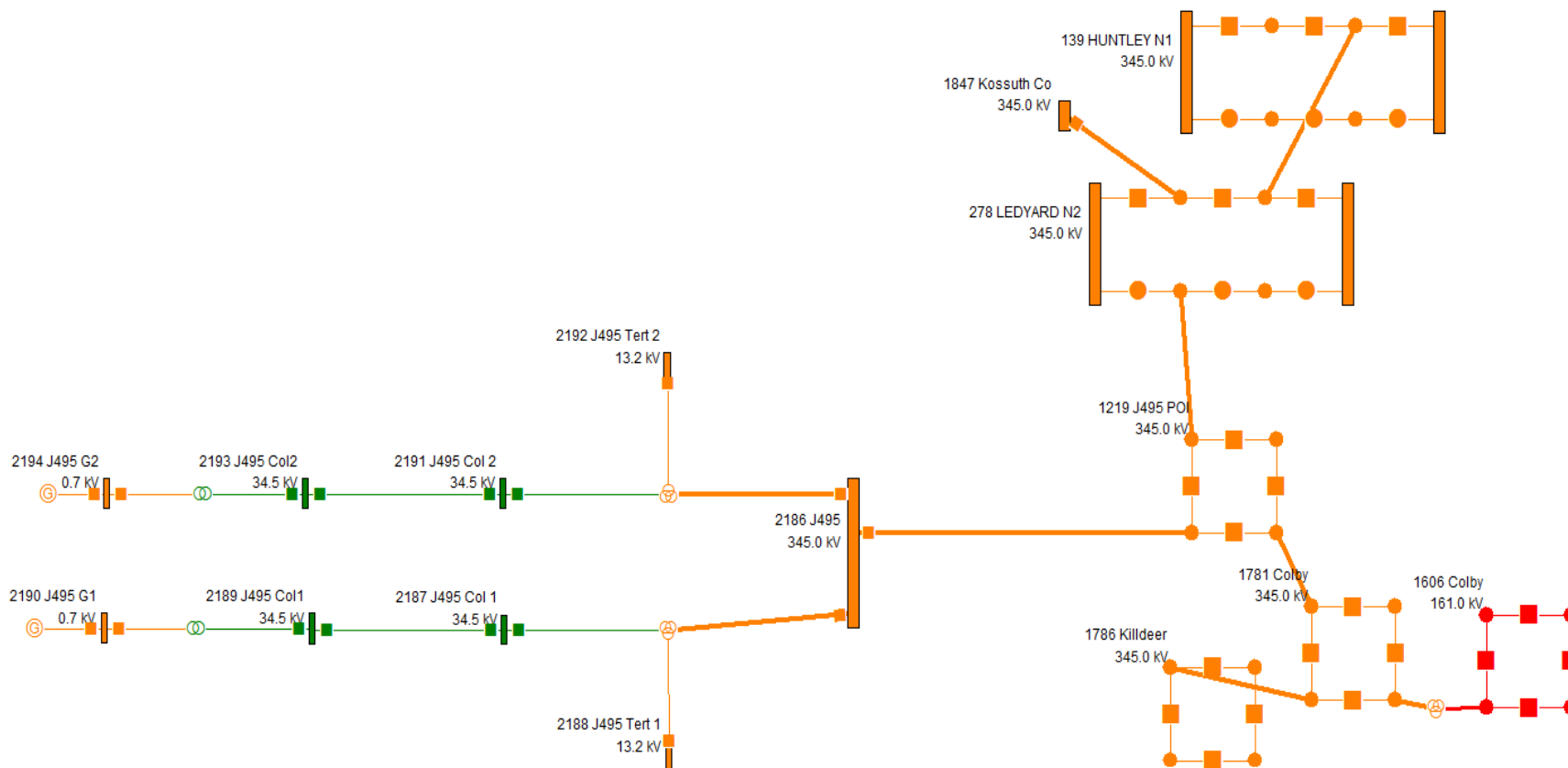


Figure1



Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J495.

J495	Ledyard 345	Colby 345	Colby 161	Huntley 345	Killdeer 345	Kossuth
SLG-pre (A)	9121.93	9313.45	20243.8	8496.7	9803.83	10370
3Ph-pre (A)	13062	11044.5	22145.6	11436.2	10822.5	14788.5
SLG-post (A)	10356.4	9530.15	20410.2	8795.42	9901.52	10673.7
3Ph-post (A)	13094.9	11059	22153.7	11449.9	10833	14804.2
Contribution SLG (A)	1234.47	216.7	166.4	298.72	97.69	303.7
Contribution 3Ph (A)	32.9	14.5	8.1	13.7	10.5	15.7

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J495. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. New circuit breakers at the POI switch station were assumed to have 50kA interrupting capability.

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J498 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J498, a proposed 340 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J498 was assumed to interconnect off the Lehigh-Grimes 345 kV line. MidAmerican is the operating agent of this jointly-owned line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 11,055 Amps without the Interconnection Customer included and 12,498 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J498

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12083	J498_TAP	J498 POI	345	MEC	7,799	9,576	1,776	11,055	12,498	1,443
11798	J498_COLHI	J498 Collector Bus	34.5		NA	31,416	NA	NA	28,289	NA
636010	LEHIGH 3	Lehigh	345	MEC	9,962	10,784	822	12,680	13,501	821
635600	GRIMES 3	Grimes	345	MEC	18,987	19,347	360	21,429	21,860	431
636000	WEBSTER3	Webster	345	MEC	10,304	10,683	379	12,492	12,946	454
6201	J411 POI	Ida Co	345	MEC	7,133	7,197	65	9,527	9,621	94
635700	SYCAMOR3	Sycamore	345	MEC	19,630	19,888	259	20,933	21,270	337
12084	J499_TAP	J499 POI	345	MEC	9,223	9,253	31	11,409	11,462	53
635630	BOONVIL3	Booneville	345	MEC	15,390	15,501	111	18,010	18,186	176
635601	GRIMES 5	Grimes	161	MEC	23,008	23,148	140	24,022	24,182	160
635369	KOSSUTH 3	Kossuth Co	345	MEC	12,312	12,389	77	14,979	15,093	113
636001	WEBSTER5	Webster	161	MEC	17,538	17,777	239	18,015	18,222	207
635200	RAUN 3	Raun	345	MEC	27,097	27,157	60	25,447	25,509	61
635680	BONDRNT3	Bondurant	345	MEC	17,504	17,629	125	19,708	19,907	200
635701	SYCAMOR5	Sycamore	161	MEC	33,629	33,864	235	33,694	33,974	281
635590	FALLOW 3	Fallow Ave	345	MEC	8,708	8,725	17	10,215	10,244	29
12085	J500_TAP	J500 POI	345	MEC	13,423	13,454	31	17,020	17,083	63
635635	MADISON3	Madison Co	345	MEC	13,218	13,279	61	16,259	16,374	115
635631	BOONVIL5	Booneville	161	MEC	18,692	18,754	63	21,728	21,825	97
635602	142ST 5	142nd St	161	MEC	17,682	17,762	80	21,324	21,447	123
635614	100 54 5	100th St & 54th Ave	161	MEC	20,682	20,778	97	25,122	25,280	158
635609	141 44 5	Hwys 141 & 44	161	MEC	15,237	15,286	49	19,093	19,182	89

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J499 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J499, a proposed 340 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J499 was assumed to interconnect off the Fallow-Grimes 345 kV line. MidAmerican is the operating agent for this jointly-owned line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,017 Amps without the Interconnection Customer included and 11,462 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J499

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12084	J499_TAP	J499 POI	345	MEC	7,553	9,253	1,700	10,017	11,462	1,445
11802	J499_COLHI	J499 Collector Bus	34.5		NA	31,310	NA	NA	28,180	NA
635590	FALLOW 3	Fallow Ave	345	MEC	7,984	8,725	741	9,476	10,244	768
635600	GRIMES 3	Grimes	345	MEC	18,785	19,347	562	21,164	21,860	695
635589	ECLIPSE3	Eclipse	345	MEC	6,642	7,086	443	7,914	8,428	514
635014	POTTCO 3	Overland Trail	345	MEC	24,626	24,800	173	24,686	24,909	222
635700	SYCAMOR3	Sycamore	345	MEC	19,481	19,888	408	20,727	21,270	543
12083	J498_TAP	J498 POI	345	MEC	9,543	9,576	33	12,434	12,498	64
635630	BOONVIL3	Booneville	345	MEC	15,317	15,501	185	17,889	18,186	297
635601	GRIMES 5	Grimes	161	MEC	22,931	23,148	217	23,928	24,182	255
635000	CBLUFFS3	Council Bluffs	345	MEC	31,629	31,829	199	28,966	29,192	226
635015	POTTCO 5	Overland Trail	161	MEC	25,444	25,496	52	26,439	26,516	77
635680	BONDRNT3	Bondurant	345	MEC	17,430	17,629	199	19,583	19,907	324
635701	SYCAMOR5	Sycamore	161	MEC	33,503	33,864	361	33,534	33,974	440
636010	LEHIGH 3	Lehigh	345	MEC	10,764	10,784	20	13,464	13,501	37
12085	J500_TAP	J500 POI	345	MEC	13,393	13,454	61	16,957	17,083	126
635635	MADISON3	Madison Co	345	MEC	13,176	13,279	102	16,177	16,374	197
635631	BOONVIL5	Booneville	161	MEC	18,650	18,754	105	21,664	21,825	161
635602	142ST 5	142nd St	161	MEC	17,637	17,762	124	21,252	21,447	196
635614	100 54 5	100th St & 54th Ave	161	MEC	20,629	20,778	149	25,031	25,280	249
635609	141 44 5	Hwys 141 & 44	161	MEC	15,211	15,286	76	19,043	19,182	139

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J500 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J500, a proposed 500 MW wind farm, and nearby substations both with and without the Interconnection Customer interconnected. J500 was assumed to interconnect off the Booneville-Atchison County and Rolling Hills-Madison County 345 kV lines. MidAmerican is the sole owner or operating agent for these line facilities. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 14,936 Amps without the Interconnection Customer included and 17,083 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than one bus away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J500

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
12085	J500_TAP	J500 POI	345	MEC	10,841	13,454	2,613	14,936	17,083	2,147
11806	J500_COLHI	J500 Collector Bus	34.5		NA	31,689	NA	NA	28,412	NA
635100	RLHILLS3	Rolling Hills	345	MEC	9,295	10,160	865	12,196	12,955	759
635017	ATCHSN 3	Atchison Co	345	MEC	11,181	11,313	132	13,827	14,022	195
635630	BOONVIL3	Booneville	345	MEC	14,857	15,501	645	17,368	18,186	818
635635	MADISON3	Madison Co	345	MEC	12,711	13,279	568	15,633	16,374	741
635013	PNYCRK 3	Pony Creek	345	MEC	27,296	27,490	194	26,333	26,579	246
640139	COOPER 3	Cooper	345	NPPD	22,334	22,450	116	21,758	21,912	154
635600	GRIMES 3	Grimes	345	MEC	18,989	19,347	358	21,312	21,860	548
635631	BOONVIL5	Booneville	161	MEC	18,471	18,754	284	21,493	21,825	332
635640	NORWLK3	Norwalk	345	MEC	13,834	14,137	304	15,932	16,365	433
635016	STHLND 3	Southland	345	MEC	27,266	27,459	193	26,317	26,562	245
635000	CBLUFFS3	Council Bluffs	345	MEC	31,608	31,829	221	28,936	29,192	256
645458	S3458 3	Sub 3458 (Neb City)	345	OPPD	17,438	17,478	40	18,418	18,483	65
640277	MOORE 3	Moore	345	NPPD	11,222	11,231	10	12,317	12,334	17
541199	ST JOE 3	St Joseph	345	GMO	12,113	12,120	8	14,541	14,557	16
300039	7FAIRPT	Fairport	345	AECI	8,838	8,844	6	10,509	10,522	13
640140	COOPER 5	Cooper	161	NPPD	11,210	11,222	12	9,931	9,945	13
635700	SYCAMOR3	Sycamore	345	MEC	19,566	19,888	323	20,790	21,270	480
12083	J498_TAP	J498 POI	345	MEC	9,552	9,576	24	12,447	12,498	51
12084	J499_TAP	J499 POI	345	MEC	9,211	9,253	43	11,377	11,462	85
635601	GRIMES 5	Grimes	161	MEC	22,956	23,148	192	23,917	24,182	266
635621	SUGRCRK5	Sugar Creek	161	MEC	14,764	14,883	119	19,689	19,914	225
635632	EARLHAM5	Earlham	161	MEC	6,936	6,958	22	8,482	8,516	35
635641	NORWLK5	Norwalk	161	MEC	21,027	21,255	228	22,740	23,041	301
635650	SE POLK3	SE Polk	345	MEC	18,175	18,463	288	18,592	18,985	393



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J504

7-17-2017

Summary

Generator Interconnection Project J504 is a 50 MW solar powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Duane Arnold Energy Center (DAEC) Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- A 50 MW equivalent current source
- 34.5kV to 600V equivalent transformer
- 34.5kV to 161 kV step up transformer

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

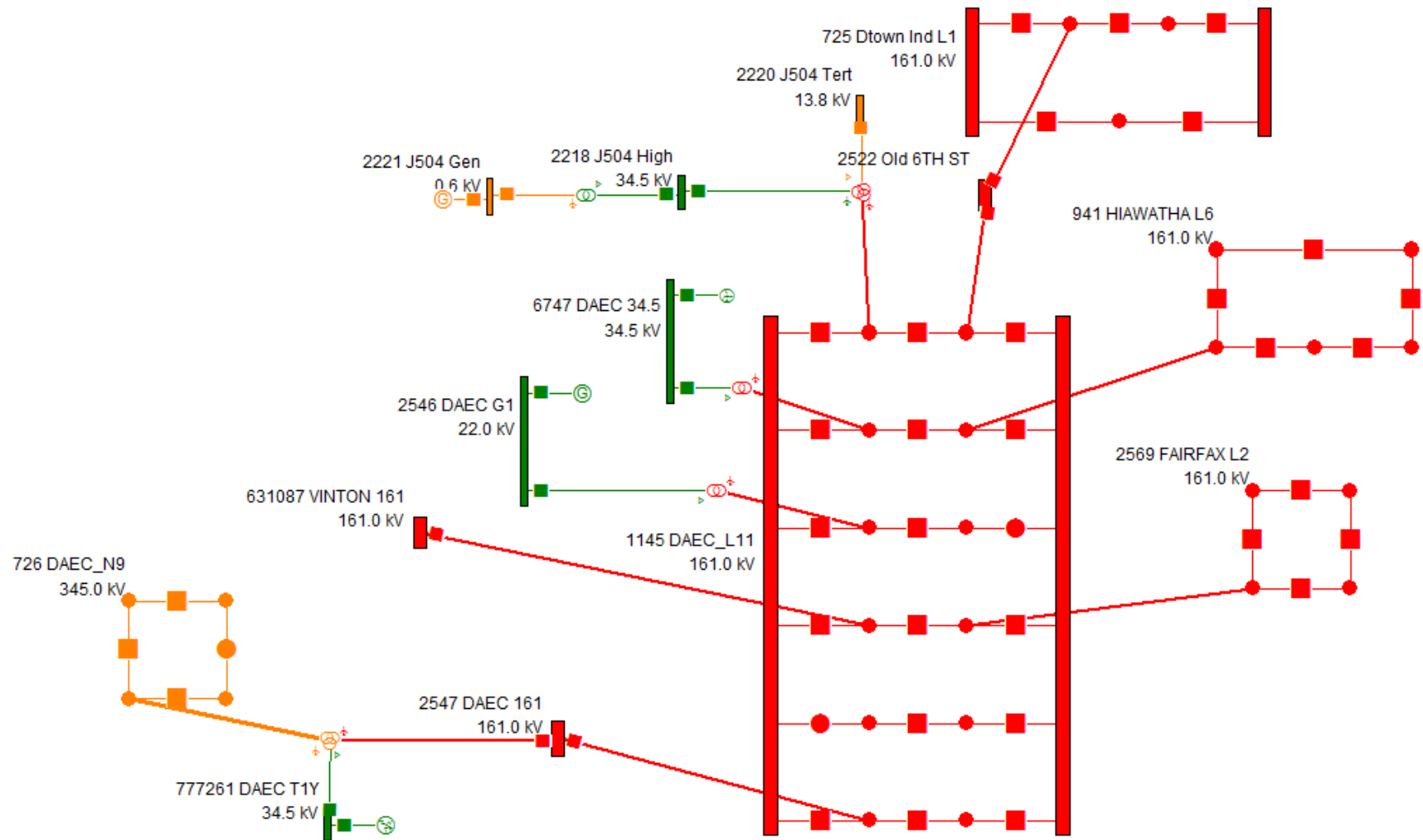


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J504.

J504	DAEC 161	DAEC 345	Hiawatha 161	Fairfax 161	DTI 161	Vinton 161
SLG-pre (A)	29067.2	11730.1	18425.9	11886.1	13286	8448.33
3Ph-pre (A)	25349.7	13413.1	14772.8	12818.1	14390.9	10401.1
SLG-post (A)	29391.4	11771.3	18450.6	11898.1	13301.5	8454.19
3Ph-post (A)	25461.9	13428.7	14795.8	12834.3	14410.7	10408.3
Contribution SLG (A)	324.2	41.2	24.7	12	15.5	5.86
Contribution 3Ph (A)	112.2	15.6	23	16.2	19.8	7.2

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J504. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. There are several existing breakers at DAEC 161 kV substation that are expected to be replaced as part of planned work at this location, as well as new breakers being installed with the bus expansion. 40kA interrupting capability was assumed for the new and replaced breakers.

J506 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J506 generating facility (100 Vestas V110 2.0 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J506 short circuit study model was developed as follows:

- Added J524, J527, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J506 generating facility with its sequence parameters at approximately 35 miles from Raun on Raun – Lakefield Jct 345 kV line. The associated 345 kV interconnection generation tie line (3 miles), two 345-34.5-13.2 kV transformers ($Z_+ = 9\%$, $Z_0 = 8.01\%$ at 75 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z_+ = 9.79\%$, $Z_0 = 9.07\%$ at 105 MVA, D-YG) were also added.
- The J506 generation was represented using the conventional generator model with the impedance of the machines set as follows per the data provided in the Vestas manual.
 - Positive saturated sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.1669 p.u.
 - Negative saturated sequence impedance (Z_2): j0.1669 p.u. (same as positive sequence impedances)
 - Zero sequence saturated impedance (Z_0): j0.1669 p.u.

The J506 short circuit benchmark model was developed as follows:

- Turned off J506 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J506 project) and benchmark case (without the J506 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J506 (study case) and without J506 (benchmark case) are listed in Table 1. With the addition of the J506 generating facility, the maximum increase in the 3PH short circuit current is 7.07% (806.3 A), and the maximum increase in the SLG short circuit current is 20.57% (1698.8 A), which are both observed at the J506 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 12205.7 A (increased by 806.3 A) and SLG fault current is 9955.5 A (increased by 1698.8 A) at the J506 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J506 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J506

Substation	Faulted Bus			Benchmark Case without J506 Project		Study Case with J506 Project		Ampere Change with J506 vs Benchmark		% Change with J506 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J506	16030	J506-COL2	34.5	-	-	16639.9	16313.0	-	-	-	-
J506	16020	J506_COL2	34.5	-	-	23877.6	18046.6	-	-	-	-
J506	16011	J506_TR2_TER	13.2	-	-	0.0	31964.5	-	-	-	-
J506	15030	J506-COL1	34.5	-	-	16639.9	16313.0	-	-	-	-
J506	15020	J506_COL1	34.5	-	-	23877.6	18046.6	-	-	-	-
J506	15011	J506_TR1_TER	13.2	-	-	0.0	31964.5	-	-	-	-
J506	15010	J506	345	-	-	9166.4	11157.3	-	-	-	-
J506	65400	J506_POI	345	8256.7	11399.4	9955.5	12205.7	1698.8	806.3	20.57%	7.07%
SHD HIGHLAND WIND FA	14839	SHD GEN BUS1	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	10839	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	473	SHD GEN BUS3	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	472	SHD GEN BUS2	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	471	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	470	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	469	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
SHD HIGHLAND WIND FA	468	SHD-345	345	10182.3	12515.4	10642.8	12849.7	460.5	334.3	4.52%	2.67%
ROCK ISLAND CLEAN LI	10838	RICL-345	345	10711.0	13314.3	11019.8	13577.2	308.8	262.9	2.88%	1.97%
MVP SBC OBRIEN COUNT	500001	OBRIEN-345	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
SBC OBRIEN COLLECTOR	11830	COLECTOR-GEN	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
SBC OBRIEN COLLECTOR	10830	COLECTOR-345	345	11415.5	14032.0	11674.8	14268.0	259.3	236.0	2.27%	1.68%
RAU RAUN	10840	RAU-345-A	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	485	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	484	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	483	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	482	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	481	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	479	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	478	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	477	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	476	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	475	RAU-345	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
RAU RAUN	474	RAU-345-B	345	27508.7	25787.3	27911.1	26135.3	402.4	348.0	1.46%	1.35%
NLN NEAL NORTH	11542	NLN 3 345	345	26055.5	24691.2	26396.1	25004.5	340.6	313.3	1.31%	1.27%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J506

Substation	Faulted Bus			Benchmark Case without J506 Project		Study Case with J506 Project		Ampere Change with J506 vs Benchmark		% Change with J506 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
NL4 NEAL SOUTH	11207	NEAL 4 345	345	25102.2	23337.7	25384.8	23606.8	282.6	269.1	1.13%	1.15%
UPLAND PRARIE WIND F	500036	J455-345	345	9346.1	12092.4	9468.7	12228.8	122.6	136.4	1.31%	1.13%
RED ROCK NORTH STAR	500030	RED ROCK TAP	345	9632.6	11451.1	9689.0	11518.8	56.4	67.7	0.59%	0.59%



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J514

7-19-2017

Summary

Generator Interconnection Project J514 is an increase of 30MW summer and 60 MW winter to an existing generating facility interconnected to ITC Midwest's Marshalltown 161 kV substation. Three phase and single line to ground fault current values were evaluated and compared to the previous values from the original J233 interconnection study using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- Two 262 MVA generators (combustion turbines)
- One 306 MVA generator (steam turbine)
- Three 18kV to 161 kV step up transformers

Sutherland Steam units 1 and 3 have approved Attachment Y retirements, and were taken out of service in the analysis of J514 which changes the fault currents as well.

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

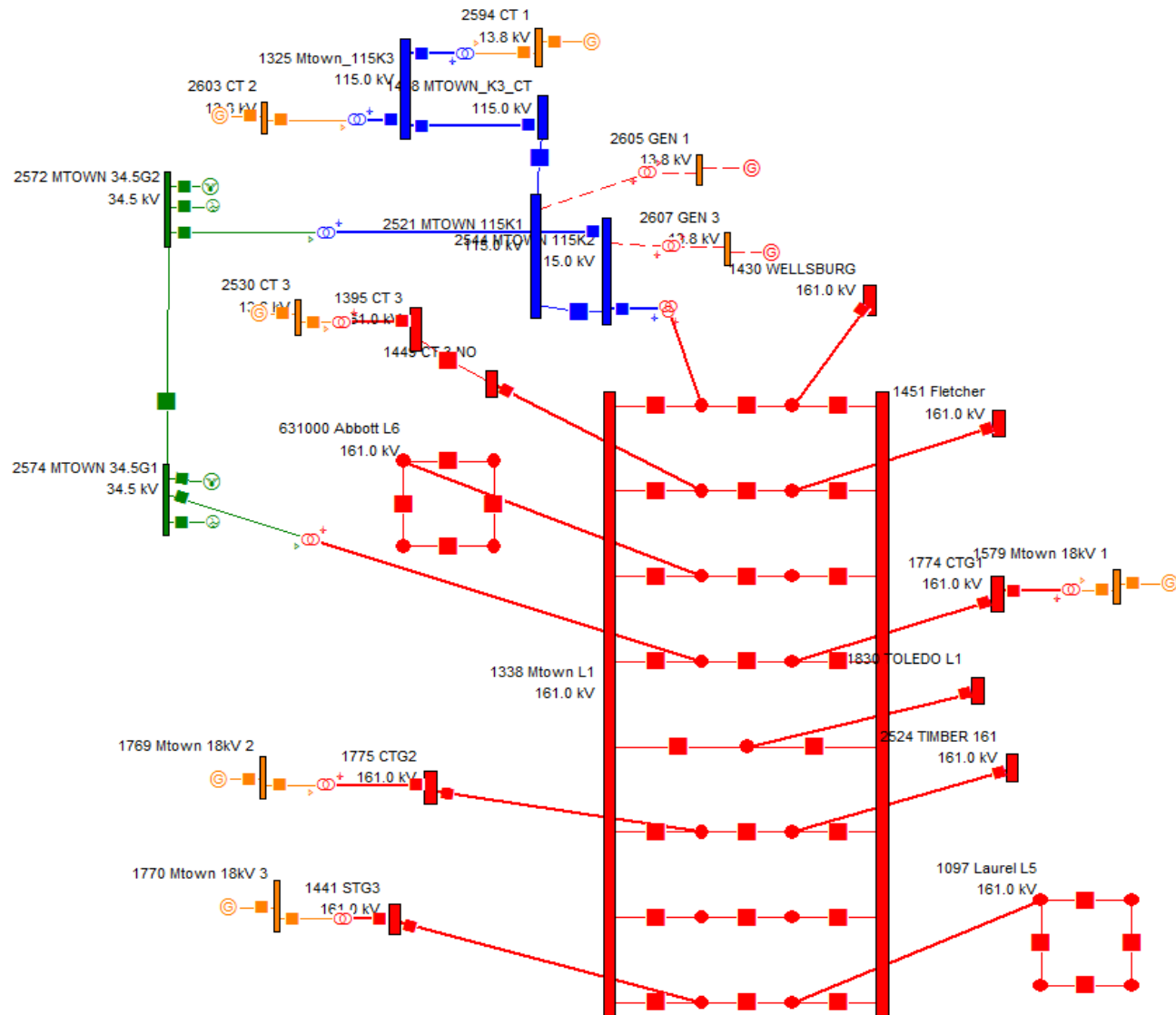


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents comparing the results from the J233 study and J514.

J514	Marshalltown 161	Marshalltown 115	Marshalltown 34.5	Wellsburg 161	Abbott 161	Toledo 161	Timber Creek 161	Laurel 161
SLG-J233 (A)	28504.7	19667.1	3372.16		7223.32		11473	9971.34
3Ph-J233 (A)	24795.4	16976.8	15773.3		10640.5		13556.6	12807.7
SLG-J514 (A)	28730.9	16514.7	3381.4	7462.9	9278.54	6641.68	12510.3	9955.3
3Ph-J514 (A)	24202.4	14536.9	15466.2	8312.51	10941.6	7754.12	13534.4	12145.9
Difference SLG (A)	226.2	-3152.4	9.24	N/A	2055.22	N/A	1037.3	-16.04
Difference 3Ph (A)	-593	-2439.9	-307.1	N/A	301.1	N/A	-22.2	-661.8

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J514. Equipment not owned by ITC Midwest was not evaluated for interrupting capability. Circuit breaker 2680 at Marshalltown 115 kV was previously identified as a constraint for J233, but has since been removed.



Evaluation of Fault Currents and Interrupting Capabilities for Generation Interconnection Project J523

7-17-2017

Summary

Generator Interconnection Project J523 is a 50 MW solar powered generating facility with proposed interconnection to the 161 kV bus at ITC Midwest's Adams Substation. Three phase and single line to ground fault current values were evaluated before and after interconnection of the generating facility using Computer Aided Protection Engineering (CAPE) software by Electrocon International. The base CAPE model maintained and used by ITC Midwest's system protection group was updated to include planned BES future system topology representative of a 2019 system was used in the analysis.

Generating Facility Assumptions

The generating facility was modeled based on available information, and included:

- A 50 MW equivalent current source
- 34.5kV to 600V equivalent transformer
- 34.5kV to 161 kV step up transformer
- 0.8 mile 161 kV interconnection line

Figure 1 below shows a one-line diagram of the modeled generating facility and nearby system.

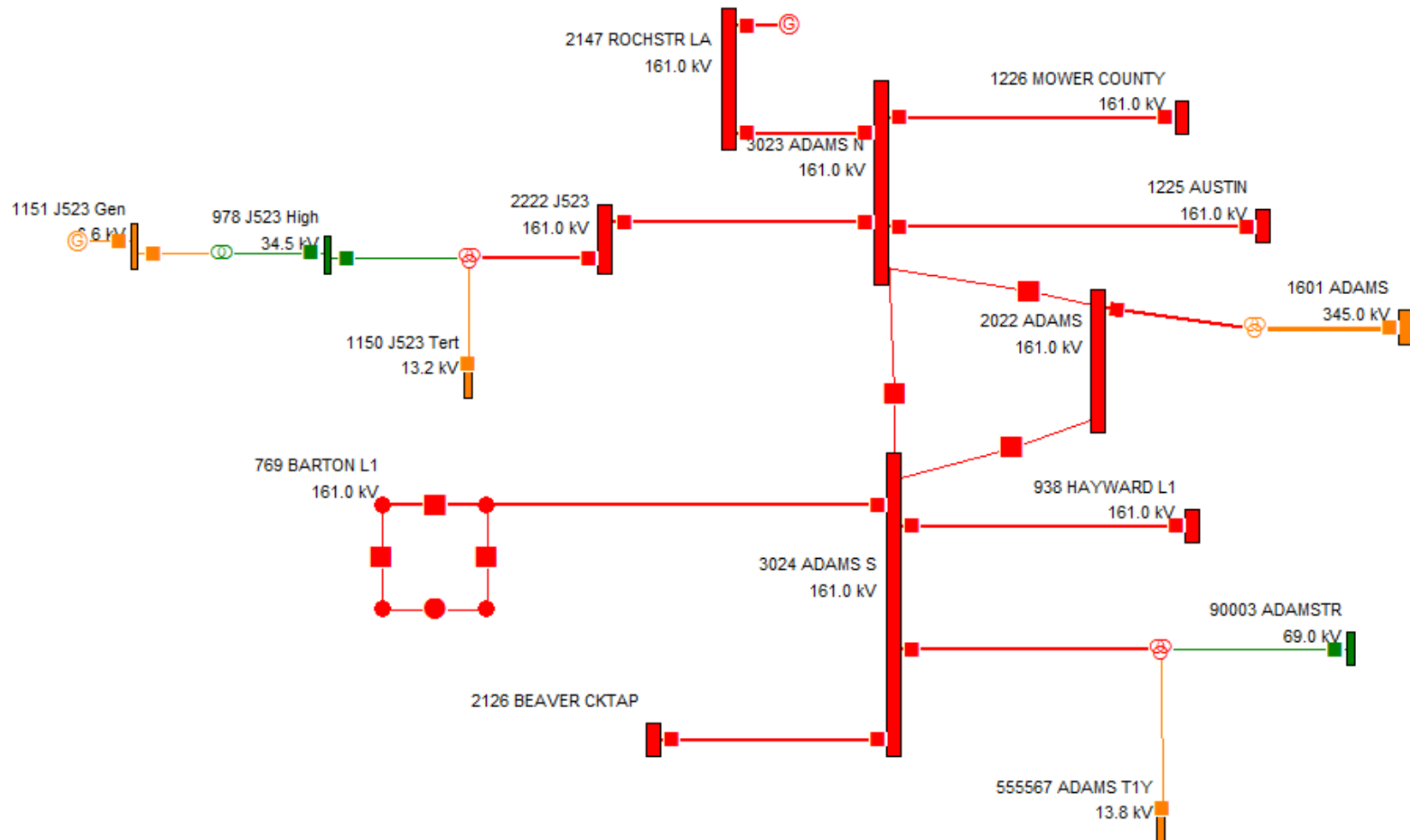


Figure1

Fault Currents

Table 1 below has the three phase and single line to ground fault currents before and after the addition of Project J523.

J523	Adams 161	Adams 345	Adams 69	Mower Co 161	Rochester 161	Austin 161	Barton 161	Beaver Creek 161	Hayward 161
SLG-pre (A)	17100.7	9853.96	12922.1	8904.79	5240.59	10452.2	9788.32	6865.9	11568.2
3Ph-pre (A)	17353.5	10595.8	12845	9170.35	4482.69	12145.4	11117	8544.83	14138.6
SLG-post (A)	17790.5	9918.36	13011.4	8971.37	5244.7	10486.1	9803.9	6901.44	11584.3
3Ph-post (A)	17464	10607.2	12860.9	9197.78	4486.18	12161.8	11126	8559.99	14151.5
Contribution SLG (A)	689.8	64.4	89.3	66.58	4.11	33.9	15.58	35.54	16.1
Contribution 3Ph (A)	110.5	11.4	15.9	27.43	3.49	16.4	9	15.16	12.9

Conclusion

Based on the analysis performed, ITC Midwest equipment has adequate interrupting capability to accommodate the interconnection of Project J523. Equipment not owned by ITC Midwest was not evaluated for interrupting capability.

J524 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J524 Solar PV generating facility (40 SMA SC 2.5 MVA inverters) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J524 short circuit study model was developed as follows:

- Added J506, J527, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J524 generating facility with its sequence parameters at Webster 161 kV substation. The associated 161 kV interconnection generation tie line, 161-34.5-13.2 kV transformer ($Z_+ = 9.0\%$, $Z_0 = 8.0\%$ at 75 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.550 kV pad mount transformer ($Z_+ = 6\%$, $Z_0 = 5.4\%$ at 112 MVA, D-YG) were also added.
- Per CAPE's manual, solar generation project should be modeled the same way as Type 4 wind generation. The J524 solar project was modeled as Type 4 controlled power and current-limited generator. It is assumed that the project has the maximum short circuit contribution of 1.5 p.u. This corresponds to a maximum fault current limit of 598 amps at 161 kV side of the GSU transformer.

The J524 short circuit benchmark model was developed as follows:

- Turned off J524 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J524 project) and benchmark case (without the J524 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to "1.00 per unit with multiplier of 1"
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J524 (study case) and without J524 (benchmark case) are

listed in Table 1. With the addition of the J524 generating facility, the maximum increase in the 3PH short circuit current is 3.63% (1028.1 A) observed at Webster 13.8 kV bus. The maximum increase in the SLG short circuit current is 6.82% (1244.2 A) observed at the Webster 161 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 19095 A (increased by 580.4 A) and SLG fault current is 19487 A (increased by 1244.2 A) at the Webster 161 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J524 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J524	97000	J524_TR_TER	13.2	-	-	0.0	28842.8	-	-	-	-
J524	96998	J524-COL1	34.5	-	-	19577.9	14160.9	-	-	-	-
J524	96997	J524_COL1	34.5	-	-	20175.1	14679.4	-	-	-	-
J524	96996	J524	161	-	-	17620.2	17752.8	-	-	-	-
WEB WEBSTER	10663	WEB-13.8-9T1	13.8	0.0	28298.2	0.0	29326.3	0.0	1028.1	0.00%	3.63%
WEB WEBSTER	10652	WEB-161	161	18242.8	18515.0	19487.0	19095.4	1244.2	580.4	6.82%	3.13%
WEB WEBSTER	10642	WEB-69	69	9937.1	9570.4	10246.7	9837.1	309.6	266.7	3.12%	2.79%
WEB WEBSTER	10645	WEB-12.4-8T1	12.5	0.0	13579.2	0.0	13949.8	0.0	370.6	0.00%	2.73%
CBPC-HOPE	10600	CB HOPE 161	161	7023.7	8906.4	7234.1	9138.7	210.4	232.3	3.00%	2.61%
CBPC-VINCENT JCT	10362	VNCNT JCT 69	69	3659.4	5108.1	3754.8	5238.2	95.4	130.1	2.61%	2.55%
CBPC-SNELL	10364	SNELL 69	69	3485.2	4818.2	3573.7	4938.3	88.5	120.1	2.54%	2.49%
FDD SUB D FD	401100	FDD-13.8	13.8	2033.2	2833.5	2033.2	2903.9	0.0	70.4	0.00%	2.48%
FDT SUB T FD	4607	FDT-8T6-13.2	13.2	9677.2	9467.4	9917.5	9702.6	240.3	235.2	2.48%	2.48%
FDT SUB T FD	2101	FDT-8T2-13.2	13.2	8180.0	8272.4	8180.0	8477.9	0.0	205.5	0.00%	2.48%
FDT SUB T FD	1100	FDT-8T1-13.2	13.2	8026.7	8039.2	8026.7	8238.9	0.0	199.7	0.00%	2.48%
FDH HARVEST AVE	629	FDH-8T2-13.2	13.2	7765.5	7608.8	7765.5	7797.3	0.0	188.5	0.00%	2.48%
FDH HARVEST AVE	628	FDH-8T1-13.2	13.2	7765.5	7608.8	7765.5	7797.3	0.0	188.5	0.00%	2.48%
FTL TATE AND LYLE	6262	FTL-8T2-13.2	13.2	10892.3	10500.3	11162.1	10760.4	269.8	260.1	2.48%	2.48%
FTL TATE AND LYLE	6261	FTL-8T1-13.2	13.2	10892.3	10500.3	11162.1	10760.4	269.8	260.1	2.48%	2.48%
FDT SUB T FD	3610	FDT-8T5-13.8	13.8	0.0	7317.9	0.0	7498.9	0.0	181.0	0.00%	2.47%
FDT SUB T FD	610	FDT-69-BUS#1	69	6394.9	6300.0	6556.7	6455.8	161.8	155.8	2.53%	2.47%
FDQ SUB Q FD	2520	FDQ-6T2-13.2	13.2	9335.5	8543.8	9566.0	8754.8	230.5	211.0	2.47%	2.47%
FDQ SUB Q FD	620	FDQ-69-B2	69	5144.9	6305.1	5274.2	6460.8	129.3	155.7	2.51%	2.47%
FDQ SUB Q FD	448	FDQ-69-B1	69	5144.9	6305.1	5274.2	6460.8	129.3	155.7	2.51%	2.47%
FDQ SUB Q FD	1620	FDQ-6T1-13.2	13.2	9517.3	8695.5	9752.3	8910.2	235.0	214.7	2.47%	2.47%
FDQ SUB Q FD	2529	FDQ-FD-FOODS	13.2	2787.1	3775.1	2787.1	3868.3	0.0	93.2	0.00%	2.47%
FDN SUB N FD	402520	FDN-13.8	13.8	2540.2	3431.2	2540.2	3515.9	0.0	84.7	0.00%	2.47%
FHA HAYES	617	FHA-69	69	10652.1	9552.7	10920.9	9788.5	268.8	235.8	2.52%	2.47%
FHA HAYES	613	FHA-13.8-8T1	13.8	0.0	17504.2	0.0	17936.2	0.0	432.0	0.00%	2.47%
FHA HAYES	619	FHA-6T1-13.2	13.2	6516.1	6252.2	6516.1	6406.5	0.0	154.3	0.00%	2.47%
FDG SUB G FD	1618	FDG-6T1-13.2	13.2	5377.1	5145.8	5377.1	5272.7	0.0	126.9	0.00%	2.47%
FDB SUB B BARNUM FD	630	FDB-69	69	4444.3	5738.7	4554.9	5880.2	110.6	141.5	2.49%	2.47%
FDT SUB T FD	454	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
FDT SUB T FD	453	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	452	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	451	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	450	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDT SUB T FD	100	FDT-161	161	9717.5	11130.1	9999.0	11404.4	281.5	274.3	2.90%	2.46%
FDM SUB M FD	634	FDM-69	69	1923.9	3059.7	1971.5	3135.1	47.6	75.4	2.47%	2.46%
FDK SUB K FD	1615	FDK-6T1-13.2	13.2	9490.9	8883.6	9724.8	9102.5	233.9	218.9	2.46%	2.46%
FDG SUB G FD	618	FDG-69	69	6235.7	7467.3	6392.6	7651.3	156.9	184.0	2.52%	2.46%
FDK SUB K FD	615	FDK-69	69	8008.0	8632.3	8210.5	8844.9	202.5	212.6	2.53%	2.46%
FDT SUB T FD	611	FDT-69-BUS#2	69	8821.0	7517.1	9040.7	7702.1	219.7	185.0	2.49%	2.46%
FDT SUB T FD	455	FDT-69-TIE	69	8821.0	7517.1	9040.7	7702.1	219.7	185.0	2.49%	2.46%
FDT SUB T FD	1610	FDT-8T3-13.8	13.8	0.0	13494.4	0.0	13826.5	0.0	332.1	0.00%	2.46%
FTL TATE AND LYLE	626	FTL-161-B1	161	8793.0	10958.0	9040.3	11227.6	247.3	269.6	2.81%	2.46%
FTL TATE AND LYLE	449	FTL-161-B2	161	8793.0	10958.0	9040.3	11227.6	247.3	269.6	2.81%	2.46%
FDH HARVEST AVE	627	FDH-161	161	8866.0	11023.1	9115.5	11294.3	249.5	271.2	2.81%	2.46%
FDT SUB T FD	2510	FDT-8T4-13.8	13.8	0.0	13349.1	0.0	13677.5	0.0	328.4	0.00%	2.46%
FHA HAYES	616	FHA-161	161	9277.0	11381.1	9539.0	11660.8	262.0	279.7	2.82%	2.46%
FDK-FDT-XFR TAP POIN	612	FDK-FDT-XFR	69	7788.8	7755.9	7983.3	7946.5	194.5	190.6	2.50%	2.46%
XFR FRONTIER ETHANOL	641	XFR-69	69	1962.1	2737.5	2009.5	2803.3	47.4	65.8	2.42%	2.40%
CBPC-SNELL SW STR	10365	SNSLL SW 69	69	3517.3	4552.4	3602.1	4660.9	84.8	108.5	2.41%	2.38%
FDV SUB V FD TAP	10370	SUB V FD TAP	69	3204.4	4040.2	3279.7	4134.5	75.3	94.3	2.35%	2.33%
WRI WRIGHT	10653	WRI-161	161	8459.7	10040.3	8683.2	10274.3	223.5	234.0	2.64%	2.33%
CBPC-DUNCOMBE	10373	DUNCOMBE-69	69	2958.9	3824.5	3027.3	3912.2	68.4	87.7	2.31%	2.29%
CBPC-HOPE	10361	CB HOPE 69	69	9993.2	12653.3	10239.2	12942.3	246.0	289.0	2.46%	2.28%
HUM HUMBOLDT CENTRAL	11369	HUM-6T1-12.4	12.5	8088.4	7554.9	8088.4	7726.4	0.0	171.5	0.00%	2.27%
HUM HUMBOLDT CENTRAL	10369	HUM-69	69	4391.0	6691.8	4493.5	6843.3	102.5	151.5	2.33%	2.26%
HUMBOLDT CENTRAL TAP	10368	HUM-69-TAP	69	5496.9	7969.2	5626.2	8149.6	129.3	180.4	2.35%	2.26%
HUE HUMBOLDT EAST	11371	HUE-6T1-12.4	12.5	4996.5	4802.4	4996.5	4911.0	0.0	108.6	0.00%	2.26%
GMC GILMORE CITY	10322	GMC-69	69	1156.2	1846.3	1156.4	1888.0	0.2	41.7	0.02%	2.26%
BOD BODE	10326	BOD-69	69	1047.1	1700.2	1047.3	1738.6	0.2	38.4	0.02%	2.26%
CBPC-SWEAZY JCT	10366	SWEAZY J 69	69	3534.5	3928.2	3614.7	4016.9	80.2	88.7	2.27%	2.26%
HUE HUMBOLDT EAST	10371	HUE-69	69	4866.7	7192.7	4980.2	7355.0	113.5	162.3	2.33%	2.26%
RUTLAND TAP	10324	RUT-69-TAP	69	4728.5	7021.8	4838.7	7180.2	110.2	158.4	2.33%	2.26%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
RUTLAND JCT	10323	RUT-69-JCT	69	1951.2	3098.9	1995.9	3168.8	44.7	69.9	2.29%	2.26%
CBPC-BODE	10325	CB BODE 69	69	1258.4	2035.0	1258.7	2080.9	0.3	45.9	0.02%	2.26%
HUM HUMBOLDT CENTRAL	19369	HUM-TIE	69	3075.6	4760.1	3146.5	4867.4	70.9	107.3	2.31%	2.25%
CBPC-HOPE	10363	CB HOPE MP	99	40344.1	61437.5	41854.5	62819.5	1510.4	1382.0	3.74%	2.25%
WRI WRIGHT	10610	WRI-69	69	10152.2	9079.0	10384.8	9282.9	232.6	203.9	2.29%	2.25%
XFR FRONTIER ETHANOL	642	XFR-6T1-13.8	13.8	5782.8	4972.5	5782.8	5083.9	0.0	111.4	0.00%	2.24%
CBPC-PASSWATER	11360	PASSWATER-13	13.2	0.0	7630.5	0.0	7801.0	0.0	170.5	0.00%	2.23%
CBPC-PASSWATER	10360	PASSWATER-69	69	4012.3	4076.2	4102.1	4167.0	89.8	90.8	2.24%	2.23%
THO THOR	11332	THO-6T1-12.4	12.5	4423.3	4198.2	4423.3	4290.9	0.0	92.7	0.00%	2.21%
THO THOR	10332	THO-69	69	3006.0	4671.2	3073.4	4774.0	67.4	102.8	2.24%	2.20%
CBPC-WEBSTER CITY MU	10609	CB WCMU 69	69	3905.4	5206.5	3991.8	5320.8	86.4	114.3	2.21%	2.20%
FGF GOLDFIELD	11343	FGF-6T1-12.4	12.5	6290.3	5823.9	6290.3	5949.6	0.0	125.7	0.00%	2.16%
CBPC-WEBSTER	10608	CB WBSTR J69	69	4421.2	5190.7	4517.0	5302.4	95.8	111.7	2.17%	2.15%
FGF GOLDFIELD	10343	FGF-69	69	2891.7	4488.8	2954.7	4585.3	63.0	96.5	2.18%	2.15%
EAG EAGLE GROVE NORT	18372	EAG TO AGP	12.5	4810.0	5114.4	4810.0	5222.4	0.0	108.0	0.00%	2.11%
EAG EAGLE GROVE NORT	12372	EAG-6T2-12.4	12.5	7165.8	6669.5	7165.8	6810.3	0.0	140.8	0.00%	2.11%
EAG EAGLE GROVE NORT	19372	EAG TO CLR	12.5	6868.3	6414.7	6868.3	6550.1	0.0	135.4	0.00%	2.11%
EAG EAGLE GROVE NORT	11372	EAG-6T1-12.4	12.5	6869.4	6415.7	6869.4	6551.1	0.0	135.4	0.00%	2.11%
EAG EAGLE GROVE NORT	10372	EAG-69	69	3817.0	5544.9	3898.2	5661.5	81.2	116.6	2.13%	2.10%
CLR CLARION	10329	CLR-69	69	2036.3	3158.1	2036.5	3215.4	0.2	57.3	0.01%	1.81%
CBPC-LAKE CORNELIA T	10328	CORNELIA TAP	69	1959.1	3041.5	1959.2	3095.4	0.1	53.9	0.01%	1.77%
CBPC-LAKE CORNELIA	10341	LKE CORNELIA	69	1497.7	2375.9	1497.8	2418.0	0.1	42.1	0.01%	1.77%
GMC GILMORE CITY	10327	GMC-69-TIE	69	2272.0	3548.2	2272.4	3611.0	0.4	62.8	0.02%	1.77%
CBPC-GILMORE JUNCTIO	10318	CB GILMJ 69	69	2479.6	3864.7	2524.1	3933.1	44.5	68.4	1.79%	1.77%
CBPC-JEWEL	10619	JEWEL-69	69	3473.1	4304.5	3534.6	4380.5	61.5	76.0	1.77%	1.77%
CBPC-RENWICK TAP	10273	CB-RENWICK69	69	906.2	1476.0	906.3	1501.8	0.1	25.8	0.01%	1.75%
LVM MAPCO	13289	LVM MAPCO 69	69	1827.0	2950.4	1827.2	3001.9	0.2	51.5	0.01%	1.75%
LUV LUVERNE	10289	LUV-69	69	1827.0	2950.4	1827.2	3001.9	0.2	51.5	0.01%	1.75%
IPS TAP LIVERMORE /	10288	IPS TAP 69	69	2488.4	4000.5	2532.4	4070.3	44.0	69.8	1.77%	1.74%
CBPC-GARDEN CITY JCT	10612	GARDEN-69	69	3546.9	4555.6	3605.8	4630.9	58.9	75.3	1.66%	1.65%
CBPC-WILLIAMS	10614	WILLIAMS-69	69	3367.4	4309.1	3422.5	4379.6	55.1	70.5	1.64%	1.64%
CBPC-BOONDOCKS	10618	BOONDOCKS-69	69	3499.1	4496.5	3555.6	4569.1	56.5	72.6	1.61%	1.61%
CBPC-SHER HUMB TAP	636	SHER HUM TAP	69	1911.2	2516.1	1911.3	2556.1	0.1	40.0	0.01%	1.59%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
CBPC-SHER HUMB TAP	635	SUB M TAP 69	69	1911.2	2516.1	1911.3	2556.1	0.1	40.0	0.01%	1.59%
FDM SUB M FD	633	FDM-69-CBPC	69	1581.5	2167.6	1581.6	2202.0	0.1	34.4	0.01%	1.59%
CBPC-BLAIRSBURG	10615	BLAIRSRBG-69	69	4913.2	5931.3	4991.1	6025.2	77.9	93.9	1.59%	1.58%
FWL WALL LAKE WIND F	18625	FWL-13.8-8T3	13.8	0.0	15624.6	0.0	15869.3	0.0	244.7	0.00%	1.57%
FWL WALL LAKE WIND F	17625	FWL-69	69	9590.8	8876.4	9741.5	9015.2	150.7	138.8	1.57%	1.56%
FWL WALL LAKE WIND F	10625	FWL-161	161	8688.3	9811.2	8827.2	9964.6	138.9	153.4	1.60%	1.56%
CBPC-WILLEMSEN	10333	WILMSEN-69	69	4568.6	6076.6	4638.1	6168.7	69.5	92.1	1.52%	1.52%
CBPC-TWIN LAKES	10198	TWN LKS 69	69	1525.2	1902.3	1525.3	1931.0	0.1	28.7	0.01%	1.51%
CBPC-TWIN LAKES	10191	TWN LKS TAP	69	1965.9	2479.0	1966.0	2516.4	0.1	37.4	0.01%	1.51%
CBPC-GALBRAITH	10291	CB GALBR 69	69	2720.6	4306.7	2721.0	4369.8	0.4	63.1	0.01%	1.47%
CBPC-ALGONA SOUTH	10391	ALGONA SOUTH	69	2670.5	4205.3	2670.8	4264.0	0.3	58.7	0.01%	1.40%
CBPC-ALGONA SOUTH	10390	ALGONA SOUTH	69	2670.5	4205.3	2670.8	4264.0	0.3	58.7	0.01%	1.40%
WEB WEBSTER	10662	WEB-345	345	11464.8	13570.3	11813.0	13759.6	348.2	189.3	3.04%	1.39%
GAMESA POCAHONTAS PR	10649	GAMESA-161	161	7585.7	8951.0	7689.9	9072.6	104.2	121.6	1.37%	1.36%
PMY POMEROY 161KV	10648	PMY-161	161	7585.7	8951.0	7689.9	9072.6	104.2	121.6	1.37%	1.36%
ITC MIDWEST-COULTER	10309	COULTER TAP	69	1975.0	3011.2	1975.1	3051.5	0.1	40.3	0.01%	1.34%
CBPC-BURT	11128	CB BURT 161	161	4864.2	6360.1	4935.1	6445.0	70.9	84.9	1.46%	1.33%
CBPC-POC-POCAHONTAS	10320	POC-69	69	2764.2	3930.8	2764.3	3982.9	0.1	52.1	0.00%	1.33%
WIL WILLIAMS PIPELIN	10287	WIL-69	69	1624.5	2491.0	1624.5	2522.9	0.0	31.9	0.00%	1.28%
LATIMER TAP	10302	LAT-69-TAP	69	2091.6	3169.2	2091.6	3209.7	0.0	40.5	0.00%	1.28%
CBPC-ALGONA EAST	10395	ALGONA EAST	69	3294.4	4580.4	3336.6	4638.8	42.2	58.4	1.28%	1.27%
CBPC-ALGONA NORTH	10397	ALGONA NORTH	69	3768.9	4770.5	3816.6	4830.4	47.7	59.9	1.27%	1.26%
IPL-KOSSUTH	10398	IPL-KOSSUTH	69	3491.9	4509.1	3536.0	4565.7	44.1	56.6	1.26%	1.26%
LAT LATIMER	10301	LAT-69	69	2209.4	3328.5	2209.4	3369.5	0.0	41.0	0.00%	1.23%
CBPC-BURT	10248	CB BURT 69	69	6334.3	6064.2	6413.8	6138.7	79.5	74.5	1.26%	1.23%
CBPC-BURT	11127	CB BURT MP	99	4189.7	4151.6	4241.6	4202.0	51.9	50.4	1.24%	1.21%
CBPC-BOONE VALLEY JU	10263	CB-BV-JCT-69	69	1995.7	3123.9	1995.8	3161.3	0.1	37.4	0.01%	1.20%
CBPC-EAST	10197	CBPC EAST 69	69	1664.3	1875.2	1664.3	1897.1	0.0	21.9	0.00%	1.17%
CBPC-EAST	10192	CBPC EASTTAP	69	2472.2	2791.1	2472.2	2823.6	0.0	32.5	0.00%	1.16%
FVR FARNHAMVILLE	10190	FVR-69	69	1563.8	1781.6	1563.8	1802.3	0.0	20.7	0.00%	1.16%
PMY POMEROY 161KV	10651	PMY-69-A27	69	5020.2	4726.7	5078.4	4781.4	58.2	54.7	1.16%	1.16%
PMY POMEROY 161KV	10650	PMY-69	69	5020.2	4726.7	5078.4	4781.4	58.2	54.7	1.16%	1.16%
FPC POCAHONTAS WIND	10430	FPC-161	161	7940.4	9119.6	8032.3	9224.6	91.9	105.0	1.16%	1.15%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J524

Substation	Faulted Bus			Benchmark Case without J524 Project		Study Case with J524 Project		Ampere Change with J524 vs Benchmark		% Change with J524 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
CBPC-RINARD	10196	RINARD 69	69	2536.4	2861.3	2536.4	2893.8	0.0	32.5	0.00%	1.14%
PMY POMEROY 161KV	10654	PMY-34.5-AB	34.5	785.4	2604.7	785.4	2634.2	0.0	29.5	0.00%	1.13%
PMY POMEROY 161KV	10644	PMY-34.5	34.5	785.4	2604.7	785.4	2634.2	0.0	29.5	0.00%	1.13%
CBPC-WHITTEMORE JUNC	10274	WHITTEMORE J	69	3393.4	5207.1	3393.6	5261.7	0.2	54.6	0.01%	1.05%
FWL WALL LAKE WIND F	12625	FWL-34.5-2	34.5	15068.5	15726.4	15226.4	15890.8	157.9	164.4	1.05%	1.05%
FWL WALL LAKE WIND F	11625	FWL-34.5-1	34.5	14896.4	15653.0	15051.9	15816.1	155.5	163.1	1.04%	1.04%
CBPC-DOVER	10164	CB-DOVER-69	69	2461.9	3760.8	2461.9	3799.1	0.0	38.3	0.00%	1.02%
FWL WALL LAKE WIND F	14625	FWL-13.8-8T2	13.8	0.0	20321.1	0.0	20526.8	0.0	205.7	0.00%	1.01%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11487.0	14089.3	11674.5	14231.8	187.5	142.5	1.63%	1.01%
LEH LEHIGH	10775	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	188	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	187	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
LEH LEHIGH	186	LEH-345	345	11518.2	14128.7	11706.5	14271.5	188.3	142.8	1.63%	1.01%
FWL WALL LAKE WIND F	13625	FWL-13.8-8T1	13.8	0.0	20260.2	0.0	20464.9	0.0	204.7	0.00%	1.01%

A short circuit analysis was performed to determine the impact of the proposed generation on the adjacent substation fault current levels. The proposed generation, as outlined in this document, was added to the existing Otter Tail Power Company's ASPEN OneLiner system model (BANG13-0503.OLR) and faults were simulated on all busses within one substation from the substation that contained the point of interconnection. As per a recommendation from MISO, the fault current from the J526 wind generation was assumed to be 1 per unit of the full nameplate rating.

Study Results

Based on the results of the study, the fault current ratings of the Otter Tail equipment in the area are not exceeded.

With the proposed J526 project additions, the fault currents are estimated to be roughly 11.7 kA at the Big Stone South 345 kV bus, 20.6 kA at the Big Stone South 230 kV bus, and 37.1 kA at the Big Stone South 34.5 kV bus. There does not appear to be any short circuit related upgrades needed for J526.

A general protective relay coordination review will need to be performed by OTP when the projects approach completion and before energizing the generation source.

J527 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J527 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J527 short circuit study model was developed as follows:

- Added J506, J524, J528, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J527 generating facility with its sequence parameters at approximately 33 miles from Booneville on Booneville - Cooper 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J527 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J527 short circuit benchmark model was developed as follows:

- Turned off J527 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J527 project) and benchmark case (without the J527 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J527 (study case) and without J527 (benchmark case) are listed in Table 1. With the addition of the J527 generating facility, the maximum increase in the 3PH short circuit current is 7.03% (1119.2 A), and the maximum increase in the SLG short circuit current is 18.94% (2339.9 A), which are both observed at the J527 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 17037 A (increased by 1119.2 A) and SLG fault current is 14696 A (increased by 2339.9 A) at the J527 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J527 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J527	65209	J527_TER2	13.8	-	-	0.0	39002.8	-	-	-	-
J527	65208	J527_TER1	13.8	-	-	0.0	39002.8	-	-	-	-
J527	65207	J527_COL2	34.5	-	-	27952.6	22515.4	-	-	-	-
J527	65206	J527_COL1	34.5	-	-	27952.6	22515.4	-	-	-	-
J527	65204	J527 COL2	34.5	-	-	31599.5	23596.0	-	-	-	-
J527	65202	J527 COL	34.5	-	-	31599.5	23596.0	-	-	-	-
J527	65201	J527	345	-	-	14510.2	16812.7	-	-	-	-
J527	65200	J527_POI	345	12356.5	15917.7	14696.4	17036.9	2339.9	1119.2	18.94%	7.03%
ORIENT WIND PROJECT	20921	J500-345-BT	345	14798.1	17781.6	16501.8	18769.8	1703.7	988.2	11.51%	5.56%
ORIENT WIND PROJECT	20911	J500-345	345	14798.1	17781.6	16501.8	18769.8	1703.7	988.2	11.51%	5.56%
J528	65300	J528_POI	345	13148.1	15972.6	14134.3	16673.3	986.2	700.7	7.50%	4.39%
J528	65301	J528	345	12987.8	15770.3	13941.5	16451.9	953.7	681.6	7.34%	4.32%
CRH ROLLING HILLS WI	21916	CRH-345-BT	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	20916	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	426	CRH-345-B1	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	425	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	424	CRH-345-B3	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	423	CRH-345	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
CRH ROLLING HILLS WI	422	CRH-345-B2	345	10426.9	13182.5	10724.1	13483.7	297.2	301.2	2.85%	2.28%
DEB BOONEVILLE	20905	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	16	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	15	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	14	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	13	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DEB BOONEVILLE	10	DEB-345	345	15855.6	18515.1	16268.6	18933.1	413.0	418.0	2.60%	2.26%
DED MADISON COUNTY	20904	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	212	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	211	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DED MADISON COUNTY	210	MADISON 345	345	13668.6	16692.6	13964.1	17034.9	295.5	342.3	2.16%	2.05%
DGN NORWALK	20955	DGN-345	345	14316.1	16542.0	14467.6	16737.6	151.5	195.6	1.06%	1.18%
DGG GRIMES	20920	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	54	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	53	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DGG GRIMES	52	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	51	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	50	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	49	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DGG GRIMES	48	DGG-345	345	19595.2	22140.7	19789.9	22401.9	194.7	261.2	0.99%	1.18%
DAG SYCAMORE EC	20900	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	185	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	184	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	183	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	182	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAG SYCAMORE EC	181	DAG-345	345	20091.6	21496.7	20259.4	21722.2	167.8	225.5	0.84%	1.05%
DAC SE POLK	20907	DAC-345	345	18605.3	19127.6	18747.6	19305.9	142.3	178.3	0.76%	0.93%
DGC GREATER DES MOIN	20893	GDMEC CT TR2	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20892	GDMEC CT TR1	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20891	GDMEC ST TR1	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	20890	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	82	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	81	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	80	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	79	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DGC GREATER DES MOIN	78	GDMEC 345	345	19211.5	19415.5	19355.0	19592.5	143.5	177.0	0.75%	0.91%
DAH BONDURANT	20901	DAH-345	345	17701.5	20025.2	17815.0	20198.8	113.5	173.6	0.64%	0.87%
DEB BOONEVILLE	20831	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	21	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	20	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	19	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	18	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DEB BOONEVILLE	17	DEB-161	161	18929.4	21979.6	19097.2	22134.6	167.8	155.0	0.89%	0.71%
DAG SYCAMORE EC	20800	DAG-161-B2	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	504	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	503	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	502	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	501	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J527

Substation	Faulted Bus			Benchmark Case without J527 Project		Study Case with J527 Project		Ampere Change with J527 vs Benchmark		% Change with J527 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DAG SYCAMORE EC	500	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	499	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	498	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	497	DAG-161	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%
DAG SYCAMORE EC	496	DAG-161-B1	161	34128.0	34232.3	34299.1	34440.9	171.1	208.6	0.50%	0.61%

J528 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J528 generating facility (80 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J528 short circuit study model was developed as follows:

- Added J506, J524, J527, J530, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J528 generating facility with its sequence parameters at approximately 32 miles from Madison on Madison – Rolling Hills 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 66 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.690 kV pad mount transformers ($Z=5.75\%$ at 112 MVA, D-YG) were also added.
- Modeled the J528 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J528 short circuit benchmark model was developed as follows:

- Turned off J528 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J528 project) and benchmark case (without the J528 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J528 (study case) and without J528 (benchmark case) are listed in Table 1. With the addition of the J528 generating facility, the maximum increase in the 3PH short circuit current is 5.50% (868.9 A), and the maximum increase in the SLG short circuit current is 14.20% (1757.2 A), which are both observed at the J528 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 16673 A (increased by 868.9 A) and SLG fault current is 14134 A (increased by 1757.2 A) at the J528 POI 345 kV bus. Based on the Transmission Owner's short circuit criteria, interconnection of the J528 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J528

Substation	Faulted Bus			Benchmark Case without J528 Project		Study Case with J528 Project		Ampere Change with J528 vs Benchmark		% Change with J528 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J528	65309	J528_TER2	13.8	-	-	0.0	29537.6	-	-	-	-
J528	65308	J528_TER1	13.8	-	-	0.0	29537.6	-	-	-	-
J528	65307	J528_COL2	34.5	-	-	21600.0	17422.8	-	-	-	-
J528	65306	J528_COL1	34.5	-	-	21600.0	17422.8	-	-	-	-
J528	65304	J528 COL2	34.5	-	-	24201.3	18134.2	-	-	-	-
J528	65302	J528 COL1	34.5	-	-	24201.3	18134.2	-	-	-	-
J528	65301	J528	345	-	-	13941.5	16451.9	-	-	-	-
J528	65300	J528_POI	345	12377.1	15804.4	14134.3	16673.3	1757.2	868.9	14.20%	5.50%
ORIENT WIND PROJECT	20921	J500-345-BT	345	15219.2	18001.0	16501.8	18769.8	1282.6	768.8	8.43%	4.27%
ORIENT WIND PROJECT	20911	J500-345	345	15219.2	18001.0	16501.8	18769.8	1282.6	768.8	8.43%	4.27%
J527	65200	J527_POI	345	13924.2	16471.8	14696.4	17036.9	772.2	565.1	5.55%	3.43%
J527	65201	J527	345	13763.9	16263.3	14510.2	16812.7	746.3	549.4	5.42%	3.38%
DED MADISON COUNTY	20904	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	212	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	211	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
DED MADISON COUNTY	210	MADISON 345	345	13666.7	16732.3	13964.1	17034.9	297.4	302.6	2.18%	1.81%
CRH ROLLING HILLS WI	21916	CRH-345-BT	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	20916	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	426	CRH-345-B1	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	425	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	424	CRH-345-B3	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	423	CRH-345	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
CRH ROLLING HILLS WI	422	CRH-345-B2	345	10502.3	13251.1	10724.1	13483.7	221.8	232.6	2.11%	1.76%
DEB BOONEVILLE	20905	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	16	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	15	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	14	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	13	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DEB BOONEVILLE	10	DEB-345	345	16007.2	18634.0	16268.6	18933.1	261.4	299.1	1.63%	1.61%
DGN NORWALK	20955	DGN-345	345	14330.9	16572.9	14467.6	16737.6	136.7	164.7	0.95%	0.99%
DGG GRIMES	20920	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	54	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	53	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J528

Substation	Faulted Bus			Benchmark Case without J528 Project		Study Case with J528 Project		Ampere Change with J528 vs Benchmark		% Change with J528 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
DGG GRIMES	52	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	51	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	50	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	49	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DGG GRIMES	48	DGG-345	345	19652.6	22207.4	19789.9	22401.9	137.3	194.5	0.70%	0.88%
DAG SYCAMORE EC	20900	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	185	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	184	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	183	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	182	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAG SYCAMORE EC	181	DAG-345	345	20136.0	21551.9	20259.4	21722.2	123.4	170.3	0.61%	0.79%
DAC SE POLK	20907	DAC-345	345	18629.9	19161.5	18747.6	19305.9	117.7	144.4	0.63%	0.75%
DGC GREATER DES MOIN	20893	GDMEC CT TR2	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20892	GDMEC CT TR1	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20891	GDMEC ST TR1	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	20890	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	82	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	81	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	80	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	79	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DGC GREATER DES MOIN	78	GDMEC 345	345	19237.3	19449.7	19355.0	19592.5	117.7	142.8	0.61%	0.73%
DAH BONDURANT	20901	DAH-345	345	17726.3	20063.0	17815.0	20198.8	88.7	135.8	0.50%	0.68%
DEB BOONEVILLE	20831	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	21	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	20	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	19	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	18	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%
DEB BOONEVILLE	17	DEB-161	161	18987.4	22019.4	19097.2	22134.6	109.8	115.2	0.58%	0.52%

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J529 Short Circuit Study Performed by MEC

The scope of this DPP Optional Study facilities study is a review of the available fault current at the proposed 345 kV interconnection substation for MISO generation queue request J529, a proposed 250 MW wind farm along with J590, a proposed 90 MW expansion at the same interconnection point, and nearby substations both with and without the Interconnection Customer interconnected. J529/J590 was assumed to interconnect off the O'Brien County-Kossuth County 345 kV line. The fault currents were used to identify if any existing MidAmerican circuit breakers become overdutied because of the proposed Interconnection Customer based on the system configuration. The study reviewed single-line-to-ground (SLG) fault current levels and three phase (3PH) fault current levels.

The Interconnection Customer is in a DPP study cycle of the MISO generation interconnection process where the system impact study is not complete. As a result, the short circuit study is preliminary and does not include changes to the transmission system and/or the generators in the area that may be required when the DPP study results are known. The results of the short circuit analysis are summarized in Table 1.

The results of the short circuit analysis showed the three phase fault current at the 345 kV interconnection substation bus to be 10,000 Amps without the Interconnection Customer included and 11,705 Amps with the Interconnection Customer included (based upon the assumed modeling information for the generator step-up transformer, wind turbines, grounding transformers, and other collector system assumptions). These assumptions affect the results. For example, generator step-up transformer information may be different from the impedances from the transformer test report. In specifying equipment or completing equipment settings such as voltage control systems, the Interconnection Customer should be aware that fault currents are subject to change and may increase or decrease at the interconnection point because of additions and/or retirements of the transmission system and/or area generation or system contingencies.

As shown in the table, the changes in fault current at buses more than a couple buses away from the point of interconnection are comparatively small. Based on MidAmerican's short circuit criteria, no MidAmerican short circuit constraints appear for the Interconnection Customer's project. The study results are subject to change based on the outcome of the DPP study or if project design considerations change from those that were studied.

A protective relay coordination review will be required if the Interconnection Customer's project proceeds, and the Interconnection Customer will be required to provide relay settings to MidAmerican. In addition, continued communication and coordination will be required for the parties to meet NERC Standard PRC-001 and PRC-005 and/or future standards.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents
with and without J529/J590

Bus Number	Bus Name	English Name	Base kV	Owner	SLG Fault Current Comparison			3 Ph Fault Current Comparison		
					Base SLG w/o new wind farm	SLG with new wind farm	SLG Difference w/ wind farm vs Base	Base 3PH w/o new wind farm	3PH with new wind farm	3PH Difference w/ wind farm vs Base
		J529J590 POI	345		7,660	9,703	2,043	10,000	11,705	1,705
		J529J590 Collector	34.5		23,730	37,957	14,227	20,331	32,060	11,730
55368	J455 POI	J455 POI	345	MEC	9,827	10,320	493	11,388	11,993	606
635369	KOSSUTH 3	Kossuth Co	345	MEC	8,502	9,587	1,085	11,804	12,852	1,048
635368	OBRIEN 3	O'Brien Co	345	MEC	11,489	11,804	315	13,740	14,192	452
631197	LEDYARD3	Ledyard	345	ITCM	8,258	8,705	448	11,552	12,112	560
636000	WEBSTER3	Webster	345	MEC	9,825	10,033	208	11,901	12,150	249
631138	LAKEFLD3	ITCM Lakefield	345	ITCM	15,041	15,234	192	16,469	16,736	267
635400	HIGHLND 3	Highland	345	MEC	10,112	10,283	171	12,330	12,588	258
631193	HUNTLEY3	Huntley	345	ITCM	7,823	8,019	196	10,432	10,725	293
631198	COLBY3	Colby	345	ITCM	7,974	8,039	65	9,386	9,484	98
636010	LEHIGH 3	Lehigh	345	MEC	9,802	9,910	109	12,385	12,541	156
636001	WEBSTER5	Webster	161	MEC	17,205	17,352	147	17,701	17,831	130
601034	NOBLES 3	Nobles Co	345	XCEL	5,670	5,687	16	8,543	8,587	43
601029	LKFLDXL3	Xcel Lakefield	345	XCEL	20,018	20,173	156	14,337	14,438	101
631041	LAKEFLD5	ITCM Lakefield	161	ITCM	17,185	17,286	101	18,941	19,081	140
635200	RAUN 3	Raun	345	MEC	27,244	27,345	101	25,743	25,839	95

J530 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J530 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J530 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J534, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J530 generating facility with its sequence parameters at approximately 26 miles from Montezuma on Montezuma – Hills 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J530 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J530 short circuit benchmark model was developed as follows:

- Turned off J530 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J530 project) and benchmark case (without the J530 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J530 (study case) and without J530 (benchmark case) are listed in Table 1. With the addition of the J530 generating facility, the maximum increase in the 3PH short circuit current is 9.32% (1118.4 A), and the maximum increase in the SLG short circuit current is 35.43% (2793.4 A), which are both observed at the J530 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 13118 A (increased by 1118.4 A) and SLG fault current is 10678 A (increased by 2793.4 A) at the J530 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J530 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J530

Substation	Faulted Bus			Benchmark Case without J530 Project		Study Case with J530 Project		Ampere Change with J530 vs Benchmark		% Change with J530 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J530	75739	J530_TER2	13.8	-	-	0.0	38565.2	-	-	-	-
J530	75738	J530_TER1	13.8	-	-	0.0	38565.2	-	-	-	-
J530	75737	J530_COL2	34.5	-	-	27565.4	22179.3	-	-	-	-
J530	75736	J530_COL1	34.5	-	-	27565.4	22179.3	-	-	-	-
J530	75734	J530 COL2	34.5	-	-	31050.5	23176.8	-	-	-	-
J530	75732	J530 COL1	34.5	-	-	31050.5	23176.8	-	-	-	-
J530	75731	J530	345	-	-	10605.1	12990.5	-	-	-	-
IOWA COUNTY WIND FAR	30143	J530-345	345	7884.2	11999.6	10677.6	13118.0	2793.4	1118.4	35.43%	9.32%
DBM MONTEZUMA	30150	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	3	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	2	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
DBM MONTEZUMA	1	DBM-345	345	9487.7	13350.5	10189.5	13789.9	701.8	439.4	7.40%	3.29%
IHL HILLS	30120	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	302	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	301	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	300	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	299	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
IHL HILLS	298	IHL-345	345	16284.1	19520.5	17023.0	20052.7	738.9	532.2	4.54%	2.73%
NORTH ENGLISH WIND F	30152	J555-345	345	7080.8	10350.4	7466.1	10612.6	385.3	262.2	5.44%	2.53%
ENGLISH FARMS WIND F	31151	J475-GEN-BT	345	7134.4	9894.0	7467.6	10123.3	333.2	229.3	4.67%	2.32%
ENGLISH FARMS WIND F	30151	J475-345	345	7134.4	9894.0	7467.6	10123.3	333.2	229.3	4.67%	2.32%
ICK SUB K IOWA CITY	30125	ICK 345	345	11112.0	14647.8	11315.5	14870.4	203.5	222.6	1.83%	1.52%
ICT SUB T HASKINS	30126	ICT-345	345	11124.8	15559.9	11284.2	15762.5	159.4	202.6	1.43%	1.30%
Q92 SUB 92	30118	SUB 92 345	345	12960.6	17700.9	13131.1	17926.8	170.5	225.9	1.32%	1.28%
OGS	602660	OGS 345	345	9121.7	10976.8	9226.6	11082.8	104.9	106.0	1.15%	0.97%
IHL HILLS	30282	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	309	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	308	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	307	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	306	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	305	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	304	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%
IHL HILLS	303	IHL-161	161	19038.2	19536.6	19305.9	19703.2	267.7	166.6	1.41%	0.85%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J530

Substation	Faulted Bus			Benchmark Case without J530 Project		Study Case with J530 Project		Ampere Change with J530 vs Benchmark		% Change with J530 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
Q93 SUB 93	30128	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	30127	Q93-345RATBT	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	317	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	316	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	315	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
Q93 SUB 93	314	Q93-345	345	19011.0	18963.7	19136.1	19116.7	125.1	153.0	0.66%	0.81%
IEN HYDROCARBON SUB	30288	IEN-161	161	11971.3	14732.5	12065.3	14825.5	94.0	93.0	0.79%	0.63%
ICP SUB P CORALVILLE	30296	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	313	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	312	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICP SUB P CORALVILLE	311	ICP-161	161	11283.3	13577.8	11353.5	13654.8	70.2	77.0	0.62%	0.57%
ICJ SUB J IOWA CITY	30290	ICJ-161	161	10866.8	13294.4	10934.5	13368.6	67.7	74.2	0.62%	0.56%
ICJ SUB J IOWA CITY	332	ICJ-161-B2	161	10866.8	13294.4	10934.5	13368.6	67.7	74.2	0.62%	0.56%
ICU SUB U IOWA CITY	30294	ICU-161-B1	161	10425.2	13260.2	10485.8	13333.7	60.6	73.5	0.58%	0.55%
ICU SUB U IOWA CITY	297	ICU-161-B2	161	10425.2	13260.2	10485.8	13333.7	60.6	73.5	0.58%	0.55%
ICR CORAL RIDGE SUBS	30298	ICR-161	161	10158.2	12727.6	10216.0	12796.1	57.8	68.5	0.57%	0.54%
QWL WALCOTT	30117	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	293	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	292	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
QWL WALCOTT	291	QWL-345	345	11081.5	15598.6	11126.0	15682.5	44.5	83.9	0.40%	0.54%
ICK SUB K IOWA CITY	30303	ICK 161	161	10535.8	12157.9	10599.9	12222.4	64.1	64.5	0.61%	0.53%
ICE SUB E IOWA CITY	30302	ICE-161-B2	161	9279.8	12250.4	9335.8	12314.6	56.0	64.2	0.60%	0.52%
ICE SUB E IOWA CITY	331	ICE-161-B1	161	9279.8	12250.4	9335.8	12314.6	56.0	64.2	0.60%	0.52%
ICY SUB Y IOWA CITY	30301	ICY-161	161	8893.4	11843.9	8943.4	11903.5	50.0	59.6	0.56%	0.50%
ICG NORTHGATE IOWA C	30300	ICG-161	161	8990.9	11938.3	9039.1	11998.3	48.2	60.0	0.54%	0.50%

J534 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J534 generating facility (100 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J534 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J530, J535 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J534 generating facility with its sequence parameters at approximately 17 miles from Kossuth on Kossuth – Webster 345 kV line. The associated 345 kV interconnection generation tie line, two 345-34.5-13.8 kV transformers ($Z+=9\%$, $Z0=6.7\%$ at 90 MVA, YG-YG-D), two 34.5 kV collector system equivalent lines and two 34.5-0.69 kV pad mount transformers ($Z=5.75\%$ at 140 MVA, D-YG) were also added.
- Modeled the J534 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) ($Z1$): j0.2 p.u.
 - Negative sequence impedance ($Z2$): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance ($Z0$): j0.2 p.u.

The J534 short circuit benchmark model was developed as follows:

- Turned off J534 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J534 project) and benchmark case (without the J534 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J534 (study case) and without J534 (benchmark case) are listed in Table 1. With the addition of the J534 generating facility, the maximum increase in the 3PH short circuit current is 9.16% (1119.5 A), and the maximum increase in the SLG short circuit current is 26.78% (2413.0 A), which are both observed at the J534 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 13335 A (increased by 1119.5 A) and SLG fault current is 11425 A (increased by 2413.0 A) at the J534 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J534 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J534

Substation	Faulted Bus			Benchmark Case without J534 Project		Study Case with J534 Project		Ampere Change with J534 vs Benchmark		% Change with J534 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J534	66009	J534_TER2	13.8	-	-	0.0	38787.0	-	-	-	-
J534	66008	J534_TER1	13.8	-	-	0.0	38787.0	-	-	-	-
J534	66007	J534_COL2	34.5	-	-	27709.5	22309.0	-	-	-	-
J534	66006	J534_COL1	34.5	-	-	27709.5	22309.0	-	-	-	-
J534	66004	J534 COL2	34.5	-	-	31209.3	23318.2	-	-	-	-
J534	66002	J534 COL1	34.5	-	-	31209.3	23318.2	-	-	-	-
J534	66001	J534	345	-	-	11331.0	13204.1	-	-	-	-
J534	66000	J534_POI	345	9011.9	12215.5	11424.9	13335.0	2413.0	1119.5	26.78%	9.16%
MVP SCS KOSSUTH COUN	500103	KOSSUTH CO	345	12236.9	15105.6	13296.1	15802.7	1059.2	697.1	8.66%	4.61%
PLUM CREEK WIND FARM	10834	J501-345-BT	345	12236.9	15105.6	13296.1	15802.7	1059.2	697.1	8.66%	4.61%
WEB WEBSTER	10662	WEB-345	345	11283.7	13342.4	11813.0	13759.6	529.3	417.2	4.69%	3.13%
MVP SCS LEDYARD	500104	LEDYARD 345	345	9597.2	13156.1	9877.2	13444.6	280.0	288.5	2.92%	2.19%
PALO ALTO CNTY WIND	500132	J529-345-BT	345	10494.1	12549.1	10777.8	12822.8	283.7	273.7	2.70%	2.18%
PALO ALTO CNTY WIND	500032	J529-345	345	10494.1	12549.1	10777.8	12822.8	283.7	273.7	2.70%	2.18%
LEH LEHIGH	10775	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	188	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	187	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LEH LEHIGH	186	LEH-345	345	11483.7	14032.2	11706.5	14271.5	222.8	239.3	1.94%	1.71%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11453.1	13993.8	11674.5	14231.8	221.4	238.0	1.93%	1.70%
RED ROCK NORTH STAR	500030	RED ROCK TAP	345	9585.3	11390.5	9689.0	11518.8	103.7	128.3	1.08%	1.13%
MVP SCS HUNTLEY	500105	HUNTLEY 345	345	8367.7	11202.8	8463.1	11327.8	95.4	125.0	1.14%	1.12%
WEB WEBSTER	10652	WEB-161	161	19220.4	18918.6	19487.0	19095.4	266.6	176.8	1.39%	0.93%
J524	96996	J524	161	17407.6	17600.3	17620.2	17752.8	212.6	152.5	1.22%	0.87%
BEAVER CREEK WIND PR	11778	J498-345-BT	345	9868.2	12803.8	9943.1	12909.4	74.9	105.6	0.76%	0.82%
BEAVER CREEK WIND PR	10778	J498-345	345	9868.2	12803.8	9943.1	12909.4	74.9	105.6	0.76%	0.82%
RED ROCK WIND FARM J	500044	RR-9T3-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500043	RR-9T2-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500042	RR-9T1-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500041	RROCK-GEN-BT	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
RED ROCK WIND FARM J	500040	RED ROCK-345	345	7998.9	8914.3	8059.3	8987.6	60.4	73.3	0.76%	0.82%
UPLAND PRARIE WIND F	500036	J455-345	345	9417.1	12147.5	9468.7	12228.8	51.6	81.3	0.55%	0.67%
NORTH STAR WIND FARM	500050	N STAR-345	345	5860.4	7247.0	5892.3	7295.0	31.9	48.0	0.54%	0.66%
MVP SBC OBRIEN COUNT	500001	OBRIEN-345	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J534

Substation	Faulted Bus			Benchmark Case without J534 Project		Study Case with J534 Project		Ampere Change with J534 vs Benchmark		% Change with J534 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
SBC OBRIEN COLLECTOR	11830	COLECTOR-GEN	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%
SBC OBRIEN COLLECTOR	10830	COLECTOR-345	345	11622.8	14191.7	11674.8	14268.0	52.0	76.3	0.45%	0.54%

J535 Short Circuit Analysis

1.1 Introduction

A short circuit analysis was performed by Siemens PTI to assess the impact of the J535 generating facility (84 GE 2.5 MW WTGs) on the adequacy of existing circuit breakers and related equipment in the study area.

1.2 Short Circuit Model

A CAPE short circuit database including positive, negative and zero sequence parameters of the MidAmerican (MEC) system and its neighboring systems was provided by the MISO. This starting short circuit model is listed below:

- Short Circuit Model: "CAPE14_7_05_2017_MEC_MISO.fdb"

The J535 short circuit study model was developed as follows:

- Added J506, J524, J527, J528, J530, J534 projects with online status.
- Turned on J475, J498, J499, J500, J501 and J529 projects.
- Added the J535 generating facility with its sequence parameters at approximately 44 miles from Lehigh on Lehigh – J411 345 kV line. The associated 345 kV interconnection generation tie line, 345-34.5-13.8 kV transformer ($Z_+ = 9\%$, $Z_0 = 6.7\%$ at 150 MVA, YG-YG-D), 34.5 kV collector system equivalent line and 34.5-0.69 kV pad mount transformer ($Z = 5.75\%$ at 235.2 MVA, D-YG) were also added.
- Modeled the J535 generator facility as a generator with impedances listed below. These values are based on the conservative assumption that the converters "crowbar" (disconnect to protect the power electronics within) for close-in faults. In this case the turbine rotor is short-circuited like that of a squirrel cage induction generator. The behavior can be approximated to $X' = 0.2$, contributing a maximum of 5 per unit fault current.
 - Positive sequence impedances (i.e. subtransient, transient and synchronous) (Z_1): j0.2 p.u.
 - Negative sequence impedance (Z_2): j0.2 p.u. (same as positive sequence impedances)
 - Zero sequence impedance (Z_0): j0.2 p.u.

The J535 short circuit benchmark model was developed as follows:

- Turned off J535 project and associated interconnection facilities.

1.3 Short Circuit Analysis

Short circuit analysis was performed on the study case (with the J535 project) and benchmark case (without the J535 project). Classical short circuit solution option was used in the calculation. Following fault simulation options were used in the short circuit analysis:

- Prefault voltage was set to “1.00 per unit with multiplier of 1”
- All loads, shunts and transformer magnetizing branches were ignored.

Buses within 5-bus distance from the POI of the project were considered as the Study Project Area. Three-phase (3PH) and Single Line to Ground (SLG) faults were simulated in the Study Project area. The results of short circuit currents with J535 (study case) and without J535 (benchmark case) are listed in Table 1. With the addition of the J535 generating facility, the maximum increase in the 3PH short circuit current is 10.66% (950.0 A), and the maximum increase in the SLG short circuit current is 32.09% (2040.0 A), which are both observed at the J535 POI 345 kV substation bus.

1.4 Summary of Short Circuit Analysis

The study results show that the 3PH fault current is 9858 A (increased by 950.0 A) and SLG fault current is 8398 A (increased by 2040.0 A) at the J535 POI 345 kV bus. Based on the Transmission Owner’s short circuit criteria, interconnection of the J535 generation project does not cause any Transmission Owner short circuit constraints.

Table 1. Single-Line-to-Ground (SLG) and Three Phase (3PH) Fault Currents with and without J535

Substation	Faulted Bus			Benchmark Case without J535 Project		Study Case with J535 Project		Ampere Change with J535 vs Benchmark		% Change with J535 vs Benchmark	
	#	Name	kV	SLG Fault Current (A)	3PH Fault Current (A)	SLG Fault Current (A)	3PH Fault Current (A)	SLG	3PH	SLG	3PH
J535	66206	J535_TER	13.8	-	-	0.0	61330.5	-	-	-	-
J535	66205	J535_COL	34.5	-	-	44527.2	35055.8	-	-	-	-
J535	66203	J535 COL	34.5	-	-	47736.3	35775.1	-	-	-	-
J535	66202	J535	345	-	-	8352.8	9788.2	-	-	-	-
J535	66201	J535_POI	345	6358.0	8908.1	8398.0	9858.1	2040.0	950.0	32.09%	10.66%
FIC IDA CO SUB IDA G	11776	GEN BUS-345	345	7273.9	9733.8	8223.5	10299.9	949.6	566.1	13.05%	5.82%
FIC IDA CO SUB IDA G	10776	IDA CNTY-345	345	7273.9	9733.8	8223.5	10299.9	949.6	566.1	13.05%	5.82%
IDA CO WEST WIND FAR	10779	J412-345	345	7218.9	10014.2	7899.1	10479.1	680.2	464.9	9.42%	4.64%
LEH LEHIGH	10775	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	188	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	187	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LEH LEHIGH	186	LEH-345	345	11256.0	13929.3	11706.5	14271.5	450.5	342.2	4.00%	2.46%
LGR LUNDGREN WIND EN	10777	LGR-345	345	11226.9	13891.5	11674.5	14231.8	447.6	340.3	3.99%	2.45%
WEB WEBSTER	10662	WEB-345	345	11624.2	13575.3	11813.0	13759.6	188.8	184.3	1.62%	1.36%
BEAVER CREEK WIND PR	11778	J498-345-BT	345	9798.7	12757.5	9943.1	12909.4	144.4	151.9	1.47%	1.19%
BEAVER CREEK WIND PR	10778	J498-345	345	9798.7	12757.5	9943.1	12909.4	144.4	151.9	1.47%	1.19%
RAU RAUN	10840	RAU-345-A	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	485	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	484	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	483	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	482	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	481	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	479	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	478	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	477	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	476	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	475	RAU-345	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
RAU RAUN	474	RAU-345-B	345	27682.8	25917.4	27911.1	26135.3	228.3	217.9	0.82%	0.84%
NLN NEAL NORTH	11542	NLN 3 345	345	26199.5	24808.5	26396.1	25004.5	196.6	196.0	0.75%	0.79%
NL4 NEAL SOUTH	11207	NEAL 4 345	345	25215.9	23438.7	25384.8	23606.8	168.9	168.1	0.67%	0.72%
J534	66000	J534_POI	345	11374.6	13268.4	11424.9	13335.0	50.3	66.6	0.44%	0.50%
J534	66001	J534	345	11281.7	13138.9	11331.0	13204.1	49.3	65.2	0.44%	0.50%

2022 Cost Allocation Results

I.1 Distribution Factor (DF) and MW Contribution Results for Cost Allocation in 2022

Table I-1: Distribution Factor and MW Contribution on Constraints for Thermal NU Cost Allocation in 2022

Table I-2: Distribution Factor and MW Contribution on Voltage / Stability Constraints for NU Cost Allocation in 2022

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I.2 Cost Allocation Details

Table I-3: Network Upgrades Cost Allocation in 2022

Table I-3: Network Upgrades Cost Allocation in 2022

Monitored Element					Cost	J432	J460	J475	J485	J488	J493	J495	J498	J499	J500	J504	J506	J510	J514	J523	J524	J526	J527	J528	J529	J530	J534	J535	Upgrade for			
75736 J530 POI	345	636400 HILLS	3	345	1	\$300,000	\$0	\$0	\$115,118	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
601002 ADAMS 9	345	631046 ADAMS	5	161	605739 ADAMS	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
601006 SPLT RK3	345	652537 WHITE	3	345	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
601031 BRKNGCO3	345	652537 WHITE	3	345	2	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
601054 HAZEL CK3	345	601053 HAZEL	CK4	230	605753 HAZEL	\$4,900,000	\$0	\$2,844,327	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$2,055,673	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
602008 MINVALT4	230	652550 GRANITF4	230	1		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
615529 GRE-PANTHER4	230	619940 HUC-MCLEOD 4	230	1		\$1,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
616002 GRE-JOHNJC77	115	620216 ORTONVL7		115	1	\$16,850,000	\$0	\$0	\$0	\$0	\$0	\$5,663,545	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$11,246,455	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
616002 GRE-JOHNJC77	115	652555 MORRIS 7		115	1	\$8,725,000	\$0	\$0	\$0	\$0	\$0	\$2,901,539	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$5,823,461	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
Big Stone 230-115-13.8 kv xfmr						\$3,250,000	\$0	\$0	\$0	\$0	\$0	\$785,521	\$821,453	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,643,026	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620314 BIGSTON4	230	620325 BROWNSV4		230	1	\$350,000	\$0	\$0	\$0	\$0	\$0	\$78,318	\$90,564	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$181,117	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620314 BIGSTON4	230	652503 BLAIR 4		230	1	\$200,000	\$0	\$0	\$0	\$0	\$0	\$48,343	\$50,552	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$101,106	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620327 HANKSON4	230	620329 WAHPETN4		230	1	\$1,050,000	\$0	\$99,674	\$0	\$0	\$0	\$248,664	\$233,879	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$467,782	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620327 HANKSON4	230	620363 FORMAN 4		230	1	\$400,000	\$0	\$0	\$0	\$0	\$0	\$400,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620329 WAHPETN4	230	658109 FEROSFL4		230	1	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620362 OAKES 4	230	620363 FORMAN 4		230	1	\$950,000	\$0	\$0	\$0	\$0	\$0	\$950,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
620362 OAKES 4	230	601098 ELLENDLMVP4		230	1	\$1,650,000	\$0	\$0	\$0	\$0	\$0	\$1,650,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631004 M-TOWN 7	115	631013 BLRSTWN7		115	1	\$600,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$600,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631013 BLRSTWN7	115	631017 PRKR CK7		115	1	\$1,700,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$1,700,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
631047 LIME CK L1 5	161	631100 LIME CK L2 5	161	2		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631051 HAZLTON L2 5	161	636210 WASHBRN5		161	1	\$16,700,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631079 BNE JCY5	161	630020 SUB T FD 5		161	1	\$7,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$7,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631003 TRAEER 5	161	631006 OVSAT 5		161	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631005 PARNELL 5	161	611113 J430 POI		161	1	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631008 ARNOLD 5	161	631094 FAIRFAXS		161	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631113 POWESHKE 5	161	631117 REASNORS		161	1	\$290,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631142 ARNOLD 3	345	631148 MORGAN VLLY 3	345	1		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$76,715	\$88,935	\$124,351	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631148 ARNOLD 3	345	631148 MORGAN VLLY 3	345	1		\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
631148 MORGAN VLLY 3	345	630420 TIFFIN 3		345	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
635200 RAUN 3	345	645451 S3451 3		345	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
635600 GRIMES 3	345	635700 SYCAMOR3		345	2	\$1,200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$519,976	\$600,024	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
635600 GRIMES 3	345	630003 BVR CRK 3		345	1	\$600,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$318,003	\$0	\$0	\$0	\$0	\$0	\$0	\$21,174	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
635605 GRNGRTP5	161	635614 100 54 5		161	1	\$2,000,000	\$0	\$0	\$0	\$0	\$0	\$889,769	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO		
635600 BONDRTN3	345	635730 MMTZUMA3		345	1	\$300,000	\$4,129	\$8,049	\$0	\$0	\$0	\$6,072	\$6,243	\$7,422	\$34,333	\$43,684	\$60,570	\$0	\$0	\$12,066	\$0	\$1,199	\$3,437	\$12,428	\$30,834	\$24,689	\$13,533	\$0	\$14,700	\$16,532	\$H MISO	
636001 WEBSTERS	161	636020 SUB T FD 5		161	1	\$6,250,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$6,250,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
636203 LUNDQST5	161	636204 DR NE 5		161	1	\$100,000	\$0	\$0	\$0	\$0	\$0	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
636204 DR NE 5	161	636210 WASHBRN5		161	1	\$100,000	\$0	\$0	\$0	\$0	\$0	\$100,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
636230 FRANKLN5	161	636235 WALL LK5		161	1	\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$135,553	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
652503 BLAIR 4	230	652550 GRANITF4		230	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
631017 PRKR CK7	115	631032 SQUARE7		115	1	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
631098 MOOKETA5	161	631099 WYOMING5		161	1	\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
631100 LIBERTY5	161	631101 DUNDEE 5		161	1	\$5,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
Twin Branch-Argenta 345 kv						\$200,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
Benton-T-094 345 kv (N5106)						\$19,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
Benton-T-094 345 kv (N5312.3)						\$1,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$H MISO	
25 Mvar switched cap bank at Lathrop 161 kv (344923)						\$2,000,000	\$37,590	\$74,730	\$53,435	\$0	\$56,218	\$57,039	\$71,687	\$175,383	\$207,536	\$340,912	\$4,232	\$97,214	\$0	\$11,000	\$6,500	\$22,816	\$113,681	\$168,141	\$134,778	\$105,646	\$47,119	\$106,084	\$106,656	Reactive Power NU		
25 Mvar switched cap bank at J430 POI 161 kv (51113)						\$2,500,000	\$29,917	\$56,326	\$22,955	\$0	\$42,960	\$44,968	\$38,168	\$290,525	\$372,767	\$533,626	\$0	\$100,935	\$0	\$28,506	\$89,320	\$270,924	\$216,876	\$100,644	\$0	\$111,693	\$140,805	\$100,644	\$0	Reactive Power NU		
100 Mvar switched cap bank at Montezuma 345 kv (635730)						\$5,000,000	\$57,662	\$112,589	\$612,929	\$0	\$84,800	\$87,186	\$100,516	\$470,728	\$578,935	\$802,791	\$8,546	\$167,016	\$0	\$16,929	\$7,134	\$47,635	\$173,588	\$408,583	\$327,174	\$188,821	\$307,033	\$205,752	\$227,591	Reactive Power NU		
150 Mvar switched cap bank at Blackhawk 345 kv (636199) and 50 Mvar switched cap bank at Midport 161 kv (636202)						\$6,000,000	\$163,941	\$337,169	\$10,811	\$0	\$245,043	\$248,985	\$945,053	\$499,180	\$234,601	\$311,406	\$5,646	\$364,509	\$0	\$47,605	\$56,507	\$113,639	\$496,528	\$155,623	\$123,681	\$640,330	\$0	\$649,340	\$350,243	Reactive Power NU		
2*5 Mvar switched cap bank at Oakes 41.6 kv (620162)						\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	OTP LPC	
Irvine-Powershiek 161 kv						\$11,000,000	\$0	\$0	\$11,000,000	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	NRIS
Total Cost Per Project for Actual NRIS Elections for each Project						\$125,070,000	\$293,239	\$3,532,864	\$11,815,248	\$0	\$4,596,000	\$10,145,953	\$21,607,170	\$12,775,548	\$5,651,436	\$4,825,432	\$841															

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